Application No	
Candidate Name	
Roll No.	
Test Date	21/01/2026
Test Time	9:00 AM - 12:00 PM
Subject	B. Tech

Section : Mathematics Section A

Q.1 If the coefficient of x in the expansion of $(ax^2+bx+c)(1-2x)^{26}$ is -56 and the coefficients of x^2 and x^3 are both zero, then $a+b+c$ is equal to :

Options

1. 1403
2. 1483
3. 1500
4. 1300

x^2: $a-52b+1300c=0$

x^3: $-52a+1300b-20800c=0$

x: $b-52c=-56$

$a=1300,\ b=100,\ c=3$

$a+b+c=1403$

*No elegant sol^n

Binomial Tough XI

Question Type : MCQ
Question ID : 8606541132
Option 1 ID : 8606543851
Option 2 ID : 8606543852
Option 3 ID : 8606543853
Option 4 ID : 8606543850
Status : Not Attempted and Marked For Review
Chosen Option : --

Q.2 Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive terms such that $a_2 \cdot a_3 \cdot a_4 = 64$ and $a_1 + a_3 + a_5 = \frac{813}{7}$.
Then $a_3 + a_5 + a_7$ is equal to :

Options

1. 3256
2. 3252
3. 3244
4. 3248

$a_3 = 4$

$4(1+r^2+r^4)$

$4(1+28+784)$

$813 \times 4 =$

$4\left(r^2+\frac{1}{r^2}+1\right) = \frac{813}{7}$

$r^2+\frac{1}{r^2} = \frac{785}{28}$

$r^2 = 28$

Easy SRS XI

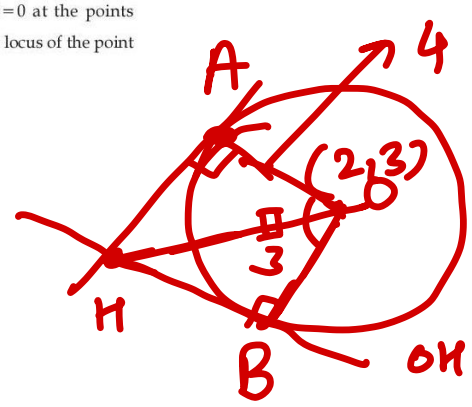
Question Type : MCQ
Question ID : 8606541130
Option 1 ID : 8606543845
Option 2 ID : 8606543844
Option 3 ID : 8606543842
Option 4 ID : 8606543843
Status : Answered
Chosen Option : 2

PRASHANT JAIN

Q.3 Let PQ and MN be two straight lines touching the circle $x^2+y^2-4x-6y-3=0$ at the points A and B respectively. Let O be the centre of the circle and $\angle AOB = \pi/3$. Then the locus of the point of intersection of the lines PQ and MN is :

Options

1. $x^2+y^2-12x-18y-25=0$
2. $3(x^2+y^2)-18x-12y+25=0$
3. $x^2+y^2-18x-12y-25=0$
4. $3(x^2+y^2)-12x-18y-25=0$ ✓



$OH = 4\sec 30^\circ = \frac{8}{\sqrt{3}}$

$(x-2)^2+(y-3)^2 = \frac{64}{3}$

Circle XI Medium

Question Type : MCQ
Question ID : 8606541136
Option 1 ID : 8606543866
Option 2 ID : 8606543868
Option 3 ID : 8606543869
Option 4 ID : 8606543867
Status : Answered
Chosen Option : 4

Q.4 The sum of all the roots of the equation $(x-1)^2-5|x-1|+6=0$, is :

Options

1. 4 ✓
2. 3
3. 1
4. 5

$|x-1| = 2, 3$

$x = 1 \pm 2$
1 ± 3

Form easy XI

Question Type : MCQ
Question ID : 8606541128
Option 1 ID : 8606543834
Option 2 ID : 8606543835
Option 3 ID : 8606543837
Option 4 ID : 8606543836
Status : Answered
Chosen Option : 1

PRASHANT JAIN

Q.5 Let $f : \mathbf{R} \to (0, \infty)$ be a twice differentiable function such that $f(3) = 18, f'(3) = 0$ and $f''(3) = 4$. Then $\lim\limits_{x \to 1} \left(\log_e \left(\frac{f(2+x)}{f(3)} \right)^{\frac{18}{(x-1)^2}} \right)$ is equal to :

Options

1. 2
2. 18
3. 1
4. 9

$\frac{18}{h^2} \times \frac{f(3+h) - f(3)}{f(3)}$

$1 \times \frac{f'(3+h) - f'(3)}{2h} = \frac{f''(3)}{2} = 2$

Limits XII easy

Question Type : MCQ
Question ID : 8606541145
Option 1 ID : 8606543903
Option 2 ID : 8606543905
Option 3 ID : 8606543902
Option 4 ID : 8606543904
Status : Not Attempted and Marked For Review
Chosen Option : --

Q.6 Let $y = y(x)$ be the solution curve of the differential equation $(1+x^2)dy + (y - \tan^{-1}x)\,dx = 0, y(0) = 1$. Then the value of $y(1)$ is :

Options

1. $\frac{2}{e^{\pi/4}} + \frac{\pi}{4} - 1$
2. $\frac{4}{e^{\pi/4}} - \frac{\pi}{2} - 1$
3. $\frac{4}{e^{\pi/4}} + \frac{\pi}{2} - 1$
4. $\frac{2}{e^{\pi/4}} - \frac{\pi}{4} - 1$

$\frac{dy}{dx} + \frac{y}{1+x^2} = \frac{\tan^{-1}x}{1+x^2}$

$(e^{\tan^{-1}x} y) = \int e^{\tan^{-1}x} \frac{(\tan^{-1}x)}{1+x^2}$

$= \int t e^t dt$

$e^t y = te^t - e^t + C$

$y = t - 1 + ce^{-t}$

$\therefore y = t - 1 + 2e^{-t}$

$y(1) = \frac{\pi}{4} - 1 + 2e^{-\pi/4}$

DE XII Tough

Question Type : MCQ
Question ID : 8606541144
Option 1 ID : 8606543899
Option 2 ID : 8606543900
Option 3 ID : 8606543901
Option 4 ID : 8606543898
Status : Answered
Chosen Option : 1

PRASHANT JAIN

Q.7

If the domain of the function $f(x) = \cos^{-1}\left(\frac{2x-5}{11-3x}\right) + \sin^{-1}(2x^2-3x+1)$ is the interval $[\alpha, \beta]$, then $\alpha+2\beta$ is equal to :

Options

1. 3
2. 5
3. 2
4. 1

$-1 \le \frac{2x-5}{11-3x} \le 1$, $-1 \le 2x^2-3x+1 \le 1$

$(-\infty, \frac{16}{5}] \cup (11/3, \infty)$ $\quad x \in [0, \frac{3}{2}]$

$x \in [0, \frac{3}{2}]$

Functions
XII
Tough

Question Type : MCQ
Question ID : 8606541127
Option 1 ID : 8606543832
Option 2 ID : 8606543833
Option 3 ID : 8606543831
Option 4 ID : 8606543830
Status : Answered
Chosen Option : 1

Q.8

Let (α, β, γ) be the co-ordinates of the foot of the perpendicular drawn from the point $(5, 4, 2)$ on the line $\vec{r} = \left(-\hat{i}+3\hat{j}+\hat{k}\right)+\lambda\left(2\hat{i}+3\hat{j}-\hat{k}\right)$.

Then the length of the projection of the vector $\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$ on the vector $6\hat{i}+2\hat{j}+3\hat{k}$ is :

Options

1. 3
2. $\frac{18}{7}$
3. 4
4. $\frac{15}{7}$

$(5,4,2) \to (2\lambda-6, 3\lambda-1, -\lambda-1)$

$(2\lambda-1, 3\lambda+3, -\lambda+1)$

$2(2\lambda-6)+3(3\lambda-1)+(-1)(-\lambda-1)=0$

$14\lambda = 14 \Rightarrow \lambda = 1$

$(\alpha, \beta, \gamma) = (-1, 6, 0)$

$\frac{6+12}{7}$

3-D
XII
easy

Question Type : MCQ
Question ID : 8606541139
Option 1 ID : 8606543879
Option 2 ID : 8606543880
Option 3 ID : 8606543878
Option 4 ID : 8606543881
Status : Not Attempted and Marked For Review
Chosen Option : --

PRASHANT JAIN

Q.9 Let $\vec{a} = -\hat{i} + 2\hat{j} + 2\hat{k}$, $\vec{b} = 8\hat{i} + 7\hat{j} - 3\hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} \times \vec{c} = \vec{b}$.

If $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = 4$, then $|\vec{a} + \vec{c}|^2$ is equal to :

$c = c_1 i + c_2 j + c_3 k$

No elegant soln's (lengthy)

Vector Tough XII

Options

1. 27
2. 30
3. 33
4. 35

Question Type : MCQ
Question ID : 8606541141
Option 1 ID : 8606543886
Option 2 ID : 8606543887
Option 3 ID : 8606543888
Option 4 ID : 8606543889
Status : Answered
Chosen Option : 1

Q.10 Let the mean and variance of 7 observations 2, 4, 10, x, 12, 14, y, $x > y$, be 8 and 16 respectively. Two numbers are chosen from $\{1, 2, 3, x-4, y, 5\}$ one after another without replacement, then the probability, that the smaller number among the two chosen numbers is less than 4, is :

Options

1. $\frac{1}{3}$
2. $\frac{2}{5}$
3. $\frac{3}{5}$
4. $\frac{4}{5}$

$\frac{16+26+x+y}{7} = 8 \Rightarrow x+y=14$

$16+64 = \frac{2^2+4^2+10^2+12^2+14^2+x^2+y^2}{7}$

$\Rightarrow x^2+y^2 = 100$

$x=8, y=6$

$\{1, 2, 3, 4, 5, 6\}$

$P(\text{smaller} \geq 4) = \frac{{}^3C_2}{{}^6C_2} = \frac{3}{15}$

$1 - \frac{3}{15} = \frac{4}{5}$

Question Type : MCQ
Question ID : 8606541133
Option 1 ID : 8606543857
Option 2 ID : 8606543856
Option 3 ID : 8606543855
Option 4 ID : 8606543854
Status : Not Attempted and Marked For Review
Chosen Option : --

Probability XII Medium

PRASHANT JAIN

Q.11

Let the foci of a hyperbola coincide with the foci of the ellipse $\frac{x^2}{36}+\frac{y^2}{16}=1$. If the eccentricity of the hyperbola is 5, then the length of its latus rectum is :

Options

1. 12
2. 16
3. $24\sqrt{5}$
4. $\frac{96}{\sqrt{5}}$

$e = 1-\frac{4}{9}$

$3e = \sqrt{5} \Rightarrow 12e = 4\sqrt{5}$

$2 \times A \times 5 = 4\sqrt{5}$

$A = \frac{2}{\sqrt{5}}$

$LR = \frac{2B^2}{A} = 48A = \frac{96}{\sqrt{5}}$

$\frac{x^2}{A^2} - \frac{y^2}{B^2} = 1$

$1 + \frac{B^2}{A^2} = 25$

$\frac{B^2}{A^2} = 24$

Conic

Easy

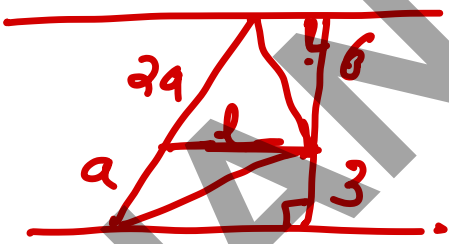
Question Type : MCQ
Question ID : 8606541134
Option 1 ID : 8606543858
Option 2 ID : 8606543859
Option 3 ID : 8606543860
Option 4 ID : 8606543861
Status : Answered
Chosen Option : 4

Q.12

Let a point A lie between the parallel lines L_1 and L_2 such that its distances from L_1 and L_2 are 6 and 3 units, respectively. Then the area (in sq. units) of the equilateral triangle ABC, where the points B and C lie on the lines L_1 and L_2, respectively, is :

Options

1. $12\sqrt{2}$
2. 27
3. $21\sqrt{3}$
4. $15\sqrt{6}$



$l = \sqrt{9a^2 - 2a^2}$

$l = \sqrt{7}\,a$

$\frac{1}{2} l(y) = \frac{\sqrt{3}}{4}(2a)^2$

$\sqrt{7} = \frac{\sqrt{3}}{2} a$

$A = \frac{\sqrt{3}}{4} \times 9a^2$

$A = \frac{\sqrt{3}}{4} \times 9 \times \frac{4}{3} \times 7 = 21\sqrt{3}$

Straight Line

Easy

Question Type : MCQ
Question ID : 8606541135
Option 1 ID : 8606543862
Option 2 ID : 8606543865
Option 3 ID : 8606543864
Option 4 ID : 8606543863
Status : Answered
Chosen Option : 3

PRASHANT JAIN

Q.13

If $x^2+x+1=0$, then the value of $\left(x+\frac{1}{x}\right)^4+\left(x^2+\frac{1}{x^2}\right)^4+\left(x^3+\frac{1}{x^3}\right)^4+\ldots+\left(x^{25}+\frac{1}{x^{25}}\right)^4$ is :

Options

1. 128
2. 145
3. 162
4. 175

$T_r = \left(x^r + \frac{1}{x^r}\right)^4$

$T_r = 1 \quad r = 3k+1, 2$

$T_r = 16 \quad r = 3k$

$16 \times 8 + 1$

145

Easy
Complex No
XII

Question Type : MCQ
Question ID : 8606541129
Option 1 ID : 8606543838
Option 2 ID : 8606543839
Option 3 ID : 8606543840
Option 4 ID : 8606543841
Status : Answered
Chosen Option : 2

Q.14

The value of $\int_{-\pi/6}^{\pi/6}\left(\frac{\pi+4x^{11}}{1-\sin\left(|x|+\pi/6\right)}\right)dx$ is equal to :

Options

1. 8π
2. 4π
3. 2π
4. 6π

$= 2^6 \int_0^{\pi/3} \frac{\pi}{1-\sin\left(x+\pi/6\right)}$

$x+\frac{\pi}{6} = t$

$I = 2\pi \int_{\pi/6}^{\pi/3} \frac{1}{1-\sin t}$

$I = 2\pi \int_{\pi/6}^{\pi/3} \left(\frac{1+\sin t}{\cos^2 t}\right) dt$

$I = 2\pi \left[\tan t + \sec t\right]_{\pi/6}^{\pi/3}$

$I = 4\pi$

DI
XII
Medium

Question Type : MCQ
Question ID : 8606541142
Option 1 ID : 8606543893
Option 2 ID : 8606543891
Option 3 ID : 8606543890
Option 4 ID : 8606543892
Status : Answered
Chosen Option : 2

PRASHANT JAIN

Q.15 The value of $\text{cosec}10^\circ - \sqrt{3}\sec 10^\circ$ is equal to :

Options
1. 4
2. 2
3. 6
4. 8

Trigo XI easy

$\frac{1}{\sin} - \frac{\sqrt{3}}{\cos}$

$\left(\frac{\cos - \sqrt{3}\sin}{2\sin\cos}\right) 4$

$\frac{\cos(10+60^\circ)}{\sin 20} 4 = 4$

Question Type : MCQ
Question ID : 8606541138
Option 1 ID : 8606543875
Option 2 ID : 8606543874
Option 3 ID : 8606543876
Option 4 ID : 8606543877
Status : Answered
Chosen Option : 1

Q.16 The number of strictly increasing functions f from the set $\{1, 2, 3, 4, 5, 6\}$ to the set $\{1, 2, 3, \ldots, 9\}$ such that $f(i) \neq i$ for $1 \leq i \leq 6$, is equal to :

Options
1. 28
2. 22
3. 27
4. 21

$f(1) = 2$ $f(2) = 3$ $f(3) = 4$

$7C_5$ $6C_5$ $5C_5$

28

Sunchony XII Medium

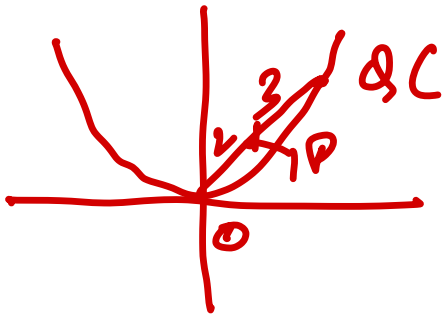
Question Type : MCQ
Question ID : 8606541131
Option 1 ID : 8606543849
Option 2 ID : 8606543847
Option 3 ID : 8606543848
Option 4 ID : 8606543846
Status : Not Attempted and Marked For Review
Chosen Option : --

PRASHANT JAIN

Q.17 Let O be the vertex of the parabola $x^2=4y$ and Q be any point on it. Let the locus of the point P, which divides the line segment OQ internally in the ratio 2 : 3 be the conic C. Then the equation of the chord of C, which is bisected at the point (1, 2), is :

Options

1. $x-2y+3=0$
2. $5x-y-3=0$
3. $4x-5y+6=0$
4. $5x-4y+3=0$



$5k = 2\left(\frac{5h}{4}\right)^2$

$8k = 5h^2$

$5h = 4t$

$5k = 2t^2$

Conic XI Medium

$5x^2 = 8y$

$5x - 4(y+2) = -11$

$5x - 4y + 3 = 0$

Question Type : MCQ
Question ID : 8606541137
Option 1 ID : 8606543873
Option 2 ID : 8606543871
Option 3 ID : 8606543870
Option 4 ID : 8606543872
Status : Answered
Chosen Option : 4

Q.18 Let $\vec{c}$ and $\vec{d}$ be vectors such that $|\vec{c}+\vec{d}| = \sqrt{29}$ and $\vec{c}\times(2\hat{i}+3\hat{j}+4\hat{k}) = (2\hat{i}+3\hat{j}+4\hat{k})\times\vec{d}$. If λ_1, λ_2 $(\lambda_1 > \lambda_2)$ are the possible values of $(\vec{c}+\vec{d})\cdot(-7\hat{i}+2\hat{j}+3\hat{k})$, then the equation $K^2x^2+(K^2-5K+\lambda_1)xy+\left(3K+\frac{\lambda_2}{2}\right)y^2-8x+12y+\lambda_2 = 0$ represents a circle, for K equal to :

Options

1. -1
2. 4
3. 2
4. 1

$(\vec{c}+\vec{d}) \times (2i+3j+4k) = 0$

$\vec{c}+\vec{d} = K(2i+3j+4k)$

$|\vec{c}+\vec{d}| = \sqrt{29}$

$\Rightarrow K = \pm 1$

$\pm[-14+6+12] = \pm 4$

λ_1, λ_2

$K^2 = 3K + \frac{\lambda_2}{2}$

$\Rightarrow \lambda_2 = -4$

$K^2 - 3K + 2 = 0$

$K = 1, 2$

$K^2 - 5K + 4 = 0$

$K = 1, 4$

Question Type : MCQ
Question ID : 8606541140
Option 1 ID : 8606543882
Option 2 ID : 8606543885
Option 3 ID : 8606543884
Option 4 ID : 8606543883
Status : Answered
Chosen Option : 4

Vectors XII Medium

PRASHANT JAIN

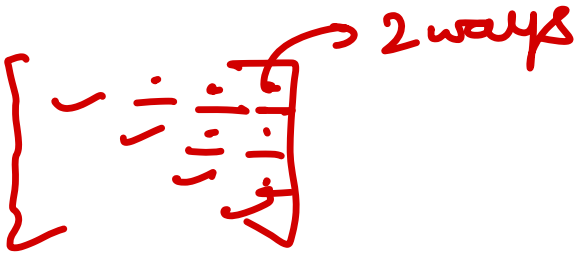
Q.19 The number of relations, defined on the set {a, b, c, d}, which are both reflexive and symmetric, is equal to :

Options

1. 1024
2. 64
3. 256
4. 16

2^6

Relations XII Easy



Question Type : MCQ
Question ID : 8606541126
Option 1 ID : 8606543829
Option 2 ID : 8606543827
Option 3 ID : 8606543828
Option 4 ID : 8606543826
Status : Answered
Chosen Option : 2

Q.20 The area of the region, inside the ellipse $x^2+4y^2=4$ and outside the region bounded by the curves $y=|x|-1$ and $y=1-|x|$, is :

Options

1. $3(\pi-1)$
2. $2\pi-\frac{1}{2}$
3. $2\pi-1$
4. $2(\pi-1)$

$2(\pi-1)$

AUC XII Easy

Question Type : MCQ
Question ID : 8606541143
Option 1 ID : 8606543897
Option 2 ID : 8606543896
Option 3 ID : 8606543894
Option 4 ID : 8606543895
Status : Answered
Chosen Option : 4

Section : Mathematics Section B

Q.21 Let $S=\{(m, n): m, n \in \{1, 2, 3, \ldots, 50\}\}$. If the number of elements (m, n) in S such that 6^m+9^n is a multiple of 5 is p and the number of elements (m, n) in S such that $m+n$ is a square of a prime number is q, then $p+q$ is equal to _____.

Given –

Answer :

$m = 1, 2, 3, \ldots, 50$

$n = 1, 3, 5, \ldots, 49$

$p = 50 \times 25 = 1250$

$6^m \equiv 1 \pmod 5$

$\Rightarrow n$ has to be odd

as $9^n \equiv -1 \pmod 5$

n is odd

Question Type : SA
Question ID : 8606541148
Status : Not Answered

$m+n=4$ → 3

$m+n=9$ → 8

$m+n=25$ → 24

$m+n>49$ → 48

$p+q=83$

$q=83$

PnC XII Tough

PRASHANT JAIN

Q.22

For some $\alpha, \beta \in \mathbf{R}$, let $A = \begin{bmatrix} \alpha & 2 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & \beta \end{bmatrix}$ be such that $A^2 - 4A + 2I = B^2 - 3B + I = O$. Then $(\det(\text{adj}(A^3 - B^3)))^2$ is equal to _____.

Given 225
Answer :

$A^3 = 14A - 8I$
$B^3 = 9B - 3I$
$\alpha = 2 \quad \beta = 2$
$A^3 - B^3 = 14A - 8B - 5I$
$|A^3 - B^3| = -15 = \begin{bmatrix} 15 & 20 \\ 6 & 7 \end{bmatrix}$
225
m&D, XII, Medium

Question Type : SA
Question ID : 8606541146
Status : Answered

Q.23

$6\int_0^{\pi} |(\sin 3x + \sin 2x + \sin x)|\,dx$ is equal to _____.

Given –
Answer :

$6\int_0^{\pi} |2\cos^2 x + \cos x)(\sin x)\,dx$
$\cos x = t$
Super lengthy
$I = 12 \int_{-1}^{1} |2t^2 + t|$
DI XII Tough
$I = 17$

Question Type : SA
Question ID : 8606541150
Status : Not Attempted and Marked For Review

Q.24

Let $f : \mathbf{R} \to \mathbf{R}$ be a twice differentiable function such that the quadratic equation $f(x)m^2 - 2f'(x)m + f''(x) = 0$ in m, has two equal roots for every $x \in \mathbf{R}$. If $f(0) = 1$, $f'(0) = 2$, and (α, β) is the largest interval in which the function $f(\log_e x - x)$ is increasing, then $\alpha + \beta$ is equal to _____.

Given –
Answer :

$(f')^2 = ff''$
$\frac{f'}{f} = \frac{f''}{f'} \Rightarrow f = ce^{2x}$
$\Rightarrow \frac{f'}{f} = 2 \Rightarrow f(x) = e^{2x}$
AoD XII med
$\alpha + \beta = 1$
$g(x) = e^{2(\ln x - x)} = x^2 e^{-2x}$
$g'(x) > 0 \Rightarrow x \in (0, 1]$

Question Type : SA
Question ID : 8606541149
Status : Not Attempted and Marked For Review

Q.25

Let $a_1 = 1$ and for $n \geq 1$, $a_{n+1} = \frac{1}{2}a_n + \frac{n^2 - 2n - 1}{n^2(n+1)^2}$. Then $\left|\sum_{n=1}^{\infty}\left(a_n - \frac{2}{n^2}\right)\right|$ is equal to _____.

Given –
Answer :

$a_{n+1} = \frac{a_n}{2} + \frac{2}{(n+1)^2} - \frac{1}{n^2}$
2
$a_{n+1} - \frac{2}{(n+1)^2} = \frac{1}{2}\left(a_n - \frac{2}{n^2}\right)$
b_n
$b_{n+1} = \frac{b_n}{2}$
$b_1 = a_1 - 2 = -1$
SAS Tough XI
Infinite GP with $a = -1, r = \frac{1}{2}$

Question Type : SA
Question ID : 8606541147
Status : Not Attempted and Marked For Review

Section : Physics Section A

PRASHANT JAIN

NTA JEE Mains Jan 2026

Application No	
Candidate Name	
Roll No.	
Test Date	21/01/2026
Test Time	3:00 PM - 6:00 PM
Subject	B. Tech

Section : **Mathematics Section A**

Q.1 A random variable X takes values 0, 1, 2, 3 with probabilities $\frac{2a+1}{30}, \frac{8a-1}{30}, \frac{4a+1}{30}, b$ respectively, where $a, b \in \mathbf{R}$. Let μ and σ respectively be the mean and standard deviation of X such that $\sigma^2+\mu^2=2$. Then $\frac{a}{b}$ is equal to :

Options

1. 3
2. 12
3. 60
4. 30

$\frac{14a+1}{30} + b = 1$

$\sigma^2+\mu^2 = \sum p_i x_i^2$

$2 = \frac{8a-1}{30} + \left(\frac{4a+1}{30}\right)4 + 9b^2$

$8a+90b=19$

$14a+30b=29$

$a=2, b=\frac{1}{30}$

$\frac{a}{b}=60$

Probability
XII
Medium

Question Type : **MCQ**
Question ID : **860654835**
Option 1 ID : **8606542845**
Option 2 ID : **8606542844**
Option 3 ID : **8606542842**
Option 4 ID : **8606542843**
Status : **Not Attempted and Marked For Review**
Chosen Option : **--**

PRASHANT JAIN

Q.2

For the matrices $A=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}$ and $B=\begin{bmatrix}-29 & 49\\ -13 & 18\end{bmatrix}$, if $(A^{15}+B)\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$, then among the following which one is **true** ?

Options

1. $x=11, y=2$
2. $x=18, y=11$
3. $x=16, y=3$
4. $x=5, y=7$

$A^2-2A+I=0$ $A^2=2A-I$

$A^{15}=15A-14I$

$A^{15}+B=\begin{bmatrix}2 & -11\\ 2 & -11\end{bmatrix}$

M & D easy XII

Question Type : **MCQ**
Question ID : **860654830**
Option 1 ID : **8606542824**
Option 2 ID : **8606542825**
Option 3 ID : **8606542823**
Option 4 ID : **8606542822**
Status : **Not Attempted and Marked For Review**
Chosen Option : **--**

Q.3

The positive integer n, for which the solutions of the equation $x(x+2)+(x+2)(x+4)+\cdots+(x+2n-2)(x+2n)=\frac{8n}{3}$ are two consecutive even integers, is :

Options

1. 6
2. 9
3. 3
4. 12

$nx^2+2x\left(\frac{n^2}{2}\right)+4\left(\frac{n(n^2-1)}{3}\right)-\frac{8n}{3}=0$

$x^2+2nx+\frac{4}{3}(n^2-3)=0$

$\frac{\sqrt{D}}{|a|}=2$

$D=4$

$4\left(n^2-\frac{4}{3}(n^2-3)\right)=4$

$-\frac{n^2}{3}=-3$

$n^2=9$

$n=\pm 3$

Quadratic XI Medium

Question Type : **MCQ**
Question ID : **860654828**
Option 1 ID : **8606542815**
Option 2 ID : **8606542816**
Option 3 ID : **8606542814**
Option 4 ID : **8606542817**
Status : **Marked For Review**
Chosen Option : **1**

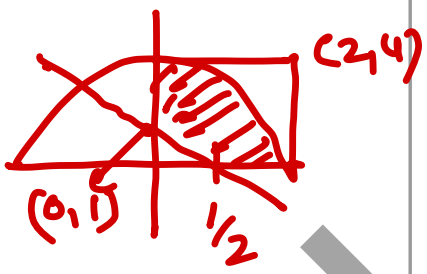
PRASHANT JAIN

Q.4 If the area of the region $\{(x, y) : 1-2x \le y \le 4-x^2, x \ge 0, y \ge 0\}$ is $\frac{\alpha}{\beta}$, $\alpha, \beta \in \mathbf{N}$, gcd $(\alpha, \beta)=1$, then the value of $(\alpha+\beta)$ is :

Options

1. 85
2. 73
3. 91
4. 67

easy AUC XII



$\frac{2}{3} \times 8 - \frac{1}{4} = \frac{61}{12}$

Question Type : **MCQ**
Question ID : **860654845**
Option 1 ID : **8606542884**
Option 2 ID : **8606542883**
Option 3 ID : **8606542885**
Option 4 ID : **8606542882**
Status : **Answered**
Chosen Option : **2**

Q.5 Let $a_1, \frac{a_2}{2}, \frac{a_3}{2^2}, \ldots, \frac{a_{10}}{2^9}$ be a G.P. of common ratio $\frac{1}{\sqrt{2}}$. If $a_1+a_2+\ldots+a_{10}=62$, then a_1 is equal to :

Options

1. $2-\sqrt{2}$
2. $\sqrt{2}-1$
3. $2(2-\sqrt{2})$
4. $2(\sqrt{2}-1)$

$[1, \frac{1}{2}, \frac{1}{2^2} - - , \frac{1}{2^9} \rightarrow r = \frac{1}{2}$

$a_1, \frac{a_2}{2} - - , \frac{a_{10}}{2^9} \rightarrow r = \frac{1}{\sqrt{2}}$

$a_1, a_2, a_3 - - , a_{10} \rightarrow r = \sqrt{2}$

$a\frac{((\sqrt{2})^{10}-1)}{\sqrt{2}-1} = 62$

$a = 2(\sqrt{2}-1)$

SBS XI Easy

Question Type : **MCQ**
Question ID : **860654832**
Option 1 ID : **8606542833**
Option 2 ID : **8606542831**
Option 3 ID : **8606542832**
Option 4 ID : **8606542830**
Status : **Answered**
Chosen Option : **4**

PRASHANT JAIN

Q.6 Let $f: \mathbf{R} \to \mathbf{R}$ be a twice differentiable function such that $f''(x) > 0$ for all $x \in \mathbf{R}$ and $f'(a-1) = 0$, where a is a real number. Let $g(x) = f(\tan^2 x - 2\tan x + a)$, $0 < x < \frac{\pi}{2}$.

Consider the following two statements :

(I) g is increasing in $\left(0, \frac{\pi}{4}\right)$

(II) g is deceasing in $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

Then,

Options

1. Only (I) is True
2. Only (II) is True
3. Neither (I) nor (II) is True
4. Both (I) and (II) are True

AOD
XII
Tough

Question Type : **MCQ**
Question ID : **860654843**
Option 1 ID : **8606542874**
Option 2 ID : **8606542875**
Option 3 ID : **8606542877**
Option 4 ID : **8606542876**
Status : **Answered**
Chosen Option : **1**

Q.7 Let z be the complex number satisfying $|z-5| \le 3$ and having maximum positive principal argument. Then $34\left|\frac{5z-12}{5iz+16}\right|^2$ is equal to :

Options

1. 26
2. 20
3. 16
4. 12

Complex
XII
Medium

Question Type : **MCQ**
Question ID : **860654827**
Option 1 ID : **8606542813**
Option 2 ID : **8606542812**
Option 3 ID : **8606542811**
Option 4 ID : **8606542810**
Status : **Answered**
Chosen Option : **2**

PRASHANT JAIN

Q.8 Let α and β be the roots of the equation $x^2+2ax+(3a+10)=0$ such that $\alpha<1<\beta$. Then the set of all possible values of a is :

Options

1. $(-\infty, \frac{-11}{5}) \cup (5, \infty)$
2. $(-\infty, -3)$
3. $(-\infty, \frac{-11}{5})$
4. $(-\infty, -2) \cup (5, \infty)$

$f(1) < 0$

$1+2a+3a+10<0$

$a < \frac{-11}{5}$

Quadratic
XI
easy

Question Type : **MCQ**
Question ID : **860654829**
Option 1 ID : **8606542821**
Option 2 ID : **8606542819**
Option 3 ID : **8606542820**
Option 4 ID : **8606542818**
Status : **Answered**
Chosen Option : **3**

PRASHANT JAIN

Q.9 If the line $\alpha x + 4y = \sqrt{7}$, where $\alpha \in \mathbf{R}$, touches the ellipse $3x^2+4y^2=1$ at the point P in the first quadrant, then one of the focal distances of P is :

Options

1. $\frac{1}{\sqrt{3}} + \frac{1}{2\sqrt{7}}$
2. $\frac{1}{\sqrt{3}} + \frac{1}{2\sqrt{5}}$
3. $\frac{1}{\sqrt{3}} - \frac{1}{2\sqrt{11}}$
4. $\frac{1}{\sqrt{3}} - \frac{1}{2\sqrt{5}}$

$3xx_1 + 4yy_1 = 1$

$\alpha x + 4y = \sqrt{7}$

$\frac{3x_1}{\alpha} = \frac{y_1}{4} = \frac{1}{\sqrt{7}}$

$3x^2+\frac{y}{7}=1$

$x=\frac{1}{\sqrt{7}}$

$\left(\frac{1}{\sqrt{7}}, \frac{1}{\sqrt{7}}\right)$

$e=1/2$

$fD = e\left[\frac{x_1}{e} \pm \frac{1}{\sqrt{7}}\right) = \frac{1}{\sqrt{3}} \pm \frac{1}{2\sqrt{7}}$

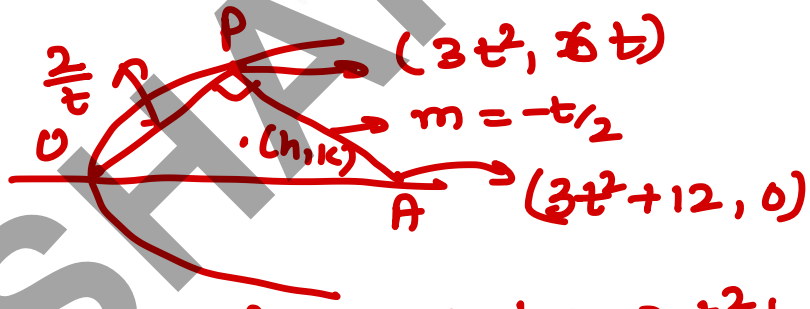
Conic XI Medium

Question Type : **MCQ**
Question ID : **860654838**
Option 1 ID : **8606542856**
Option 2 ID : **8606542854**
Option 3 ID : **8606542857**
Option 4 ID : **8606542855**
Status : **Marked For Review**
Chosen Option : **1**

Q.10 Let $y^2=12x$ be the parabola with its vertex at O. Let P be a point on the parabola and A be a point on the x-axis such that $\angle OPA = 90°$. Then the locus of the centroid of such triangles OPA is :

Options

1. $y^2-6x+4=0$
2. $y^2-2x+8=0$
3. $y^2-4x+8=0$
4. $y^2-9x+6=0$



$3h = 6t^2+12 \Rightarrow h = 2t^2+4$

$3k = 6t \Rightarrow k = 2t$

$x = 2\left(\frac{y^2}{4}\right)+4$

$y^2-2x+8=0$

Conic XI Easy

Question Type : **MCQ**
Question ID : **860654837**
Option 1 ID : **8606542851**
Option 2 ID : **8606542853**
Option 3 ID : **8606542850**
Option 4 ID : **8606542852**
Status : **Answered**
Chosen Option : **2**

PRASHANT JAIN

Q.11

Let $A=\{x:|x^2-10|\le 6\}$ and $B=\{x:|x-2|>1\}$. Then

$A \to 4 \le x^2 \le 16$

$2 \le x \le 4$

$-4 \le x \le -2$

$B: \quad x>3, \; x<1$

Options

1. $A \cup B=(-\infty,1]\cup(2,\infty)$ ✗
2. $A \cap B=[-4,-2]\cup[3,4]$ ✗
3. $B-A=(-\infty,-4)\cup(-2,1)\cup(4,\infty)$ ✓
4. $A-B=[2,3)$

FOM

XI Easy

Question Type : MCQ
Question ID : 860654833
Option 1 ID : 8606542836
Option 2 ID : 8606542835
Option 3 ID : 8606542837
Option 4 ID : 8606542834
Status : Answered
Chosen Option : 4

Q.12

Let $f(x)=x^3+x^2f'(1)+2xf''(2)+f'''(3)$, $x \in \mathbf{R}$. Then the value of $f'(5)$ is :

$f'(1)=a$

$f''(2)=b$

$f'''(3)=c$

$f(x)=x^3+ax^2+2bx+c$

$f'(x)=3x^2+2ax+2b$

$2a+2b+3=a$

$f''(x)=6x+2a$

$12+2a=b$

$a=-\frac{27}{5}$

$b=6/5$

Options

1. $\frac{117}{5}$
2. $\frac{62}{5}$
3. $\frac{2}{5}$
4. $\frac{657}{5}$

$f'(5)=75-54+\frac{12}{5}=\frac{117}{5}$

DOD

XII

Medium

Question Type : MCQ
Question ID : 860654842
Option 1 ID : 8606542873
Option 2 ID : 8606542871
Option 3 ID : 8606542870
Option 4 ID : 8606542872
Status : Answered
Chosen Option : 1

PRASHANT JAIN

Manipulation

Q.13 Let the line L pass through the point $(-3, 5, 2)$ and make equal angles with the positive coordinate axes. If the distance of L from the point $(-2, r, 1)$ is $\sqrt{\frac{14}{3}}$, then the sum of all possible values of r is :

Options

1. 16
2. 12
3. 6
4. 10

3-D
XII
Medium

$\frac{x+3}{1} = \frac{y-5}{1} = \frac{z-2}{1} = \alpha$

$(\alpha-3, \alpha+5, \alpha+2)$

$(-2, r, 1)$

$\sqrt{\frac{14}{3}}$

$(\alpha-1) + \alpha - r + 5 + \alpha + 1 = 0$

$3\alpha - r + 5 = 0$

$\frac{14}{3} = (\alpha-1)^2 + (\alpha)^2 + (\alpha+1)^2$

$\frac{14}{3} = 6\alpha^2 + 2 \Rightarrow 14 = 2(3\alpha)^2 + 6$

$(3\alpha)^2 = 4$

$r^2 - 10r + 21 = 0$

Question Type : **MCQ**
Question ID : **860654839**
Option 1 ID : **8606542861**
Option 2 ID : **8606542860**
Option 3 ID : **8606542858**
Option 4 ID : **8606542859**
Status : **Answered**
Chosen Option : **4**

Q.14 For a triangle ABC, let $\vec{p} = \vec{BC}$, $\vec{q} = \vec{CA}$ and $\vec{r} = \vec{BA}$. If $|\vec{p}| = 2\sqrt{3}$, $|\vec{q}| = 2$ and $\cos\theta = \frac{1}{\sqrt{3}}$, where θ is the angle between $\vec{P}$ and $\vec{q}$, then $\left|\vec{p} \times \left(\vec{q} - 3\vec{r}\right)\right|^2 + 3\left|\vec{r}\right|^2$ is equal to :

Options

1. 410
2. 340
3. 200
4. 220

$|p \times q - 3p \times r|^2 + 3r^2$

$p + q = r$

$p \times q = p \times r$

$4|p \times q|^2 + 3r^2$

$|p \times q|^2 = (12)(4)\left(\frac{2}{3}\right) = 32$

$4 \times 32 + 72$

$128 + 72 = 200$

C, θ, 2, $2\sqrt{3}$, A, B, $\sqrt{4+12+8}$, $\sqrt{24}$

Question Type : **MCQ**
Question ID : **860654841**
Option 1 ID : **8606542869**
Option 2 ID : **8606542868**
Option 3 ID : **8606542866**
Option 4 ID : **8606542867**
Status : **Answered**
Chosen Option : **3**

Vector
XII
Medium

PRASHANT JAIN

Q.15 Let $y = y(x)$ be the solution of the differential equation $\sec x \frac{dy}{dx} - 2y = 2 + 3\sin x$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $y(0) = -\frac{7}{4}$. Then $y\left(\frac{\pi}{6}\right)$ is equal to :

Options

1. $-3\sqrt{3} - 7$
2. $-\frac{5}{4}$
3. $-3\sqrt{2} - 7$
4. $-\frac{5}{2}$

$y' - 2\cos x\, y = 2\cos x + 3\sin x \cos x$

$e^{-2\sin x}\, y = \int e^{-2\sin x}(3\sin x + 2)\cos x\, dx$

$y = -\frac{3}{2}\cos x - \frac{7}{4}$

$y(\pi/6) = -3/4 - 7/4 = -5/2$

DE XII medium

Question Type : MCQ
Question ID : 860654844
Option 1 ID : 8606542878
Option 2 ID : 8606542880
Option 3 ID : 8606542879
Option 4 ID : 8606542881
Status : Not Answered
Chosen Option : --

Q.16 Let $A = \{2, 3, 5, 7, 9\}$. Let R be the relation on A defined by $x R y$ if and only if $2x \le 3y$. Let l be the number of elements in R, and m be the minimum number of elements required to be added in R to make it a symmetric relation. Then $l + m$ is equal to :

Options

1. 25
2. 21
3. 23
4. 27

$y \ge \frac{2x}{3}$

$x=2,\ y = 2,3,5,7,9$

$x=3,\ y = 2,3,5,7,9$

$x=5,\ y = 5,7,9$

$x=7,\ y = 7,9$

$x=9,\ y = 7,9$

$l = 18$

$m = 7$

Relations XII easy

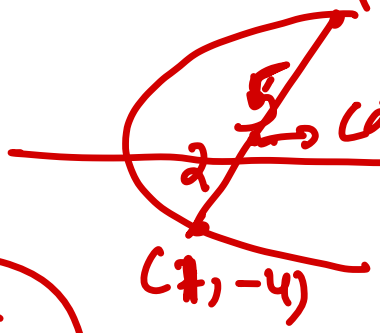
Question Type : MCQ
Question ID : 860654826
Option 1 ID : 8606542808
Option 2 ID : 8606542806
Option 3 ID : 8606542807
Option 4 ID : 8606542809
Status : Answered
Chosen Option : 1

PRASHANT JAIN

Q.17 Let one end of a focal chord of the parabola $y^2 = 16x$ be (16, 16). If P(α, β) divides this focal chord internally in the ratio 5 : 2, then the minimum value of $\alpha + \beta$ is equal to :

Options

1. 5
2. 16
3. 7
4. 22



$\alpha + \beta = 7$

$\alpha = \frac{37}{7}$ $\beta = \frac{12}{7}$

Conic XI easy

Question Type : **MCQ**
Question ID : **860654836**
Option 1 ID : **8606542846**
Option 2 ID : **8606542848**
Option 3 ID : **8606542847**
Option 4 ID : **8606542849**
Status : **Answered**
Chosen Option : **3**

Q.18 If the system of equations

$3x + y + 4z = 3$

$2x + \alpha y - z = -3$

$x + 2y + z = 4$

has no solution, then the value of α is equal to :

$$\begin{vmatrix} 3 & 1 & 4 \\ 2 & \alpha & -1 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

$\alpha = 19$

Options

1. 23
2. 19
3. 13
4. 4

M&D XII easy

Question Type : **MCQ**
Question ID : **860654831**
Option 1 ID : **8606542829**
Option 2 ID : **8606542828**
Option 3 ID : **8606542827**
Option 4 ID : **8606542826**
Status : **Answered**
Chosen Option : **2**

PRASHANT JAIN

Q.19

Let the line L_1 be parallel to the vector $3\hat{i} + 2\hat{j} + 4\hat{k}$ and pass through the point (2, 6, 7), and the line L_2 be parallel to the vector $2\hat{i} + \hat{j} + 3\hat{k}$ and pass through the point (4, 3, 5). If the line L_3 is parallel to the vector $-3\hat{i} + 5\hat{j} + 16\hat{k}$ and intersects the lines L_1 and L_2 at the points C and D, respectively, then $|\overrightarrow{CD}|^2$ is equal to :

Options

1. 89
2. 290
3. 312
4. 171

Question Type : MCQ
Question ID : 860654840
Option 1 ID : 8606542862
Option 2 ID : 8606542864
Option 3 ID : 8606542865
Option 4 ID : 8606542863
Status : Not Answered
Chosen Option : --

Q.20

The largest $n \in \mathbf{N}$, for which 7^n divides 101!, is :

Options

1. 16
2. 18
3. 19
4. 15

Question Type : MCQ
Question ID : 860654834
Option 1 ID : 8606542840
Option 2 ID : 8606542839
Option 3 ID : 8606542838
Option 4 ID : 8606542841
Status : Answered
Chosen Option : 1

Section : Mathematics Section B

PRASHANT JAIN

Q.21 If P is a point on the circle $x^2+y^2=4$, Q is a point on the straight line $5x+y+2=0$ and $x-y+1=0$ is the perpendicular bisector of PQ, then 13 times the sum of abscissa of all such points P is _____.

Given --
Answer :

Question Type : SA
Question ID : 860654847
Status : Not Answered

Q.22 If $\left(\frac{1}{{}^{15}C_0}+\frac{1}{{}^{15}C_1}\right)\left(\frac{1}{{}^{15}C_1}+\frac{1}{{}^{15}C_2}\right)\cdots\left(\frac{1}{{}^{15}C_{12}}+\frac{1}{{}^{15}C_{13}}\right)=\frac{\alpha^{13}}{{}^{14}C_0\,{}^{14}C_1\cdots{}^{14}C_{12}}$, then 30α is equal to _____.

Given --
Answer :

Question Type : SA
Question ID : 860654846
Status : Not Answered

Q.23 Let the maximum value of $(\sin^{-1}x)^2+(\cos^{-1}x)^2$ for $x\in\left[-\frac{\sqrt{3}}{2},\frac{1}{\sqrt{2}}\right]$ be $\frac{m}{n}\pi^2$, where gcd (m, n) = 1.

Then m + n is equal to _____.

Given --
Answer :

Question Type : SA
Question ID : 860654848
Status : Not Answered

Q.24 If $\int_0^1 4\cot^{-1}(1-2x+4x^2)dx = a\tan^{-1}(2)-b\log_e(5)$, where $a,b\in\mathbf{N}$, then $(2a+b)$ is equal to _____.

Given --
Answer :

Question Type : SA
Question ID : 860654850
Status : Not Answered

Q.25 Let $[\cdot]$ denote the greatest integer function and $f(x)=\lim_{n\to\infty}\frac{1}{n^3}\sum_{k=1}^{n}\left[\frac{k^2}{3^x}\right]$. Then $12\sum_{j=1}^{\infty}f(j)$ is equal to _____.

Given --
Answer :

Question Type : SA
Question ID : 860654849
Status : Not Answered

Section : Physics Section A

PRASHANT JAIN

$$I = \int_0^1 \left[\tan^{-1}(2x) - \tan^{-1}(2x-1)\right]$$

$$I = \int_0^1 \tan^{-1}(2x) - \int_0^1 \tan^{-1}(2x-1)$$

$2x-1 = t$

0 to 0

odd fn

$$I = x\tan^{-1}2x\Big|_0^1 - \int \frac{2x\,dx}{1+4x^2}$$

$$I = \tan^{-1}2 - \frac{\ln(1+4x^2)}{4}\Big|_0^1$$

$$I = \tan^{-1}2 - \frac{\ln 5}{4}$$

PRASHANT JAIN

NTA JEE Mains Jan 2026

Application No	
Candidate Name	
Roll No.	
Test Date	22/01/2026
Test Time	9:00 AM - 12:00 PM
Subject	B. Tech

Section : **Mathematics Section A**

Q.1 If the sum of the first four terms of an A.P. is 6 and the sum of its first six terms is 4, then the sum of its first twelve terms is

Options
1. –22
2. –20
3. –24
4. –26

$S_4 = 6 \quad S_6 = 4$

$S_n = \frac{-10}{24}(n-6)(n-4) + 10 - n$

$S_{12} = \frac{(-10)(6)(8)}{24} - 2$

$S_{12} = -22$

S&S XI Easy

Question Type : **MCQ**
Question ID : **444792155**
Option 1 ID : **444792528**
Option 2 ID : **444792527**
Option 3 ID : **444792529**
Option 4 ID : **444792530**
Status : **Not Answered**
Chosen Option : **--**

Q.2 Let $f(x) = x^{2025} - x^{2000}$, $x \in [0, 1]$ and the minimum value of the function $f(x)$ in the interval $[0, 1]$ be $(80)^{80}(n)^{-81}$. Then n is equal to

Options
1. – 41
2. – 81
3. – 80
4. – 40

$f(x) = x^{2000}(x^{25} - 1)$

$f' = 2025x^{2024} - 2000x^{1999}$

$x^{25} = \frac{2000}{2025} = \frac{80}{81}$

$f_{min} = \left(\frac{80}{81}\right)^{80}\left(\frac{80}{81} - 1\right)$

$= (80)^{80}(-81)^{-81}$

Max min XII Medium

Question Type : **MCQ**
Question ID : **444792166**
Option 1 ID : **444792574**
Option 2 ID : **444792572**
Option 3 ID : **444792571**
Option 4 ID : **444792573**
Status : **Answered**
Chosen Option : **2**

PRASHANT JAIN

Q.3 If $A=\begin{bmatrix}2 & 3\\ 3 & 5\end{bmatrix}$, then the determinant of the matrix $(A^{2025}-3A^{2024}+A^{2023})$ is

Options
1. 24
2. 16
3. 28
4. 12

$A^2-7A+I=0$ $\quad |A^{2025}-3A^{2024}+A^{2023}|$

$A^2-3A+I=4A$ $\quad |A^{2023}|\,|A^2-3A+I|$

$|4A| = 16$

M & D

XII

Easy

Question Type : **MCQ**
Question ID : **444792154**
Option 1 ID : **444792525**
Option 2 ID : **444792524**
Option 3 ID : **444792526**
Option 4 ID : **444792523**
Status : **Answered**
Chosen Option : **4**

Q.4 If a random variable x has the probability distribution

x	0	1	2	3	4	5	6	7
$P(x)$	0	$2k$	k	$3k$	$2k^2$	$2k$	k^2+k	$7k^2$

then $P(3 < x \le 6)$ is equal to

Options
1. 0.34
2. 0.33
3. 0.64
4. 0.22

$10k^2+9k=1$

$(k+1)(10k-1)=0 \Rightarrow k=\frac{1}{10}$

$= P(x=4)+P(x=5)+P(x=6)$

$= 3k^2+3k = 3\left(\frac{1}{10}\right)\left(\frac{11}{10}\right) = 0.33$

Probability

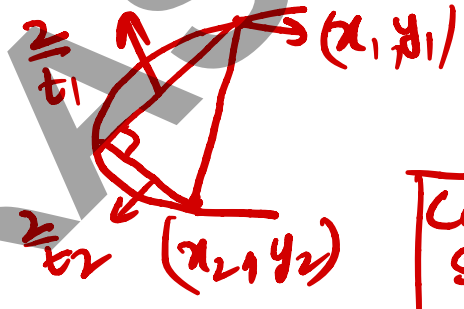
XII

Easy

Question Type : **MCQ**
Question ID : **444792158**
Option 1 ID : **444792539**
Option 2 ID : **444792542**
Option 3 ID : **444792540**
Option 4 ID : **444792541**
Status : **Not Answered**
Chosen Option : **--**

Q.5 If the chord joining the points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ on the parabola $y^2 = 12x$ subtends a right angle at the vertex of the parabola, then $x_1x_2 - y_1y_2$ is equal to

Options
1. 284
2. 292
3. 280
4. 288



$(3t_1^2, 6t_1)$

$(3t_2^2, 6t_2)$

$t_1t_2 = -4$

$x_1x_2 - y_1y_2 = 9(t_1t_2)^2 - 36(t_1t_2)^2$

$= 144 + 144 = 288$

Conic section

Easy

XI

Question Type : **MCQ**
Question ID : **444792160**
Option 1 ID : **444792548**
Option 2 ID : **444792550**
Option 3 ID : **444792547**
Option 4 ID : **444792549**
Status : **Answered**
Chosen Option : **4**

PRASHANT JAIN

Q.6 Let the relation R on the set M = {1, 2, 3,...,16} be given by R={(x, y) : 4y = 5x – 3, x, y ∈ M}.

Then the minimum number of elements required to be added in R, in order to make the relation symmetric, is equal to

Options
1. 4
2. 1
3. 2
4. 3

$5x-4y=3$

3 3

7 8 → (8,7)

11 13 → (13,11)

XII Relations Easy

Question Type : **MCQ**
Question ID : **444792151**
Option 1 ID : **444792513**
Option 2 ID : **444792514**
Option 3 ID : **444792511**
Option 4 ID : **444792512**
Status : **Answered**
Chosen Option : **3**

Q.7 Two distinct numbers *a* and *b* are selected at random from 1, 2, 3,..., 50. The probability, that their product *ab* is divisible by 3, is

Options
1. $\frac{561}{1225}$
2. $\frac{272}{1225}$
3. $\frac{664}{1225}$
4. $\frac{8}{25}$

$3K \to 16$

$3K+1 \to 17$

$3K+2 \to 17$

$1-\frac{{}^{34}C_2}{{}^{50}C_2} = \frac{1225-33\times17}{1225} =$

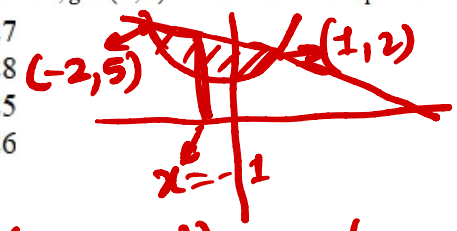
Probability XII Easy

Question Type : **MCQ**
Question ID : **444792157**
Option 1 ID : **444792537**
Option 2 ID : **444792536**
Option 3 ID : **444792535**
Option 4 ID : **444792538**
Status : **Answered**
Chosen Option : **2**

PRASHANT JAIN

Q.8 Let the line $x = -1$ divide the area of the region $\{(x, y) : 1 + x^2 \le y \le 3 - x\}$ in the ratio $m : n$, gcd $(m, n) = 1$. Then $m + n$ is equal to

Options
1. 27
2. 28
3. 25
4. 26



$x^2+1 = 3-x$

$x^2+x-2=0$

$(x+2)(x-1)=0$

Area under Curve
XII
Medium

$$\frac{\int_{-2}^{-1}(2-x-x^2)}{\int_{-1}^{1}2-x-x^2} = \frac{\left(2+\frac{3}{2}-\frac{7}{3}\right)}{4-\frac{2}{3}} = \frac{7}{20}$$

Question Type : **MCQ**
Question ID : **444792168**
Option 1 ID : **444792581**
Option 2 ID : **444792582**
Option 3 ID : **444792579**
Option 4 ID : **444792580**
Status : **Answered**
Chosen Option : **1**

Q.9 The number of solutions of $\tan^{-1}4x+\tan^{-1}6x=\frac{\pi}{6}$, where $-\frac{1}{2\sqrt{6}}<x<\frac{1}{2\sqrt{6}}$, is equal to

Options
1. 3
2. 2
3. 1
4. 0

$(x>0)$

$\frac{10x}{1-24x^2} = \frac{1}{\sqrt{3}}$

$24x^2+10\sqrt{3}x-1=0$

$x = \frac{-10\sqrt{3}\pm\sqrt{396}}{48}$

$x = \frac{\sqrt{396}-10\sqrt{3}}{48} \approx \frac{1}{16}$

I TP
XII
Medium

Question Type : **MCQ**
Question ID : **444792162**
Option 1 ID : **444792558**
Option 2 ID : **444792557**
Option 3 ID : **444792556**
Option 4 ID : **444792555**
Status : **Answered**
Chosen Option : **2**

Q.10 Let $f : [1, \infty) \to \mathbb{R}$ be a differentiable function. If $6\int_1^x f(t)dt = 3xf(x)+x^3-4$ for all $x \ge 1$, then the value of $f(2) - f(3)$ is

Options
1. -4
2. 4
3. -3
4. 3

$x=1$

$0 = 3f(1)-3=0 \Rightarrow f(1)=1$

$y=f(x)=ax^2+bx+c$

$\Rightarrow a+b=1$

$6y = 3y + 3xy' + 3x^2$

$b=2$

$3(ax^2+bx+c) = 3x(2ax+b)+3x^2$

$\Rightarrow 3a = 6a+3 \Rightarrow a=-1$

$3b=3b$

$3c=0 \Rightarrow c=0$

$f(x) = 2x-x^2$

$f(2) = 0 \quad f(3) = -3$

$f(2)-f(3)=2$

Question Type : **MCQ**
Question ID : **444792167**
Option 1 ID : **444792578**
Option 2 ID : **444792575**
Option 3 ID : **444792577**
Option 4 ID : **444792576**
Status : **Answered**
Chosen Option : **3**

DI, XII, Medium

PRASHANT JAIN

Q.11 If the image of the point P(1, 2, a) in the line $\frac{x-6}{3}=\frac{y-7}{2}=\frac{7-z}{2}$ is Q(5, b, c), then $a^2+b^2+c^2$ is equal to

Options
1. 298
2. 293
3. 264
4. 283

Question Type : **MCQ**
Question ID : **444792165**
Option 1 ID : **444792570**
Option 2 ID : **444792569**
Option 3 ID : **444792567**
Option 4 ID : **444792568**
Status : **Answered**
Chosen Option : **2**

3D XII Easy

Q.12 If the line $\alpha x + 2y = 1$, where $\alpha \in \mathbb{R}$, does not meet the hyperbola $x^2 - 9y^2 = 9$, then a possible value of α is:

Options
1. 0.7
2. 0.6
3. 0.8
4. 0.5

Question Type : **MCQ**
Question ID : **444792159**
Option 1 ID : **444792545**
Option 2 ID : **444792544**
Option 3 ID : **444792546**
Option 4 ID : **444792543**
Status : **Not Answered**
Chosen Option : **--**

Conic Medium XII

Q.13 The coefficient of x^{48} in $(1+x)+2(1+x)^2+3(1+x)^3+\ldots+100(1+x)^{100}$ is equal to

Options
1. $100\cdot{}^{100}C_{49}-{}^{100}C_{50}$
2. $100\cdot{}^{100}C_{49}-{}^{100}C_{48}$
3. ${}^{100}C_{50}+{}^{101}C_{49}$
4. $100\cdot{}^{101}C_{49}-{}^{101}C_{50}$

Question Type : **MCQ**
Question ID : **444792156**
Option 1 ID : **444792531**
Option 2 ID : **444792534**
Option 3 ID : **444792532**
Option 4 ID : **444792533**
Status : **Answered**
Chosen Option : **2**

Binomial Tough XI

PRASHANT JAIN

Q.14 Let the solution curve of the differential equation $xdy - ydx = \sqrt{x^2+y^2}\,dx$, $x > 0$, $y(1) = 0$, be $y = y(x)$. Then $y(3)$ is equal to

Options 1. 1
2. 4
3. 2
4. 6

$y = vx$

$dy = v\,dx + x\,dv$

$x(v\,dx + x\,dv) - vx\,dx = (\sqrt{1+v^2})\,x\,dx$

$x = 1 \quad v = 0$

$x\,dv = \sqrt{1+v^2}\,dx$

$\ln(v + \sqrt{1+v^2}) = \ln x + C$

$v + \sqrt{1+v^2} = x$

$y + \sqrt{y^2+9} = 9$

Question Type : **MCQ**
Question ID : **444792169**
Option 1 ID : **444792583**
Option 2 ID : **444792585**
Option 3 ID : **444792584**
Option 4 ID : **444792586**
Status : **Answered**
Chosen Option : **1**

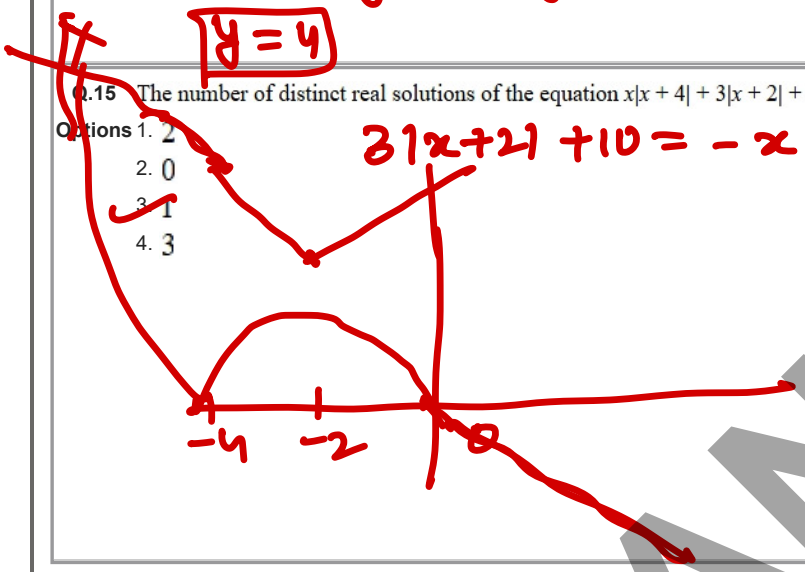
DE
XII
medium

Q.15 The number of distinct real solutions of the equation $x|x+4| + 3|x+2| + 10 = 0$ is

Options 1. 2
2. 0
3. 1
4. 3

$y = 4$

$3|x+2| + 10 = -x|x+4|$



functions
XII
Easy

Question Type : **MCQ**
Question ID : **444792153**
Option 1 ID : **444792521**
Option 2 ID : **444792519**
Option 3 ID : **444792520**
Option 4 ID : **444792522**
Status : **Not Answered**
Chosen Option : **--**

PRASHANT JAIN

Q.16

The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1}{[x]+4}\right) dx$, where $[\cdot]$ denotes the greatest integer function, is

Options

1. $\frac{7}{60}(3\pi-1)$
2. $\frac{7}{60}(\pi-3)$
3. $\frac{1}{60}(21\pi-1)$
4. $\frac{1}{60}(\pi-7)$

Question Type : **MCQ**
Question ID : **444792170**
Option 1 ID : **444792587**
Option 2 ID : **444792588**
Option 3 ID : **444792590**
Option 4 ID : **444792589**
Status : **Answered**
Chosen Option : **2**

Q.17

If the domain of the function $f(x)=\sin^{-1}\left(\frac{5-x}{3+2x}\right)+\frac{1}{\log_e(10-x)}$ is $(-\infty, \alpha] \cup [\beta, \gamma)-\{\delta\}$, then $6(\alpha+\beta+\gamma+\delta)$ is equal to

Options

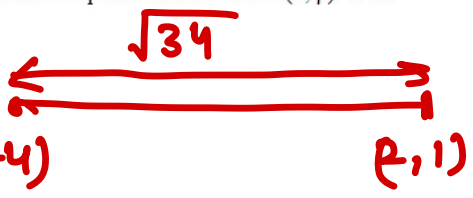
1. 70
2. 66
3. 67
4. 68

Question Type : **MCQ**
Question ID : **444792152**
Option 1 ID : **444792518**
Option 2 ID : **444792515**
Option 3 ID : **444792516**
Option 4 ID : **444792517**
Status : **Answered**
Chosen Option : **3**

PRASHANT JAIN

Q.18 Let the set of all values of r, for which the circles $(x+1)^2+(y+4)^2=r^2$ and $x^2+y^2-4x-2y-4=0$ intersect at two distinct points be the interval (α, β). Then $\alpha\beta$ is equal to

Options
1. 24
2. 21
3. 25
4. 20



Circle XI Easy

$|r-3| < \sqrt{34} < (r+3)$

$\sqrt{34}-3 < r < \sqrt{34}+3$

Question Type : **MCQ**
Question ID : **444792161**
Option 1 ID : **444792552**
Option 2 ID : **444792553**
Option 3 ID : **444792551**
Option 4 ID : **444792554**
Status : **Answered**
Chosen Option : **2**

Q.19 Let $P(\alpha, \beta, \gamma)$ be the point on the line $\frac{x-1}{2}=\frac{y+1}{-3}=z$ at a distance $4\sqrt{14}$ from the point $(1, -1, 0)$ and nearer to the origin. Then the shortest distance, between the lines $\frac{x-\alpha}{1}=\frac{y-\beta}{2}=\frac{z-\gamma}{3}$ and $\frac{x+5}{2}=\frac{y-10}{1}=\frac{z-3}{1}$, is equal to

Options
1. $2\sqrt{\frac{7}{4}}$
2. $4\sqrt{\frac{5}{7}}$
3. $7\sqrt{\frac{5}{4}}$
4. $4\sqrt{\frac{7}{5}}$

Standard method

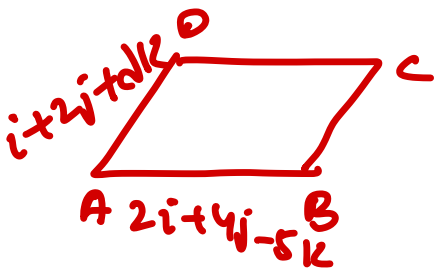
3-D XII Medium

Question Type : **MCQ**
Question ID : **444792164**
Option 1 ID : **444792564**
Option 2 ID : **444792563**
Option 3 ID : **444792565**
Option 4 ID : **444792566**
Status : **Answered**
Chosen Option : **3**

PRASHANT JAIN

Q.20 Let $\overrightarrow{AB}=2\hat{i}+4\hat{j}-5\hat{k}$ and $\overrightarrow{AD}=\hat{i}+2\hat{j}+\lambda\hat{k}, \lambda \in \mathbb{R}$. Let the projection of the vector $\vec{v}=\hat{i}+\hat{j}+\hat{k}$ on the diagonal $\overrightarrow{AC}$ of the parallelogram ABCD be of length one unit. If α, β, where $\alpha > \beta$, be the roots of the equation $\lambda^2x^2 - 6\lambda x + 5 = 0$, then $2\alpha - \beta$ is equal to

Options
1. 4
2. 6
3. 1
4. 3 ✓



$AC = 3i + 6j + (d-5)k$

$v \cdot AC = 1 \Rightarrow d = 3$

$9x^2 - 18x + 5 = 0$

$(3x-1)(3x-5)=0$

$\frac{10}{3} - \frac{1}{3} = 3$

Vector XII medium

Question Type : **MCQ**
Question ID : **444792163**
Option 1 ID : **444792561**
Option 2 ID : **444792562**
Option 3 ID : **444792559**
Option 4 ID : **444792560**
Status : **Answered**
Chosen Option : **2**

Section : **Mathematics Section B**

Q.21 If $\int (\sin x)^{\frac{-11}{2}} (\cos x)^{\frac{-5}{2}} dx = -\frac{p_1}{q_1}(\cot x)^{\frac{9}{2}} - \frac{p_2}{q_2}(\cot x)^{\frac{5}{2}} - \frac{p_3}{q_3}(\cot x)^{\frac{1}{2}} + \frac{p_4}{q_4}(\cot x)^{\frac{-3}{2}} + C$, where p_i and q_i are positive integers with $gcd(p_i, q_i) = 1$ for $i = 1, 2, 3, 4$ and C is the constant of integration, then $\frac{15p_1p_2p_3p_4}{q_1q_2q_3q_4}$ is equal to _____

16

Given --
Answer :

$\int t^{-11/2}(1+t^2)^3 dx$ $\tan x = t$

$\int t^{-11/2} + 3t^{-7/2} + 3t^{-3/2} + t^{1/2}\, dt$

$= -\frac{2}{9}c^{9/2} - \frac{6}{5}c^{5/2} - 6c^{1/2} + \frac{2}{3}c^{-3/2} + C_1$

$c \to \cot x$

IIT XII Tough

Question Type : **SA**
Question ID : **444792175**
Status : **Not Answered**

Q.22 Let $\alpha = \frac{-1+i\sqrt{3}}{2}$ and $\beta = \frac{-1-i\sqrt{3}}{2}$, $i = \sqrt{-1}$. If

$(7 - 7\alpha + 9\beta)^{20} + (9 + 7\alpha - 7\beta)^{20} + (-7 + 9\alpha + 7\beta)^{20} + (14 + 7\alpha + 7\beta)^{20} = m^{10}$,

then m is _____

49

Given --
Answer :

$\alpha = \omega \quad \beta = \omega^2$

$(7 + 9\omega^2 - 7\omega)^{20} + (9 + 7\omega - 7\omega^2)^{20} \ldots + 7^{20} = m^{10}$

$(7 + 9\omega^2 - 7\omega) = \omega[7\omega^2 + 9\omega - 7]$

$(7 + 9\omega^2 - 7\omega)(1 + \omega^{20} + \omega^{39}) = 0$

$m^{10} = 49^{10}$

Complex NO XII medium

Question Type : **SA**
Question ID : **444792171**
Status : **Not Answered**

PRASHANT JAIN

Q.23 If $\frac{\cos^2 48° - \sin^2 12°}{\sin^2 24° - \sin^2 6°} = \frac{\alpha + \beta\sqrt{5}}{2}$, where $\alpha, \beta \in N$, then $\alpha + \beta$ is equal to _____

Given Answer : **5**

Question Type : **SA**
Question ID : **444792174**
Status : **Answered**

Q.24 Let ABC be a triangle. Consider four points p_1, p_2, p_3, p_4 on the side AB, five points p_5, p_6, p_7, p_8, p_9 on the side BC, and four points $p_{10}, p_{11}, p_{12}, p_{13}$ on the side AC. None of these points is a vertex of the triangle ABC. Then the total number of pentagons, that can be formed by taking all the vertices from the points $p_1, p_2, \cdots, p_{13}$, is _____

Given Answer : **660**

Question Type : **SA**
Question ID : **444792173**
Status : **Not Answered**

Q.25 Let A be a 3×3 matrix such that $A + A^T = O$. If $A\begin{bmatrix}1\\-1\\0\end{bmatrix} = \begin{bmatrix}3\\3\\2\end{bmatrix}, A^2\begin{bmatrix}1\\-1\\0\end{bmatrix} = \begin{bmatrix}-3\\19\\-24\end{bmatrix}$ and $det(adj(2\ adj\ (A + I))) = (2)^\alpha \cdot (3)^\beta \cdot (11)^\gamma$, α, β, γ are non-negative integers, then $\alpha + \beta + \gamma$ is equal to _____

Given Answer : **18**

Question Type : **SA**
Question ID : **444792172**
Status : **Not Answered**

Section : **Physics Section A**

PRASHANT JAIN

NTA JEE Mains Jan 2026

Application No	
Candidate Name	
Roll No.	
Test Date	22/01/2026
Test Time	3:00 PM - 6:00 PM
Subject	B. Tech

Section : **Mathematics Section A**

Q.1 Let α, β be the roots of the quadratic equation $12x^2 - 20x + 3\lambda = 0, \lambda \in \mathbf{Z}$. If $\frac{1}{2} \leq |\beta - \alpha| \leq \frac{3}{2}$, then the sum of all possible values of λ is :

Options

1. 3
2. 1
3. 4
4. 6

Question Type : **MCQ**
Question ID : **860654978**
Option 1 ID : **8606543325**
Option 2 ID : **8606543324**
Option 3 ID : **8606543326**
Option 4 ID : **8606543327**
Status : **Answered**
Chosen Option : **1**

Q.2 The number of elements in the relation $R = \{(x, y) : 4x^2 + y^2 < 52, x, y \in \mathbf{Z}\}$ is

Options

1. 67
2. 86
3. 77
4. 89

Question Type : **MCQ**
Question ID : **860654976**
Option 1 ID : **8606543316**
Option 2 ID : **8606543318**
Option 3 ID : **8606543317**
Option 4 ID : **8606543319**
Status : **Answered**
Chosen Option : **3**

PRASHANT JAIN

Q.3 Let $f(x)=[x]^2-[x+3]-3$, $x\in\mathbf{R}$, where $[\cdot]$ is the greatest integer funtion. Then

Options

1. $f(x)=0$ for finitely many values of x
2. $f(x)<0$ only for $x\in[-1,3)$
3. $f(x)>0$ only for $x\in[4,\infty)$
4. $\int_0^2 f(x)dx=-6$

Functions XII Medium

Question Type : **MCQ**
Question ID : **860654993**
Option 1 ID : **8606543384**
Option 2 ID : **8606543385**
Option 3 ID : **8606543386**
Option 4 ID : **8606543387**
Status : **Not Answered**
Chosen Option : **--**

Q.4 Let S and S′ be the foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$ and P(α, β) be a point on the ellipse in the first quadrant. If $(SP)^2+(S'P)^2-SP\cdot S'P=37$, then $\alpha^2+\beta^2$ is equal to :

Options

1. 13
2. 15
3. 11
4. 17

Conic XI Medium

Question Type : **MCQ**
Question ID : **860654985**
Option 1 ID : **8606543353**
Option 2 ID : **8606543354**
Option 3 ID : **8606543352**
Option 4 ID : **8606543355**
Status : **Answered**
Chosen Option : **1**

PRASHANT JAIN

Q.5 Let C_r denote the coefficient of x^r in the binomial expansion of $(1+x)^n$, $n \in \mathbf{N}$, $0 \le r \le n$. If $P_n = C_0 - C_1 + \frac{2^2}{3}C_2 - \frac{2^3}{4}C_3 + \ldots\ldots + \frac{(-2)^n}{n+1}C_n$, then the value of $\sum_{n=1}^{25} \frac{1}{P_{2n}}$ equals.

Options

1. 580
2. 650
3. 675
4. 525

Binomial XI Medium

$T_r = (-1)^r {}^nC_r \frac{2^r}{r+1}$

$T_r = \frac{(-1)^r}{2} {}^{n+1}C_{r+1} 2^{r+1}$

$S = \frac{-1}{2(n+1)}(1-(-1)^{n+1})$

$S_{2k} = \frac{1}{2k+1}$

$\sum_{k=1}^{25}(2k+1)$

$(3+5--+51)$

$= 26^2 - 1$

$= 675$

Question Type : **MCQ**
Question ID : **860654984**
Option 1 ID : **8606543351**
Option 2 ID : **8606543350**
Option 3 ID : **8606543348**
Option 4 ID : **8606543349**
Status : **Not Answered**
Chosen Option : **--**

Q.6 Let n be the number obtained on rolling a fair die. If the probability that the system

$x - ny + z = 6$
$x + (n-2)y + (n+1)z = 8$
$(n-1)y + z = 1$

has a unique solution is $\frac{k}{6}$, then the sum of k and all possible values of n is :

Options

1. 22
2. 21
3. 20
4. 24

Probability XI Medium

$\begin{vmatrix} 1 & -n & 1 \\ 1 & n-2 & n+1 \\ 0 & n-1 & 1 \end{vmatrix} \neq 0$

$n^2 - 3n + 2 \neq 0$

$n \neq 1, 2$

$n = 3, 4, 5, 6$

$\frac{4}{6} = \frac{2}{3} \to K$

$4 + 3 + 4 + 5 + 6 = 22$

Question Type : **MCQ**
Question ID : **860654982**
Option 1 ID : **8606543342**
Option 2 ID : **8606543341**
Option 3 ID : **8606543340**
Option 4 ID : **8606543343**
Status : **Not Answered**
Chosen Option : **--**

PRASHANT JAIN

Q.7

Let P(10, $2\sqrt{15}$) be a point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, whose foci are S and S'. If the length of its latus rectum is 8, then the square of the area of Δ PSS' is equal to :

Options

1. 2700
2. 4200
3. 1462
4. 900

$2ae = 6\sqrt{5}$

$\Delta = \frac{1}{2} \times 6\sqrt{5} \times 2\sqrt{15}$

$\Delta^2 = 75 \times 36 = 2700$

$\frac{100}{a^2} - \frac{60}{b^2} = 1$

$\frac{2b^2}{a} = 8 \quad 4 \quad b^2 = 4a$

$\frac{100}{a^2} - \frac{60}{4a} = 1$

$a^2 + 15a - 100 = 0$

$a = 5$

$b = \sqrt{20}$

Conic Section XI Medium

Question Type : **MCQ**
Question ID : **860654986**
Option 1 ID : **8606543357**
Option 2 ID : **8606543359**
Option 3 ID : **8606543358**
Option 4 ID : **8606543356**
Status : **Not Answered**
Chosen Option : **--**

Q.8

Let L be the line $\frac{x+1}{2}=\frac{y+1}{3}=\frac{z+3}{6}$ and let S be the set of all points (a, b, c) on L, whose distance from the line $\frac{x+1}{2}=\frac{y+1}{3}=\frac{z-9}{0}$ along the line L is 7. Then $\sum_{(a,b,c)\in S}(a+b+c)$ is equal to :

Options

1. 28
2. 34
3. 6
4. 40

$= d$

$= k$

$(2d-1, 3d-1, 6d-3)$

$(2k-1, 3k-1, 9)$

$d = 2$

$(3, 5, 9)$

$(a_1, b_1, c_1) \xrightarrow{7} \xrightarrow{7} (a_2, b_2, c_2)$

$a_1 + a_2 = 6$

$b_1 + b_2 = 10$

$c_1 + c_2 = 18$

$\Sigma(a_1 + b_1 + c_1) = 34$

30 Medium XII

Question Type : **MCQ**
Question ID : **860654990**
Option 1 ID : **8606543373**
Option 2 ID : **8606543374**
Option 3 ID : **8606543372**
Option 4 ID : **8606543375**
Status : **Not Answered**
Chosen Option : **--**

PRASHANT JAIN

Q.9

If $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is a solution of the system of equations $AX = B$, where $\text{adj } A = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$, then $|x+y+z|$ is equal to :

Options

1. $\frac{3}{2}$
2. 2
3. 1
4. 3

$A\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1}\begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$

$= \frac{\text{adj } A}{|A|}\begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{\pm 10}\begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}\begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$

$= \pm \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$

$|x+y+z| = 2$

M&D XII Tough

Question Type : **MCQ**
Question ID : **860654981**
Option 1 ID : **8606543337**
Option 2 ID : **8606543338**
Option 3 ID : **8606543336**
Option 4 ID : **8606543339**
Status : **Not Answered**
Chosen Option : **--**

Q.10 If the mean deviation about the median of the numbers k, 2k, 3k, ..., 1000k is 500, then k^2 is equal to :

Options

1. 4
2. 9
3. 1
4. 16

$M = 500.5K$

$MD = \frac{K + 3K + 5K - - + 999K}{1000}$

$\frac{(500)(500)K}{1000} = 500$

$K = 2$

$K^2 = 4$

Statistics XII Easy

Question Type : **MCQ**
Question ID : **860654983**
Option 1 ID : **8606543346**
Option 2 ID : **8606543344**
Option 3 ID : **8606543347**
Option 4 ID : **8606543345**
Status : **Answered**
Chosen Option : **1**

PRASHANT JAIN

Q.11

Let the domain of the function $f(x) = \log_3\log_5\left(7 - \log_2\left(x^2 - 10x + 85\right)\right) + \sin^{-1}\left(\left|\frac{3x-7}{17-x}\right|\right)$ be $(\alpha, \beta]$. Then $\alpha+\beta$ is equal to :

Options

1. 9
2. 8
3. 12
4. 10

Functions XII Medium

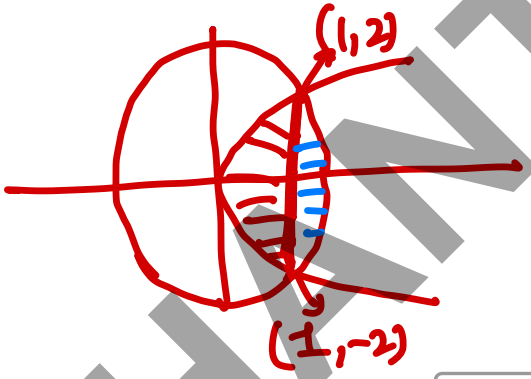
Question Type : MCQ
Question ID : **860654979**
Option 1 ID : **8606543329**
Option 2 ID : **8606543328**
Option 3 ID : **8606543331**
Option 4 ID : **8606543330**
Status : **Not Answered**
Chosen Option : **--**

Q.12

The area of the region $A = \{(x, y) : 4x^2 + y^2 \leq 8 \text{ and } y^2 \leq 4x\}$ is :

Options

1. $\pi + \frac{2}{3}$
2. $\frac{\pi}{2} + \frac{1}{3}$
3. $\pi + 4$
4. $\frac{\pi}{2} + 2$



Question Type : MCQ
Question ID : **860654994**
Option 1 ID : **8606543389**
Option 2 ID : **8606543388**
Option 3 ID : **8606543391**
Option 4 ID : **8606543390**
Status : **Not Answered**
Chosen Option : **--**

AUC XII Medium

PRASHANT JAIN

Q.13

Let $S = \{z \in \mathbb{C} : 4z^2 + \bar{z} = 0\}$. Then $\sum_{z \in S} |z|^2$ is equal to :

Options

1. $\frac{3}{16}$
2. $\frac{5}{64}$
3. $\frac{7}{64}$
4. $\frac{1}{16}$

$4z^2 = -\bar{z} \Rightarrow 4|z|^2 = |z| \Rightarrow |z| = 0$
$|z| = \frac{1}{4}$

$|z| = \frac{1}{4} \rightarrow$ 3 sol^{n}s

$\frac{1}{16} + \frac{1}{16} + \frac{1}{16}$

Complex No.
XII
Easy

Question Type : **MCQ**
Question ID : **860654980**
Option 1 ID : **8606543333**
Option 2 ID : **8606543332**
Option 3 ID : **8606543335**
Option 4 ID : **8606543334**
Status : **Not Answered**
Chosen Option : **--**

Q.14

If $\lim_{x \to 0} \frac{e^{(a-1)x} + 2\cos bx + (c-2)e^{-x}}{x\cos x - \log_e(1+x)} = 2$, then $a^2 + b^2 + c^2$ is equal to :

Options

1. 3
2. 5
3. 9
4. 7

$1 + (a-1)x + \frac{(a-1)^2x^2}{2} + 2\left(1 - \frac{b^2x^2}{2}\right) + (c-2)\left(1 - x + \frac{x^2}{2}\right)$

$x\left(1 - \frac{x^2}{2}\right) - x + \frac{x^2}{2}$

$= 2$

$c + 1 = 0 \Rightarrow c = -1$
$a - c = -1 \Rightarrow a = -2$
$b^2 = 2$

$a^2 + b^2 + c^2 = 7$

Question Type : **MCQ**
Question ID : **860654992**
Option 1 ID : **8606543380**
Option 2 ID : **8606543381**
Option 3 ID : **8606543383**
Option 4 ID : **8606543382**
Status : **Answered**
Chosen Option : **3**

Limits
XII
Medium

PRASHANT JAIN

Q.15 Among the statements

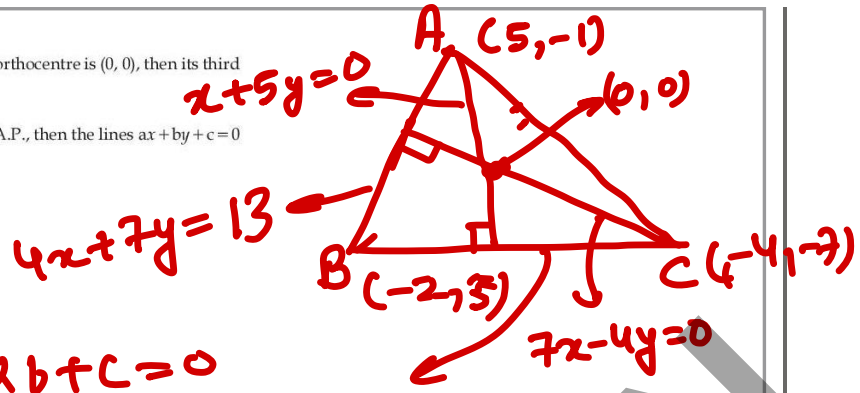
(S1) : If A(5, −1) and B(−2, 3) are two vertices of a triangle, whose orthocentre is (0, 0), then its third vertex is (−4, −7)

and

(S2) : If positive numbers 2a, b, c are three consecutive terms of an A.P., then the lines $ax+by+c=0$ are concurrent at (2, −2),

Options

1. only (S1) is correct
2. both are correct
3. only (S2) is correct
4. both are incorrect



2a-2b+c=0

(2,-2)

straight line

XI

Medium

Question Type : **MCQ**
Question ID : **860654988**
Option 1 ID : **8606543366**
Option 2 ID : **8606543364**
Option 3 ID : **8606543367**
Option 4 ID : **8606543365**
Status : **Not Answered**
Chosen Option : **--**

Q.16 Let f and g be functions satisfying $f(x+y)=f(x)f(y)$, $f(1)=7$ and $g(x+y)=g(xy)$, $g(1)=1$, for all $x, y \in N$. If $\sum_{x=1}^{n}\left(\frac{f(x)}{g(x)}\right) = 19607$, then n is equal to :

Options

1. 6
2. 7
3. 5
4. 4

$f(x) = 7^x$

$g(x+1) = g(x) \Rightarrow g(x)=1$

$\sum_{x=1}^{n} 7^x = 19607$

$7+7^2+7^3+7^4+7^5 = 19607$

n=5

Functions

XI

medium

Question Type : **MCQ**
Question ID : **860654977**
Option 1 ID : **8606543322**
Option 2 ID : **8606543323**
Option 3 ID : **8606543321**
Option 4 ID : **8606543320**
Status : **Answered**
Chosen Option : **3**

PRASHANT JAIN

Q.17 Let $\vec{a}=2\hat{i}-\hat{j}+\hat{k}$ and $\vec{b}=\lambda\hat{j}+2\hat{k}$, $\lambda\in Z$ be two vectors. Let $\vec{c}=\vec{a}\times\vec{b}$ and $\vec{d}$ be a vector of magnitude 2 in yz-plane. If $|\vec{c}|=\sqrt{53}$, then the maximum possible value of $\left(\vec{c}\cdot\vec{d}\right)^2$ is equal to :

Options

1. 26
2. 104
3. 52
4. 208

Vector XII Tough

Question Type : **MCQ**
Question ID : **860654989**
Option 1 ID : **8606543368**
Option 2 ID : **8606543370**
Option 3 ID : **8606543369**
Option 4 ID : **8606543371**
Status : **Not Answered**
Chosen Option : **--**

Q.18 Let $[\cdot]$ denote the greatest integer function, and let $f(x)=\min\{\sqrt{2}x, x^2\}$.

Let $S=\{x\in(-2,2) : \text{the function } g(x)=|x|[x^2] \text{ is discontinuous at } x\}$.

Then $\sum_{x\in S} f(x)$ equals

Options

1. $2\sqrt{6}-3\sqrt{2}$
2. $\sqrt{6}-2\sqrt{2}$
3. $2-\sqrt{2}$
4. $1-\sqrt{2}$

AOD XII Medium

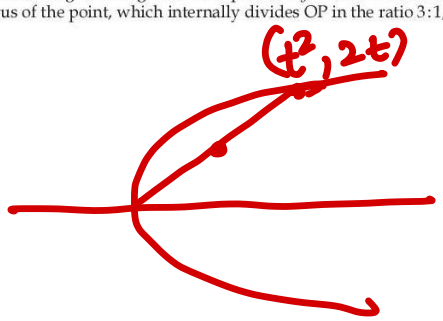
Question Type : **MCQ**
Question ID : **860654991**
Option 1 ID : **8606543379**
Option 2 ID : **8606543378**
Option 3 ID : **8606543376**
Option 4 ID : **8606543377**
Status : **Not Answered**
Chosen Option : **--**

PRASHANT JAIN

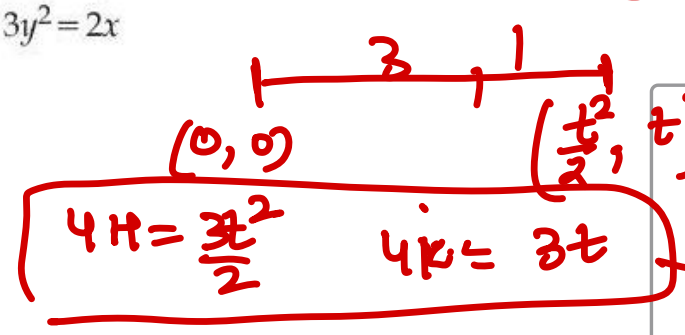
Q.19 Let the locus of the mid-point of the chord through the origin O of the parabola $y^2 = 4x$ be the curve S. Let P be any point on S. Then the locus of the point, which internally divides OP in the ratio 3:1, is :

Options

1. $2x^2 = 3y$
2. $2y^2 = 3x$
3. $3x^2 = 2y$
4. $3y^2 = 2x$



$2h = t^2$
$2k = 2t$
$2x = y^2 \rightarrow S$



$4H = \frac{3t^2}{2}$ $4k = 3t$

$8y^2 = 9x$

Conic XI Medium

Question Type : **MCQ**
Question ID : **860654987**
Option 1 ID : **8606543362**
Option 2 ID : **8606543360**
Option 3 ID : **8606543363**
Option 4 ID : **8606543361**
Status : **Answered**
Chosen Option : **2**

Q.20 If $y = y(x)$ satisfies the differential equation

$16\left(\sqrt{x + 9\sqrt{x}}\right)\left(4 + \sqrt{9 + \sqrt{x}}\right)\cos y \, dy = (1 + 2\sin y)dx, x > 0$ and $y(256) = \frac{\pi}{2}$, $y(49) = \alpha$, then $2\sin\alpha$ is equal to :

Options

1. $2\sqrt{2} - 1$
2. $2(\sqrt{2} - 1)$
3. $\sqrt{2} - 1$
4. $3(\sqrt{2} - 1)$

$\frac{16\cos y \, dy}{1 + 2\sin y} = \frac{dx}{\left(\sqrt{x + 9\sqrt{x}}\right)\left(4 + \sqrt{9 + \sqrt{x}}\right)}$

$8\ln|1 + 2\sin y| = \int \frac{4\,dx}{4\sqrt{x + 9\sqrt{x}}\left(4 + \sqrt{9 + \sqrt{x}}\right)}$

$= \int \frac{4\,dt}{t} = 4\ln(t) + C$

$2\ln|1 + 2\sin y| = \ln|4 + \sqrt{9 + \sqrt{x}}| + C$

$2\ln|1 + 2\sin y| = \ln|8|$

$\Rightarrow C = 0$

Question Type : **MCQ**
Question ID : **860654995**
Option 1 ID : **8606543392**
Option 2 ID : **8606543394**
Option 3 ID : **8606543393**
Option 4 ID : **8606543395**
Status : **Not Answered**
Chosen Option : **--**

Section : **Mathematics Section B**

$4 + \sqrt{9 + \sqrt{x}} = t$

$\frac{1}{4\sqrt{9 + \sqrt{x}}} \times \frac{dx}{\sqrt{x}} = dt$

$2\sin\alpha + 1 = 2\sqrt{2}$

$2\sin\alpha = 2\sqrt{2} - 1$

DE Tough XII

PRASHANT JAIN

Q.21 Let $[\cdot]$ be the greatest integer function. If $\alpha = \int_0^{64}\left(x^{1/3} - [x^{1/3}]\right)dx$, then $\frac{1}{\pi}\int_0^{\pi}\left(\frac{\sin^2\theta}{\sin^6\theta + \cos^6\theta}\right)d\theta$ is equal to _____.

Given --
Answer :

Question Type : **SA**
Question ID : **8606541000**
Status : **Not Answered**

Q.22 Let a vector $\vec{a} = \sqrt{2}\hat{i} - \hat{j} + \lambda\hat{k}$, $\lambda > 0$, make an obtuse angle with the vector $\vec{b} = -\lambda^2\hat{i} + 4\sqrt{2}\hat{j} + 4\sqrt{2}\hat{k}$ and an angle θ, $\frac{\pi}{6} < \theta < \frac{\pi}{2}$, with the positive z-axis. If the set of all possible values of λ is (α, β), then $\alpha + \beta + \gamma$ is equal to _____.

Given --
Answer :

Question Type : **SA**
Question ID : **860654999**
Status : **Not Answered**

Q.23 Let S be the set of the first 11 natural numbers. Then the number of elements in $A = \{B \subseteq S : n(B) \geq 2$ and the product of all elements of B is even$\}$ is _____.

Given --
Answer :

Question Type : **SA**
Question ID : **860654997**
Status : **Not Answered**

Q.24 Let $\cos(\alpha + \beta) = -\frac{1}{10}$ and $\sin(\alpha - \beta) = \frac{3}{8}$, where $0 < \alpha < \frac{\pi}{3}$ and $0 < \beta < \frac{\pi}{4}$.

If $\tan 2\alpha = \frac{3(1 - r\sqrt{5})}{\sqrt{11}(s + \sqrt{5})}$, $r, s \in N$, then $r + s$ is equal to _____.

Given --
Answer :

Question Type : **SA**
Question ID : **860654998**
Status : **Not Answered**

Q.25 Suppose a, b, c are in A.P. and $a^2, 2b^2, c^2$ are in G.P. If $a < b < c$ and $a + b + c = 1$, then $9(a^2 + b^2 + c^2)$ is equal to _____.

Given --
Answer :

Question Type : **SA**
Question ID : **860654996**
Status : **Not Answered**

Section : **Physics Section A**

PRASHANT JAIN

NTA JEE Mains Jan 2026

Application No	
Candidate Name	
Roll No.	
Test Date	23/01/2026
Test Time	9:00 AM - 12:00 PM
Subject	B. Tech

Section : **Mathematics Section A**

Q.1 If α and β $(\alpha < \beta)$ are the roots of the equation

$(-2+\sqrt{3})\left(|\sqrt{x}-3|\right)+(x-6\sqrt{x})+(9-2\sqrt{3})=0,\ x \geq 0$, then $\sqrt{\frac{\beta}{\alpha}}+\sqrt{\alpha\beta}$ is equal to :

Quadratic XI Medium

$(\sqrt{3}-2)|\sqrt{x}-3|+(\sqrt{x}-3)^2-2\sqrt{3}=0$

$t^2+(\sqrt{3}-2)t-2\sqrt{3}=0$

$(t+\sqrt{3})(t-2)=0$

$t=2$

$\sqrt{x}=5,1$

$x=25,1$

Options

1. ~~10~~
2. 8
3. 9
4. 11

~~Medium~~

Question Type : **MCQ**
Question ID : **8606541352**
Option 1 ID : **8606544597**
Option 2 ID : **8606544595**
Option 3 ID : **8606544596**
Option 4 ID : **8606544598**
Status : **Answered**
Chosen Option : **1**

PRASHANT JAIN

Q.2 Let $\vec{a} = -\hat{i} + \hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} - \hat{j} - 3\hat{k}$, $\vec{c} = \vec{a} \times \vec{b}$ and $\vec{d} = \vec{c} \times \vec{a}$.

Then $(\vec{a} - \vec{b}) \cdot \vec{d}$ is equal to :

Options

1. 4
2. −2
3. −4
4. 2

Vector XII Medium

Question Type : **MCQ**
Question ID : **8606541365**
Option 1 ID : **8606544647**
Option 2 ID : **8606544650**
Option 3 ID : **8606544648**
Option 4 ID : **8606544649**
Status : **Answered**
Chosen Option : **2**

Q.3 Let the direction cosines of two lines satisfy the equations : $4l + m - n = 0$ and $2mn + 10nl + 3lm = 0$.
Then the cosine of the acute angle between these lines is :

Options

1. $\frac{10}{3\sqrt{38}}$
2. $\frac{20}{3\sqrt{38}}$
3. $\frac{10}{\sqrt{38}}$
4. $\frac{10}{7\sqrt{38}}$

3-D XII medium

Question Type : **MCQ**
Question ID : **8606541366**
Option 1 ID : **8606544654**
Option 2 ID : **8606544651**
Option 3 ID : **8606544653**
Option 4 ID : **8606544652**
Status : **Not Answered**
Chosen Option : **--**

PRASHANT JAIN

Q.4

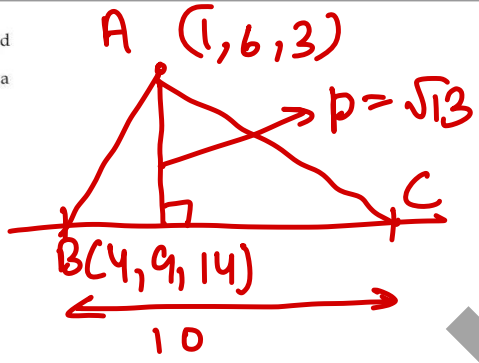
The vertices B and C of a triangle ABC lie on the line $\frac{x}{1} = \frac{1-y}{-2} = \frac{z-2}{3}$. The coordinates of A and B are (1, 6, 3) and (4, 9, α) respectively and C is at a distance of 10 units from B. The area (in sq. units) of ΔABC is :

Options

1. $15\sqrt{13}$
2. $5\sqrt{13}$
3. $20\sqrt{13}$
4. $10\sqrt{13}$

$\frac{1}{2} \times \sqrt{13} \times 10$

3D medium XII



Question Type : **MCQ**
Question ID : **8606541364**
Option 1 ID : **8606544646**
Option 2 ID : **8606544644**
Option 3 ID : **8606544645**
Option 4 ID : **8606544643**
Status : **Answered**
Chosen Option : **2**

Q.5

Let the mean and variance of 8 numbers $-10, -7, -1, x, y, 9, 2, 16$ be $\frac{7}{2}$ and $\frac{293}{4}$, respectively.

Then the mean of 4 numbers $x, y, x+y+1, |x-y|$ is :

Options

1. 11
2. 9
3. 10
4. 12

$x + y = 19$

$\frac{293}{4} + \frac{49}{4} = \frac{x^2+y^2+150+85+14y}{8}$

$x^2 + y^2 = 193$

$x = 12, y = 7$

7, 12, 20, 5

Medium Statistics XI

Question Type : **MCQ**
Question ID : **8606541358**
Option 1 ID : **8606544621**
Option 2 ID : **8606544619**
Option 3 ID : **8606544620**
Option 4 ID : **8606544622**
Status : **Answered**
Chosen Option : **1**

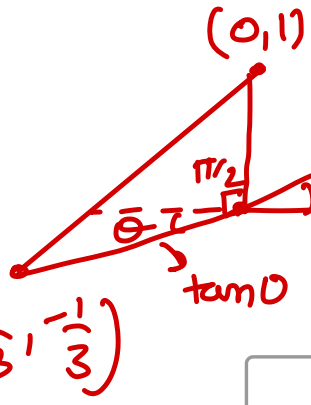
PRASHANT JAIN

Q.6

Let the line $y-x=1$ intersect the ellipse $\frac{x^2}{2}+\frac{y^2}{1}=1$ at the points A and B. Then the angle made by the line segment AB at the center of the ellipse is :

Options

1. $\frac{\pi}{2}-\tan^{-1}\left(\frac{1}{4}\right)$
2. $\pi-\tan^{-1}\left(\frac{1}{4}\right)$
3. $\frac{\pi}{2}+2\tan^{-1}\left(\frac{1}{4}\right)$
4. $\frac{\pi}{2}+\tan^{-1}\left(\frac{1}{4}\right)$



$y=x+1$

$\frac{x^2}{2}+(x+1)^2=1$

$x^2+2(x+1)^2=2$

$3x^2+4x=0 \quad x=0, x=-\frac{4}{3}$

$\tan\theta=\frac{1}{4}$

correction medium XI

Question Type : **MCQ**
Question ID : **8606541362**
Option 1 ID : **8606544637**
Option 2 ID : **8606544635**
Option 3 ID : **8606544638**
Option 4 ID : **8606544636**
Status : **Answered**
Chosen Option : **2**

Q.7

Let α and β respectively be the maximum and the minimum values of the function $f(\theta)=4\left(\sin^4\left(\frac{7\pi}{2}-\theta\right)+\sin^4(11\pi+\theta)\right)-2\left(\sin^6\left(\frac{3\pi}{2}-\theta\right)+\sin^6(9\pi-\theta)\right), \theta\in\mathbf{R}$.

Then $\alpha+2\beta$ is equal to :

Options

1. 3
2. 6
3. 4
4. 5

$f(\theta)=4(c^4+s^4)-2(s^2+c^2)$

$f(\theta)=4(1-2s^2c^2)-2=2-2s^2c^2$

$=2-\frac{\sin^2 2\theta}{2}$

$\frac{3}{2}\to\min \qquad \max=2$

$2+3=5$

Trigo XI easy

Question Type : **MCQ**
Question ID : **8606541363**
Option 1 ID : **8606544639**
Option 2 ID : **8606544640**
Option 3 ID : **8606544641**
Option 4 ID : **8606544642**
Status : **Not Answered**
Chosen Option : **--**

PRASHANT JAIN

Q.8

Number of solutions of $\sqrt{3}\cos 2\theta + 8\cos\theta + 3\sqrt{3} = 0, \theta \in [-3\pi, 2\pi]$ is :

Options

1. 5
2. 3
3. 0
4. 4

$\sqrt{3}(2c^2-1) + 8c + 3\sqrt{3} = 0$

$2\sqrt{3}c^2 + 8c + 2\sqrt{3} = 0$

$\sqrt{3}c^2 + 4c + \sqrt{3} = 0$

$(c+\sqrt{3})(\sqrt{3}c+1) = 0$

$\cos\theta = -\frac{1}{\sqrt{3}}$ → 5 times

Trigo XI easy

Question Type : **MCQ**
Question ID : **8606541361**
Option 1 ID : **8606544631**
Option 2 ID : **8606544633**
Option 3 ID : **8606544634**
Option 4 ID : **8606544632**
Status : **Answered**
Chosen Option : **1**

Q.9

The value of $\frac{{}^{100}C_{50}}{51} + \frac{{}^{100}C_{51}}{52} + \ldots\ldots + \frac{{}^{100}C_{100}}{101}$ is :

$= \frac{2^{100}}{101}$

Options

1. $\frac{2^{101}}{101}$
2. $\frac{2^{100}}{101}$
3. $\frac{2^{100}}{100}$
4. $\frac{2^{101}}{100}$

Binomial XI easy

Question Type : **MCQ**
Question ID : **8606541357**
Option 1 ID : **8606544618**
Option 2 ID : **8606544616**
Option 3 ID : **8606544615**
Option 4 ID : **8606544617**
Status : **Answered**
Chosen Option : **2**

PRASHANT JAIN

Q.10

Let $f(x) = \int \frac{(2-x^2).e^x}{(\sqrt{1+x})(1-x)^{3/2}}dx$. If $f(0)=0$, then $f\left(\frac{1}{2}\right)$ is equal to :

Back

Options

1. $\sqrt{2e}-1$
2. $\sqrt{3e}+1$
3. $\sqrt{2e}+1$
4. $\sqrt{3e}-1$

$f(x) = e^x\sqrt{\frac{1+x}{1-x}} - 1$

$f\left(\frac{1}{2}\right) = \sqrt{e}\times\sqrt{3} - 1$

Tough
II
XII

Question Type : **MCQ**
Question ID : **8606541368**
Option 1 ID : **8606544662**
Option 2 ID : **8606544659**
Option 3 ID : **8606544661**
Option 4 ID : **8606544660**
Status : **Not Answered**
Chosen Option : **--**

Q.11

Among the statements :

I : If $\begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\gamma \\ \cos\beta & \cos\gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos\alpha & \cos\beta \\ \cos\alpha & 0 & \cos\gamma \\ \cos\beta & \cos\gamma & 0 \end{vmatrix}$, then $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{3}{2}$, and

II : If $\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = px + q$, then $p^2 = 196q^2$,

$\alpha \to \cos\alpha$

$\Rightarrow 1 - \gamma^2 - \alpha^2 - \beta^2 + 2\alpha\beta\sigma = 2\alpha\beta\sigma$

$\alpha^2+\beta^2+\sigma^2 = 1$

$x=0 \Rightarrow q = \begin{vmatrix} 0 & 1 & -2 \\ -1 & 3 & -3 \\ 3 & -1 & -1 \end{vmatrix} = -12$

$x=-1 \quad \begin{vmatrix} 0 & 0 & -3 \\ -2 & -3 & \\ 2 & -3 & \end{vmatrix} = 9 - p = -36$

$p = 24$

$24\times24 \neq 196\times12^2$

Options

1. both are false
2. only I is true
3. both are true
4. only II is true

MSD
XII
Tough

Question Type : **MCQ**
Question ID : **8606541355**
Option 1 ID : **8606544608**
Option 2 ID : **8606544609**
Option 3 ID : **8606544607**
Option 4 ID : **8606544610**
Status : **Not Answered**
Chosen Option : **--**

PRASHANT JAIN

$$I = \int \frac{(2-x^2)\, e^x\, dx}{(\sqrt{1+x})\,(1-x)^{3/2}} = e^x \sqrt{\frac{1+x}{1-x}} + C$$

$$\frac{1 + 1-x^2}{(1-x)\,(\sqrt{1-x^2})} = \frac{1}{(1-x)(\sqrt{1-x^2})} + \frac{1+x}{\sqrt{1-x^2}}$$

$$\frac{1+x}{\sqrt{1-x^2}} \to \frac{1}{\sqrt{1-x^2}} + \frac{(1+x)\left(-\frac{1}{2}\right) \times (-2x)}{(1-x^2)^{3/2}}$$

$$=$$

PRASHANT JAIN

Q.12 The sum of all possible values of $n \in \mathbf{N}$, so that the coefficients of x, x^2 and x^3 in the expansion of $(1+x^2)^2(1+x)^n$, are in arithmetic progression is :

Options

1. 9
2. 12
3. 3
4. 7

$(1+2x^2+x^4)(1+x)^n$

x: n

x^2: $2+{}^nC_2$

x^3: $2n+{}^nC_3$

$2(2+{}^nC_2) = 3n+{}^nC_3$

$n^3-9n^2+26n-24=0$

$n=2,3,4$

Binomial Medium XI

Question Type : **MCQ**
Question ID : **8606541356**
Option 1 ID : **8606544613**
Option 2 ID : **8606544614**
Option 3 ID : **8606544611**
Option 4 ID : **8606544612**
Status : **Answered**
Chosen Option : **4**

Q.13 Let $f(x) = \begin{cases} \dfrac{ax^2+2ax+3}{4x^2+4x-3} & , x \neq -\dfrac{3}{2}, \dfrac{1}{2} \\ b & , x = -\dfrac{3}{2}, \dfrac{1}{2} \end{cases}$

be continuous at $x = -\dfrac{3}{2}$. If $fof(x) = \dfrac{7}{5}$, then x is equal to :

Options

1. 1
2. 2
3. 1.4
4. 0

$\frac{9a}{4} - 3a + 3 = 0$

$-\frac{3a}{4} + 3 = 0 \Rightarrow a = 4$

$f(x) \to \dfrac{4x^2+8x+3}{4x^2+4x-3} = \dfrac{(2x+3)(2x+1)}{(2x+3)(2x-1)} \to b = 1/2$

$f(f(x)) = \frac{7}{5}$

$\frac{2x+1}{2x-1} = \frac{7}{5}$

$2x = 6$

$x = 3$

$\frac{2x_1+1}{2x_1-1} = 3$

$2x_1 = 2$

Very smart calculation needed

Medium XII LCD

Question Type : **MCQ**
Question ID : **8606541367**
Option 1 ID : **8606544656**
Option 2 ID : **8606544658**
Option 3 ID : **8606544657**
Option 4 ID : **8606544655**
Status : **Not Answered**
Chosen Option : **--**

PRASHANT JAIN

Q.14 Let the domain of the function $f(x) = \log_3 \log_5 \log_7 (9x - x^2 - 13)$ be the interval (m, n). Let the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ have eccentricity $\frac{n}{3}$ and the length of the latus rectum $\frac{8m}{3}$. Then $b^2 - a^2$ is equal to :

Options

1. 11
2. 7
3. 9
4. 5

Question Type : **MCQ**
Question ID : **8606541360**
Option 1 ID : **8606544630**
Option 2 ID : **8606544628**
Option 3 ID : **8606544629**
Option 4 ID : **8606544627**
Status : **Answered**
Chosen Option : **2**

Q.15 Let $A = \{-2, -1, 0, 1, 2, 3, 4\}$. Let R be a relation on A defined by xRy if and only if $2x + y \leq 2$. Let l be the number of elements in R. Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations respectively. Then $l + m + n$ is equal to :

Options

1. 34
2. 33
3. 32
4. 35

Question Type : **MCQ**
Question ID : **8606541351**
Option 1 ID : **8606544594**
Option 2 ID : **8606544593**
Option 3 ID : **8606544592**
Option 4 ID : **8606544591**
Status : **Not Answered**
Chosen Option : **--**

PRASHANT JAIN

Q.16

Let $y = y(x)$ be the solution of the differential equation $x^4dy + (4x^3y + 2\sin x)dx = 0$, $x > 0$, $y\left(\frac{\pi}{2}\right) = 0$.

Then $\pi^4 y\left(\frac{\pi}{3}\right)$ is equal to :

Options

1. 64
2. 72
3. 92
4. 81

$x^4dy + 4x^3y\,dx + 2\cos x\,dx = 0$

$x^4y - 2\cos x = 0$

$\frac{\pi^4}{81} x\,y - 1 = 0$

$81 = y\pi^4$

DE easy XII

Question Type : **MCQ**
Question ID : **8606541370**
Option 1 ID : **8606544669**
Option 2 ID : **8606544668**
Option 3 ID : **8606544670**
Option 4 ID : **8606544667**
Status : **Answered**
Chosen Option : **4**

Q.17

The value of the integral $\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{dx}{1 + \sqrt[3]{\tan 2x}}$ is :

Options

1. $\frac{\pi}{3}$
2. $\frac{\pi}{18}$
3. $\frac{\pi}{6}$
4. $\frac{\pi}{12}$

$2I = \int_{\pi/24}^{5\pi/24} 1\,dt$

$2I = \frac{\pi}{6}$

$I = \frac{\pi}{12}$

DI easy XII

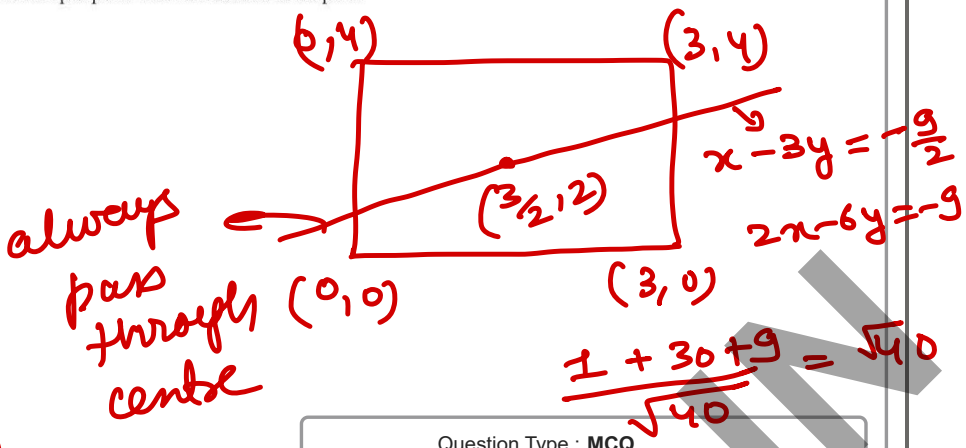
Question Type : **MCQ**
Question ID : **8606541369**
Option 1 ID : **8606544663**
Option 2 ID : **8606544666**
Option 3 ID : **8606544664**
Option 4 ID : **8606544665**
Status : **Answered**
Chosen Option : **4**

PRASHANT JAIN

Q.18 A rectangle is formed by the lines $x=0$, $y=0$, $x=3$ and $y=4$. Let the line L be perpendicular to $3x+y+6=0$ and divide the area of the rectangle into two equal parts. Then the distance of the point $\left(\frac{1}{2}, -5\right)$ from the line L is equal to :

Options

1. $\sqrt{10}$
2. $3\sqrt{10}$
3. $2\sqrt{5}$
4. $2\sqrt{10}$



straight line
XI
medium

Question Type : **MCQ**
Question ID : **8606541359**
Option 1 ID : **8606544625**
Option 2 ID : **8606544624**
Option 3 ID : **8606544623**
Option 4 ID : **8606544626**
Status : **Not Answered**
Chosen Option : **--**

Q.19 Let $S = \{z : 3 \leq |2z - 3(1+i)| \leq 7\}$ be a set of complex numbers.

Then $\underset{z \in S}{\text{Min}} \left|\left(z + \frac{1}{2}(5+3i)\right)\right|$ is equal to :

Options

1. 2
2. $\frac{3}{2}$
3. $\frac{5}{2}$
4. $\frac{1}{2}$

Complex
XII
Medium

Question Type : **MCQ**
Question ID : **8606541354**
Option 1 ID : **8606544605**
Option 2 ID : **8606544604**
Option 3 ID : **8606544606**
Option 4 ID : **8606544603**
Status : **Answered**
Chosen Option : **2**

PRASHANT JAIN

Q.20 A building construction work can be completed by two masons A and B together in 22.5 days. Mason A alone can complete the construction work in 24 days less than mason B alone. Then mason A alone will complete the construction work in :

Options

1. 30 days
2. 36 days
3. 42 days
4. 24 days

Question Type : **MCQ**
Question ID : **8606541353**
Option 1 ID : **8606544600**
Option 2 ID : **8606544601**
Option 3 ID : **8606544602**
Option 4 ID : **8606544599**
Status : **Not Answered**
Chosen Option : **--**

Section : **Mathematics Section B**

Q.21 Let $|A| = 6$, where A is a 3×3 matrix. If $|\text{adj}(3\text{adj}(A^2 \cdot \text{adj}(2A)))| = 2^m \cdot 3^n$, $m, n \in \mathbf{N}$, then $m + n$ is equal to _____.

Given Answer : **62**

Question Type : **SA**
Question ID : **8606541371**
Status : **Answered**

Q.22 Let f be a twice differentiable non-negative function such that $(f(x))^2 = 25 + \int_0^x \left((f(t))^2 + (f'(t))^2\right) dt$.

Then the mean of $f(\log_e(1)), f(\log_e(2)), \ldots, f(\log_e(625))$ is equal to _____.

Given Answer : **--**

Question Type : **SA**
Question ID : **8606541374**
Status : **Not Answered**

Q.23 The number of 4-letter words, with or without meaning, which can be formed using the letters PQRPQRSTUVP, is _____.

Given Answer : **1422**

Question Type : **SA**
Question ID : **8606541372**
Status : **Answered**

PRASHANT JAIN

Q.24

Let the area of the region bounded by the curve $y = \max\{\sin x, \cos x\}$, lines $x = 0$, $x = \frac{3\pi}{2}$, and the x-axis be A. Then, $A + A^2$ is equal to _____.

Given --
Answer :

Question Type : **SA**
Question ID : **8606541375**
Status : **Not Answered**

Q.25

From the first 100 natural numbers, two numbers first a and then b are selected randomly without replacement. If the probability that $a - b \geq 10$ is $\frac{m}{n}$, gcd $(m, n) = 1$, then $m + n$ is equal to _____.

Given --
Answer :

Question Type : **SA**
Question ID : **8606541373**
Status : **Not Answered**

Section : **Physics Section A**

PRASHANT JAIN

NTA JEE Mains Jan 2026

Application No	[redacted]
Candidate Name	[redacted]
Roll No.	[redacted]
Test Date	23/01/2026
Test Time	3:00 PM - 6:00 PM
Subject	B. Tech

Section : **Mathematics Section A**

Q.1 Let $I(x)=\int\frac{3dx}{(4x+6)\left(\sqrt{4x^2+8x+3}\right)}$ and $I(0)=\frac{\sqrt{3}}{4}+20$. If

$I\left(\frac{1}{2}\right)=\frac{a\sqrt{2}}{b}+c$, where $a, b, c \in N$, $\gcd(a, b)=1$, then $a+b+c$ is equal to

Options 1. 30
2. 31
3. 29
4. 28

Question Type : **MCQ**
Question ID : **444792469**
Option 1 ID : **4447921605**
Option 2 ID : **4447921606**
Option 3 ID : **4447921604**
Option 4 ID : **4447921603**
Status : **Not Answered**
Chosen Option : **--**

Q.2 Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $\vec{a}\times\vec{b}=2(\vec{a}\times\vec{c})$. If $|\vec{a}|=1, |\vec{b}|=4, |\vec{c}|=2$, and the angle between $\vec{b}$ and $\vec{c}$ is 60°, then $|\vec{a}\cdot\vec{c}|$ is equal to

Options 1. 0
2. 1
3. 2
4. 4

Question Type : **MCQ**
Question ID : **444792466**
Option 1 ID : **4447921591**
Option 2 ID : **4447921592**
Option 3 ID : **4447921593**
Option 4 ID : **4447921594**
Status : **Answered**
Chosen Option : **2**

PRASHANT JAIN

Q.3 The system of linear equations

$x + y + z = 6$

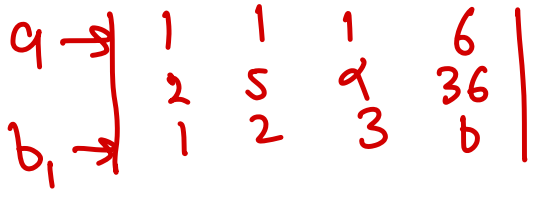
$2x + 5y + az = 36$

$x + 2y + 3z = b$

has

Options

1. infinitely many solutions for $a = 8$ and $b = 14$
2. unique solution for $a = 8$ and $b = 14$
3. infinitely many solutions for $a = 8$ and $b = 16$
4. unique solution for $a = 8$ and $b = 16$



$a + b_1 = 2$

$a + 2b_1 = 5$

$b_1 = 3$, $a = -1$

$\alpha = a + 3b_1 = 8$

$36 = -6 + 3b$

$b = 14$

M & D XII easy

Question Type : **MCQ**
Question ID : **444792455**
Option 1 ID : **4447921549**
Option 2 ID : **4447921547**
Option 3 ID : **4447921550**
Option 4 ID : **4447921548**
Status : **Answered**
Chosen Option : **4**

Q.4 Consider two sets $A = \{x \in \mathbb{Z} : |(|x-3|-3)| \le 1\}$ and $B = \left\{x \in \mathbb{R} - \{1,2\} : \frac{(x-2)(x-4)}{x-1} \log_e(|x-2|) = 0\right\}$. Then the number of onto functions $f : A \to B$ is equal to

Options

1. 81
2. 79
3. 62
4. 32

$2 \le |x-3| \le 4$

$5 \le x \le 7$ or $-1 \le x \le 1$

$n(A) = 6$

$B = \{2, 3\}$

$2^6 - 2 = 62$

functions medium XII

Question Type : **MCQ**
Question ID : **444792452**
Option 1 ID : **4447921538**
Option 2 ID : **4447921537**
Option 3 ID : **4447921536**
Option 4 ID : **4447921535**
Status : **Answered**
Chosen Option : **1**

PRASHANT JAIN

Q.5 If the mean and the variance of the data

Class	4–8	8–12	12–16	16–20
Frequency	3	λ	4	7

are μ and 19 respectively, then the value of λ + μ is

Options
1. 21
2. 20
3. 19
4. 18

Question Type : MCQ
Question ID : 444792459
Option 1 ID : 4447921566
Option 2 ID : 4447921565
Option 3 ID : 4447921564
Option 4 ID : 4447921563
Status : Answered
Chosen Option : 1

Q.6 The sum of all the real solutions of the equation $\log_{(x+3)}\left(6x^2+28x+30\right)=5-2\log_{(6x+10)}\left(x^2+6x+9\right)$ is equal to

Options
1. 2
2. 0
3. 1
4. 4

Question Type : MCQ
Question ID : 444792453
Option 1 ID : 4447921541
Option 2 ID : 4447921539
Option 3 ID : 4447921540
Option 4 ID : 4447921542
Status : Answered
Chosen Option : 3

PRASHANT JAIN

Q.7 The number of ways, in which 16 oranges can be distributed to four children such that each child gets at least one orange, is

Options
1. 455
2. 384
3. 403
4. 429

Question Type : **MCQ**
Question ID : **444792457**
Option 1 ID : **4447921557**
Option 2 ID : **4447921555**
Option 3 ID : **4447921556**
Option 4 ID : **4447921558**
Status : **Answered**
Chosen Option : **1**

Q.8 Let $\frac{\pi}{2} < \theta < \pi$ and $\cot\theta = -\frac{1}{2\sqrt{2}}$. Then the value of

$$\sin\left(\frac{15\theta}{2}\right)(\cos 8\theta + \sin 8\theta) + \cos\left(\frac{15\theta}{2}\right)(\cos 8\theta - \sin 8\theta)$$

is equal to

Options
1. $\frac{\sqrt{2}-1}{\sqrt{3}}$
2. $\frac{\sqrt{2}}{\sqrt{3}}$
3. $-\frac{\sqrt{2}}{\sqrt{3}}$
4. $\frac{1-\sqrt{2}}{\sqrt{3}}$

Question Type : **MCQ**
Question ID : **444792464**
Option 1 ID : **4447921583**
Option 2 ID : **4447921585**
Option 3 ID : **4447921586**
Option 4 ID : **4447921584**
Status : **Not Answered**
Chosen Option : **--**

PRASHANT JAIN

Q.9 If the points of intersection of the ellipses $x^2 + 2y^2 - 6x - 12y + 23 = 0$ and $4x^2 + 2y^2 - 20x - 12y + 35 = 0$ lie on a circle of radius r and centre (a, b), then the value of $ab + 18r^2$ is

Options
1. 55
2. 53
3. 51
4. 52

$2(x^2+2y^2-6x-12y+23)=0$
$+ \; 4x^2+2y^2-20x-12y+35=0$
$6x^2+6y^2-32x-36y+81=0$

$\left(\frac{32}{12}, \frac{36}{12}\right)$
$\left(\frac{8}{3}, 3\right) = (a, b)$
$r = \sqrt{\frac{47}{18}}$
$18r^2 + ab = 47 + 8 = 55$

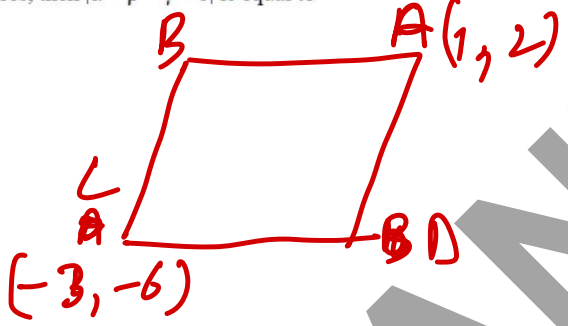
conic XI Medium

eqn of circle

Question Type : MCQ
Question ID : 444792460
Option 1 ID : 4447921568
Option 2 ID : 4447921569
Option 3 ID : 4447921567
Option 4 ID : 4447921570
Status : Answered
Chosen Option : 2

Q.10 Let A(1, 2) and C(−3, −6) be two diagonally opposite vertices of a rhombus, whose sides AD and BC are parallel to the line $7x - y = 14$. If B (α, β) and D (γ, δ) are the other two vertices, then $|\alpha + \beta + \gamma + \delta|$ is equal to

Options
1. 1
2. 6
3. 9
4. 3



$\alpha+\beta+\gamma+\delta = 1+2-3-6$
$= -6$

Question Type : MCQ
Question ID : 444792463
Option 1 ID : 4447921579
Option 2 ID : 4447921581
Option 3 ID : 4447921582
Option 4 ID : 4447921580
Status : Not Answered
Chosen Option : --

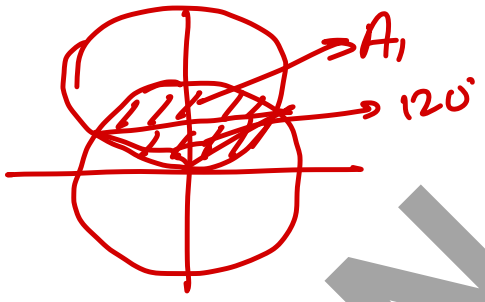
straight line easy XI

PRASHANT JAIN

Q.11 The area of the region enclosed between the circles $x^2 + y^2 = 4$ and $x^2 + (y-2)^2 = 4$ is:

Options

1. $\frac{4}{3}\left(2\pi - \sqrt{3}\right)$
2. $\frac{4}{3}\left(2\pi - 3\sqrt{3}\right)$
3. $\frac{2}{3}\left(4\pi - 3\sqrt{3}\right)$
4. $\frac{2}{3}\left(2\pi - 3\sqrt{3}\right)$



$A_1 = \frac{4\pi}{3} - \frac{1}{2} \times 2 \times 2 \times \frac{\sqrt{3}}{2}$

$2A_1 = \frac{8\pi}{3} - 2\sqrt{3} = \frac{2}{3}(4\pi - 3\sqrt{3})$

AUC XII medium

Question Type : **MCQ**
Question ID : **444792470**
Option 1 ID : **4447921610**
Option 2 ID : **4447921607**
Option 3 ID : **4447921608**
Option 4 ID : **4447921609**
Status : **Not Answered**
Chosen Option : **--**

Q.12 Let A = {0,1,2,...,9}. Let R be a relation on A defined by $(x, y) \in R$ if and only if $|x - y|$ is a multiple of 3.

Given below are two statements:

Statement I: $n(R) = 36$

Statement II: R is an equivalence relation.

In the light of the above statements, choose the ***correct*** answer from the options given below

$3k+1 \to 3$
$3k \to 4$
$3k+2 \to 3$
$n(R) \to 3\times3 + 4\times4 + 3\times3 = 34$

Options

1. Statement I is correct but Statement II is incorrect
2. Both Statement I and Statement II are correct
3. Both Statement I and Statement II are incorrect
4. Statement I is incorrect but Statement II is correct

Relations XII easy

Question Type : **MCQ**
Question ID : **444792451**
Option 1 ID : **4447921533**
Option 2 ID : **4447921531**
Option 3 ID : **4447921532**
Option 4 ID : **4447921534**
Status : **Answered**
Chosen Option : **4**

PRASHANT JAIN

Q.13 Let $\vec{a}=\hat{i}-2\hat{j}+3\hat{k}$, $\vec{b}=2\hat{i}+\hat{j}-\hat{k}$, $\vec{c}=\lambda\hat{i}+\hat{j}+\hat{k}$ and $\vec{v}=\vec{a}\times\vec{b}$. If $\vec{v}\cdot\vec{c}=11$ and the length of the projection of $\vec{b}$ on $\vec{c}$ is p, then $9p^2$ is equal to

Options:

1. 12
2. 4
3. 9
4. 6

Vector easy XII

$\frac{2+\lambda-\lambda}{\sqrt{3}}=p$

$\frac{4}{3}=p^2$

$9p^2=12$

$[a\,b\,c]=11$

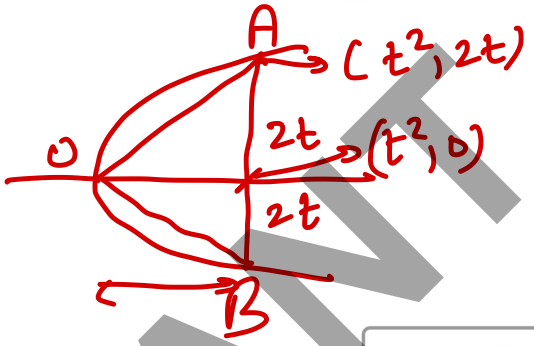
$\begin{vmatrix} 1 & -2 & 3 \\ 2 & 1 & -1 \\ a & 1 & 1 \end{vmatrix}=11 \Rightarrow a=1$

Question Type : **MCQ**
Question ID : **444792465**
Option 1 ID : **4447921590**
Option 2 ID : **4447921587**
Option 3 ID : **4447921589**
Option 4 ID : **4447921588**
Status : **Not Answered**
Chosen Option : **--**

Q.14 An equilateral triangle OAB is inscribed in the parabola $y^2=4x$ with the vertex O at the vertex of the parabola. Then the minimum distance of the circle having AB as a diameter from the origin is

Options:

1. $4\left(3-\sqrt{3}\right)$
2. $2\left(3+\sqrt{3}\right)$
3. $2\left(8-3\sqrt{3}\right)$
4. $4\left(6+\sqrt{3}\right)$



$t^4+4t^2=16t^2$

$t^2=12$

min distance $=|t^2-2t|$

$=|12-4\sqrt{3}|$

$=4(3-\sqrt{3})$

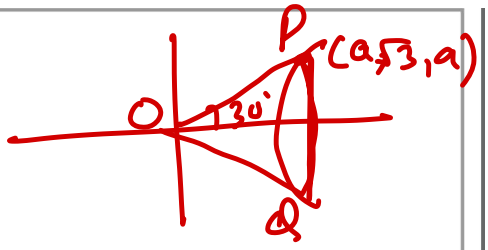
Conic XI Medium

Question Type : **MCQ**
Question ID : **444792461**
Option 1 ID : **4447921574**
Option 2 ID : **4447921573**
Option 3 ID : **4447921571**
Option 4 ID : **4447921572**
Status : **Not Answered**
Chosen Option : **--**

PRASHANT JAIN

Q.15

Let PQ be a chord of the hyperbola $\frac{x^2}{4}-\frac{y^2}{b^2}=1$, perpendicular to the x-axis such that OPQ is an equilateral triangle, O being the centre of the hyperbola. If the eccentricity of the hyperbola is $\sqrt{3}$, then the area of the triangle OPQ is



Options

1. $\frac{11}{5}$
2. $\frac{8\sqrt{3}}{5}$
3. $\frac{9}{5}$
4. $2\sqrt{3}$

$\frac{x^2}{4}-\frac{y^2}{8}=1$

$OP=2a$

$\frac{a^2\times 3}{4}-\frac{a^2}{8}=1$

$5a^2=8 \quad a^2=\frac{8}{5}$

$\frac{\sqrt{3}}{4}\times 4a^2=\frac{2\sqrt{3}}{5}$

Conic
XI
Medium

Question Type : **MCQ**
Question ID : **444792462**
Option 1 ID : **4447921578**
Option 2 ID : **4447921575**
Option 3 ID : **4447921577**
Option 4 ID : **4447921576**
Status : **Not Answered**
Chosen Option : **--**

Q.16 The least value of $(\cos^2\theta-6\sin\theta\cos\theta+3\sin^2\theta+2)$ is

Options

1. $4+\sqrt{10}$
2. -1
3. 1
4. $4-\sqrt{10}$

$3+2s^2-3s2\theta$

$3+1-c2\theta-3s2\theta$

$4-\sqrt{10}$

Trigo
XI
easy

Question Type : **MCQ**
Question ID : **444792468**
Option 1 ID : **4447921599**
Option 2 ID : **4447921601**
Option 3 ID : **4447921602**
Option 4 ID : **4447921600**
Status : **Answered**
Chosen Option : **4**

PRASHANT JAIN

Q.17 Let $\sum_{k=1}^{n} a_k = \alpha n^2 + \beta n$. If $a_{10} = 59$ and $a_6 = 7a_1$, then $\alpha + \beta$ is equal to

Options
1. 5
2. 7
3. 12
4. 3

$a+5d=7a$

$5d=6a$

$a+9d=59$

$6a+54d=354$

$59d=354$

$a=5$ $d=6$

$S_n = 3n^2+2n$

S&S easy XI

Question Type : **MCQ**
Question ID : **444792456**
Option 1 ID : **4447921552**
Option 2 ID : **4447921553**
Option 3 ID : **4447921554**
Option 4 ID : **4447921551**
Status : **Not Answered**
Chosen Option : **--**

Q.18 Bag A contains 9 white and 8 black balls, while bag B contains 6 white and 4 black balls. One ball is randomly picked up from the bag B and mixed up with the balls in the bag A. Then a ball is randomly drawn from the bag A. If the probability, that the ball drawn is white, is $\frac{p}{q}$, $gcd(p, q) = 1$, then $p + q$ is equal to

Options
1. 23
2. 22
3. 24
4. 21

	W	B
A	9	8
B	6	4

$\frac{6}{10} \times \frac{10}{18} + \frac{4}{10} \times \frac{9}{18}$

$\frac{96}{180} = \frac{8}{15}$

Probability easy XII

Question Type : **MCQ**
Question ID : **444792458**
Option 1 ID : **4447921559**
Option 2 ID : **4447921560**
Option 3 ID : **4447921562**
Option 4 ID : **4447921561**
Status : **Not Answered**
Chosen Option : **--**

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Q.19

If $f(x)=\begin{cases}\dfrac{a|x|+x^2-2(\sin|x|)(\cos|x|)}{x} & , x\neq 0\\ b & , x=0\end{cases}$

is continuous at $x = 0$, then $a + b$ is equal to

Options
1. 4
2. 0
3. 1
4. 2

$a-2=b=-a+2$

$\Rightarrow a+b=2$

LCD XII easy

Question Type : **MCQ**
Question ID : **444792467**
Option 1 ID : **4447921598**
Option 2 ID : **4447921595**
Option 3 ID : **4447921596**
Option 4 ID : **4447921597**
Status : **Answered**
Chosen Option : **4**

Q.20

If $z=\dfrac{\sqrt{3}}{2}+\dfrac{i}{2}, i=\sqrt{-1}$, then $(z^{201}-i)^8$ is equal to

Options
1. 256
2. -1
3. 0
4. 1

$z=e^{i\pi/6}$

$z^{201}=e^{i\frac{67\pi}{2}}=-i$

$(2i)^8=256$

Complex No. XII easy

Question Type : **MCQ**
Question ID : **444792454**
Option 1 ID : **4447921546**
Option 2 ID : **4447921544**
Option 3 ID : **4447921543**
Option 4 ID : **4447921545**
Status : **Answered**
Chosen Option : **1**

Section : **Mathematics Section B**

PRASHANT JAIN

Q.21 The number of elements in the

set $S=\left\{x: x \in [0,100] \text{ and } \int_0^x t^2 \sin(x-t)\,dt = x^2\right\}$ is ____

Given --
Answer :

Question Type : **SA**
Question ID : **444792474**
Status : **Not Answered**

Q.22 Let S denote the set of 4-digit numbers $abcd$ such that $a > b > c > d$ and P denote the set of 5-digit numbers having product of its digits equal to 20. Then $n(S) + n(P)$ is equal to ____

Given --
Answer :

Question Type : **SA**
Question ID : **444792472**
Status : **Not Answered**

Q.23 Let $A=\begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 1 \\ 3 & -1 & 0 \end{bmatrix}$ and B be a matrix such that $B(I-A) = I + A$. Then the sum of the diagonal elements of B^TB is equal to ____

Given --
Answer :

Question Type : **SA**
Question ID : **444792471**
Status : **Not Attempted and Marked For Review**

Q.24 If the image of the point P$(a, 2, a)$ in the line $\frac{x}{2}=\frac{y+a}{1}=\frac{z}{1}$ is Q and the image of Q in the line $\frac{x-2b}{2}=\frac{y-a}{1}=\frac{z+2b}{-5}$ is P, then $a + b$ is equal to ____.

Given --
Answer :

Question Type : **SA**
Question ID : **444792473**
Status : **Not Answered**

PRASHANT JAIN

Q.25 If the solution curve $y = f(x)$ of the differential equation
$(x^2 - 4)\,y' - 2xy + 2x\,(4 - x^2)^2 = 0\,,\ x > 2,$
passes through the point (3, 15), then the local maximum value of f is_____

IF = $\frac{1}{x^2-4}$

Given --
Answer :

$y' - \frac{2x}{x^2-4}y = -2x(4-x^2)$

$\frac{y}{x^2-4} = -x^2 + c \Rightarrow 15 = (-9)(-5) + c \Rightarrow c = 12$

$y = (x^2-4)(12-x^2)$

$y(3) = (4)(4) = 16$

DE
x1
medium

Question Type : **SA**
Question ID : **444792475**
Status : **Not Answered**

Section : **Physics Section A**

Options 1. 3 : 2

4 $\sqrt{\ }$

$\int_0^x t^2 \sin(x-t)\,dt = x^2$

$t^2\cos(x-t)\Big|_0^x - 2\int_0^x t\cos(x-t)$

$x^2 - 2\left[-t\sin(x-t)\Big|_0^x + \int_0^x \sin(x-t)\right] = x^2$

$x^2 + 2\left[\cos(x-t)\right]_0^x = x^2$

$1 - \cos x = 0$

$\cos x = 1$

$x = 2n\pi$

$0, 2\pi, 4\pi, \ldots, -1.30\pi$

16

Question Type : **MCQ**
Question ID : **444792491**
Option 1 ID : **4447921676**
Option 2 ID : **4447921677**
Option 3 ID : **4447921678**
Option 4 ID : **4447921679**
Status : **Answered**
Chosen Option : **4**

Options 1. 1.45
2. 1.55
3. 14.5
4. 15.5

Question Type : **MCQ**
Question ID : **444792484**
Option 1 ID : **4447921651**
Option 2 ID : **4447921650**
Option 3 ID : **4447921649**
Option 4 ID : **4447921648**
Status : **Answered**
Chosen Option : **2**

PRASHANT JAIN

NTA JEE Mains Jan 2026

Application No	
Candidate Name	
Roll No.	
Test Date	24/01/2026
Test Time	9:00 AM - 12:00 PM
Subject	B. Tech

Section : **Mathematics Section A**

Q.1 Let the lines $L_1 : \vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k}), \lambda \in \mathbb{R}$ and $L_2 : \vec{r} = (4\hat{i} + \hat{j}) + \mu(5\hat{i} + 2\hat{j} + \hat{k}), \mu \in \mathbb{R}$, intersect at the point R. Let P and Q be the points lying on lines L_1 and L_2, respectively, such that $|\overrightarrow{PR}| = \sqrt{29}$ and $|\overrightarrow{PQ}| = \sqrt{\frac{47}{3}}$. If the point P lies in the first octant, then $27(QR)^2$ is equal to

Options
1. 360
2. 340
3. 348
4. 320

Question Type : **MCQ**
Question ID : **444792541**
Option 1 ID : **4447921849**
Option 2 ID : **4447921847**
Option 3 ID : **4447921848**
Option 4 ID : **4447921846**
Status : **Not Answered**
Chosen Option : **--**

lengthy
30 min
tough

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Q.2 Let $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$, $\vec{b}=\hat{i}+\hat{j}$ and $\vec{c}=\vec{a}\times\vec{b}$. Let $\vec{d}$ be a vector such that $|\vec{d}-\vec{a}|=\sqrt{11}$, $|\vec{c}\times\vec{d}|=3$ and the angle between $\vec{c}$ and $\vec{d}$ is $\frac{\pi}{4}$. Then $\vec{a}\cdot\vec{d}$ is equal to

Options
1. 11
2. 1
3. 3
4. 0

$\vec{a}\cdot\vec{b} = 3 = 3\times\sqrt{2}\times\cos\theta$

$\cos\theta=\frac{1}{\sqrt{2}}$

$a\hat{}b=\pi/4$

$|c| = |a||b|\times\frac{1}{\sqrt{2}} = 3$

$|d|=\sqrt{2}$

$|c| = |d|\times\frac{1}{\sqrt{2}} = 3$

$= 3$

$a^2+d^2-2a\cdot d = 11$

$11 - 2a\cdot d = 11$

$a\cdot d = 0$

vector medium X11

Question Type : **MCQ**
Question ID : **444792540**
Option 1 ID : **4447921845**
Option 2 ID : **4447921843**
Option 3 ID : **4447921844**
Option 4 ID : **4447921842**
Status : **Not Answered**
Chosen Option : **--**

Q.3 Let $\alpha, \beta \in \mathbb{R}$ be such that the function $f(x)=\begin{cases}2\alpha\left(x^2-2\right)+2\beta x & , x<1\\ (\alpha+3)x+(\alpha-\beta) & , x\geq 1\end{cases}$ be differentiable at all $x \in \mathbb{R}$. Then $34(\alpha+\beta)$ is equal to

Options
1. 84
2. 24
3. 36
4. 48

$2\beta - 2\alpha = (\alpha+3)+(\alpha-\beta)$

$3\beta - 4\alpha = 3$

$f'(x)$ → $4\alpha x+2\beta$; $\alpha+3$

$\alpha+3 = 4\alpha+2\beta$

$\alpha = \frac{3}{17}$, $\beta = \frac{21}{17}$

$34(\alpha+\beta) = 48$

LCD X11 easy

Question Type : **MCQ**
Question ID : **444792543**
Option 1 ID : **4447921857**
Option 2 ID : **4447921854**
Option 3 ID : **4447921855**
Option 4 ID : **4447921856**
Status : **Answered**
Chosen Option : **4**

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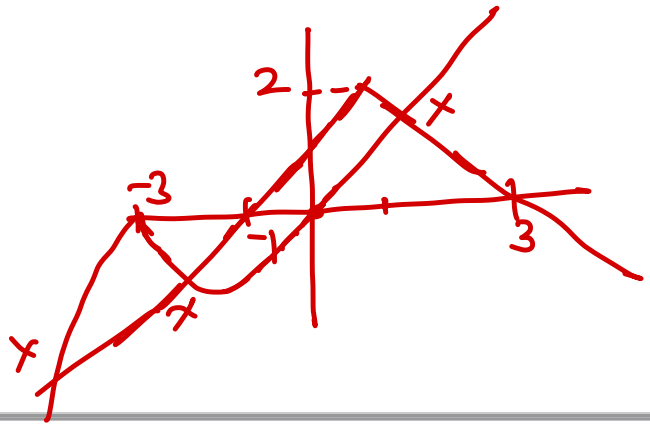
Q.4 The number of the real solutions of the equation:

$x|x+3|+|x-1|-2=0$ is

$x|x+3| = 2-|x-1|$

3 sol^ns

Options:
1. 3
2. 2
3. 4
4. 5



Question Type : **MCQ**
Question ID : **444792530**
Option 1 ID : **4447921803**
Option 2 ID : **4447921802**
Option 3 ID : **4447921805**
Option 4 ID : **4447921804**
Status : **Answered**
Chosen Option : **1**

Q.5 The mean and variance of a data of 10 observations are 10 and 2, respectively. If an observations α in this data is replaced by β, then the mean and variance become 10.1 and 1.99, respectively. Then α + β equals

$100+\beta-\alpha=101$

$\beta-\alpha=1$

$\beta+\alpha=0$

$1.99+102.01=\frac{\Sigma a_i^2+\beta^2}{10}$

$1040=\Sigma a_i^2+\beta^2$

$1620=\Sigma a_i^2+\alpha^2$

$\beta^2-\alpha^2=20$

$\beta+\alpha=20$

$\beta-\alpha=1$

Options:
1. 20
2. 15
3. 5
4. 10

Question Type : **MCQ**
Question ID : **444792533**
Option 1 ID : **4447921817**
Option 2 ID : **4447921816**
Option 3 ID : **4447921814**
Option 4 ID : **4447921815**
Status : **Not Answered**
Chosen Option : --

Statistics
XII
medium

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Q.6 If the domain of the function

$f(x) = \log_{(10x^2 - 17x + 7)}(18x^2 - 11x + 1)$

is $(-\infty, a) \cup (b, c) \cup (d, \infty) - \{e\}$, then

$90(a + b + c + d + e)$ equals:

Options
1. 316
2. 307
3. 170
4. 177

Medium
function
XII

Question Type : MCQ
Question ID : 444792527
Option 1 ID : 4447921790
Option 2 ID : 4447921791
Option 3 ID : 4447921792
Option 4 ID : 4447921793
Status : Answered
Chosen Option : 1

Q.7 From a lot containing 10 defective and 90 non-defective bulbs, 8 bulbs are selected one by one with replacement. Then the probability of getting at least 7 defective bulbs is

Options
1. $\frac{81}{10^8}$
2. $\frac{7}{10^7}$
3. $\frac{73}{10^8}$
4. $\frac{67}{10^8}$

Probability Easy
XII

Question Type : MCQ
Question ID : 444792534
Option 1 ID : 4447921820
Option 2 ID : 4447921819
Option 3 ID : 4447921818
Option 4 ID : 4447921821
Status : Not Answered
Chosen Option : --

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Q.8 If $\cot x = \frac{5}{12}$ for some $x \in \left(\pi, \frac{3\pi}{2}\right)$, then

$\sin 7x\left(\cos\frac{13x}{2}+\sin\frac{13x}{2}\right)+\cos 7x\left(\cos\frac{13x}{2}-\sin\frac{13x}{2}\right)$ is equal to

Options

1. $\frac{4}{\sqrt{26}}$
2. $\frac{6}{\sqrt{26}}$
3. $\frac{1}{\sqrt{13}}$
4. $\frac{5}{\sqrt{13}}$

sin x is -ve

$\cos\frac{x}{2}+\sin\frac{x}{2}=E$

$E^2 = 1+\sin x$

$E^2 = 1-\frac{12}{13}=\frac{1}{13}$

Trigo XI easy

Question Type : MCQ
Question ID : 444792537
Option 1 ID : 4447921831
Option 2 ID : 4447921830
Option 3 ID : 4447921833
Option 4 ID : 4447921832
Status : Not Answered
Chosen Option : --

Q.9 Let R be a relation defined on the set $\{1, 2, 3, 4\} \times \{1, 2, 3, 4\}$ by

$R = \{((a, b), (c, d)) : 2a + 3b = 3c + 4d\}$.

Then the number of elements in R is

Options

1. 18
2. 6
3. 15
4. 12

$3(b-c) = 2(2d-a)$

$a=d=3 \quad 3(b-c)=6 \Rightarrow b-c=2$

3 1
4 2 → 2

$d=1 \quad 3(b-c)=2(2-a)$

$\Rightarrow a=2 \quad b=c \rightarrow$ 4 cases

Relations Tough XII

$d=4 \quad 3(b-c)=2(8-a)$

$a=2 \quad 3(b-c)=12$

$b-c=4$ ✗

Question Type : MCQ
Question ID : 444792526
Option 1 ID : 4447921789
Option 2 ID : 4447921786
Option 3 ID : 4447921788
Option 4 ID : 4447921787
Status : Not Answered
Chosen Option : --

$d=2 \quad 3(b-c)=2(4-a)$

$a=1$

$b-c=2 \rightarrow$ 2 cases

$a=4$

$3(b-c)=0 \rightarrow$ 4 cases

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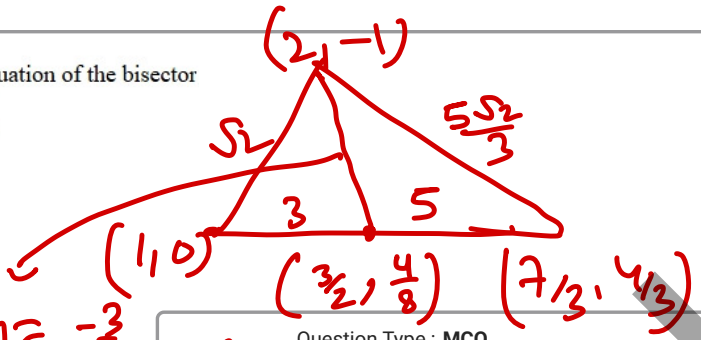
Q.10 Let A(1, 0), B(2, −1) and $C\left(\frac{7}{3},\frac{4}{3}\right)$ be three points. If the equation of the bisector of the angle ABC is $\alpha x + \beta y = 5$, then the value of $\alpha^2 + \beta^2$ is

Options
1. 13
2. 5
3. 10
4. 8



$3x+y=5$

straight line

Medium

$m = \frac{-\frac{3}{2}}{\frac{1}{2}} = -3$

Question Type : **MCQ**
Question ID : **444792538**
Option 1 ID : **4447921837**
Option 2 ID : **4447921834**
Option 3 ID : **4447921836**
Option 4 ID : **4447921835**
Status : **Not Answered**
Chosen Option : **--**

Q.11 Let each of the two ellipses $E_1 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, (a > b)$ and $E_2 : \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1, (A < B)$ have eccentricity $\frac{4}{5}$. Let the lengths of the latus recta of E_1 and E_2 be l_1 and l_2, respectively, such that $2l_1^2 = 9l_2$. If the distance between the foci of E_1 is 8, then the distance between the foci of E_2 is

Options
1. $\frac{96}{5}$
2. $\frac{32}{5}$
3. $\frac{16}{5}$
4. $\frac{8}{5}$

$1-\frac{b^2}{a^2}=\frac{16}{25}$ $\quad \frac{2b^2}{a}=l_1=\frac{18}{25}$

$\frac{b^2}{a^2}=\frac{9}{25}$ $\quad \frac{2A^2}{B}=l_2$

$l_2=\frac{18B}{25}$

$2\times\left(\frac{18}{25}\right)a^2 = 9\left(\frac{18}{25}\right)B$

$\frac{4}{25}a^2 = B$

$a=5$

$2ae=8$

$2Be = 8\times\frac{4}{5} = \frac{32}{5}$

$B=4$

Conic

Medium

Question Type : **MCQ**
Question ID : **444792535**
Option 1 ID : **4447921825**
Option 2 ID : **4447921824**
Option 3 ID : **4447921823**
Option 4 ID : **4447921822**
Status : **Answered**
Chosen Option : **2**

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Q.12 Consider an A.P.: $a_1, a_2, \ldots, a_n; a_1 > 0$. If $a_2 - a_1 = \frac{-3}{4}$, $a_n = \frac{1}{4}a_1$, and $\sum_{i=1}^{n} a_i = \frac{525}{2}$, then $\sum_{i=1}^{17} a_i$ is equal to

Options
1. 136
2. 952
3. 476
4. 238

Question Type : MCQ
Question ID : 444792531
Option 1 ID : 4447921806
Option 2 ID : 4447921809
Option 3 ID : 4447921808
Option 4 ID : 4447921807
Status : Not Answered
Chosen Option : --

Q.13 The value of $\frac{\sqrt{3}\,\text{cosec}20^\circ - \sec 20^\circ}{\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ}$ is equal to

Options
1. 64
2. 12
3. 16
4. 32

Question Type : MCQ
Question ID : 444792539
Option 1 ID : 4447921841
Option 2 ID : 4447921839
Option 3 ID : 4447921838
Option 4 ID : 4447921840
Status : Answered
Chosen Option : 1

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Q.14

If the function $f(x)=\dfrac{e^{x}\left(e^{\tan x-x}-1\right)+\log_{e}(\sec x+\tan x)-x}{\tan x-x}$ is

continuous at $x=0$, then the value of $f(0)$ is equal to

Options
1. 2
2. $\dfrac{2}{3}$
3. $\dfrac{1}{2}$
4. $\dfrac{3}{2}$

Question Type : MCQ
Question ID : 444792542
Option 1 ID : 4447921853
Option 2 ID : 4447921851
Option 3 ID : 4447921850
Option 4 ID : 4447921852
Status : Not Answered
Chosen Option : --

Q.15

Let $S=\left\{z\in\mathbb{C}:\left|\dfrac{z-6i}{z-2i}\right|=1 \text{ and } \left|\dfrac{z-8+2i}{z+2i}\right|=\dfrac{3}{5}\right\}$.

Then $\sum_{z\in S}|z|^{2}$ is equal to

Options
1. 423
2. 385
3. 413
4. 398

Question Type : MCQ
Question ID : 444792528
Option 1 ID : 4447921797
Option 2 ID : 4447921794
Option 3 ID : 4447921796
Option 4 ID : 4447921795
Status : Not Answered
Chosen Option :

PRASHANT SAIN

Q.16

Let $f(t)=\int\left(\frac{1-\sin\left(\log_e t\right)}{1-\cos\left(\log_e t\right)}\right)dt,\ t>1.$

If $f\left(e^{\pi/2}\right)=-e^{\pi/2}$ and $f\left(e^{\pi/4}\right)=\alpha\, e^{\pi/4}$, then α equals

Options
1. $1+\sqrt{2}$
2. $-1+\sqrt{2}$
3. $-1-2\sqrt{2}$
4. $-1-\sqrt{2}$

Question Type : **MCQ**
Question ID : **444792544**
Option 1 ID : **4447921860**
Option 2 ID : **4447921861**
Option 3 ID : **4447921858**
Option 4 ID : **4447921859**
Status : **Not Answered**
Chosen Option : **--**

Q.17 Let 729, 81, 9, 1, … be a sequence and P_n denote the product of the first n terms of this sequence.

If $2\sum_{n=1}^{40}\left(P_n\right)^{\frac{1}{n}}=\frac{3^{\alpha}-1}{3^{\beta}}$ and $gcd\,(\alpha,\beta)=1$, then

$\alpha+\beta$ is equal to

Options
1. 76
2. 73
3. 75
4. 74

Question Type : **MCQ**
Question ID : **444792529**
Option 1 ID : **4447921801**
Option 2 ID : **4447921798**
Option 3 ID : **4447921800**
Option 4 ID : **4447921799**
Status : **Not Answered**
Chosen Option : **--**

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Q.18 Let $S = \frac{1}{25!} + \frac{1}{3!23!} + \frac{1}{5!21!} + \ldots$ up to 13 terms. If $13S = \frac{2^k}{n!}$, $k \in N$, then $n + k$ is equal to

Options
1. 49
2. 52
3. 50
4. 51

$26!\,S = {}^{26}C_1 + {}^{26}C_3 - \ldots + {}^{26}C_{25}$

$26!\,S = 2^{25}$

$26\,S = \frac{2^{25}}{25!}$

$13S = \frac{2^{24}}{25!}$

Binomial ✓ easy

Question Type : **MCQ**
Question ID : **444792532**
Option 1 ID : **4447921810**
Option 2 ID : **4447921813**
Option 3 ID : **4447921811**
Option 4 ID : **4447921812**
Status : **Not Answered**
Chosen Option : --

Q.19 Let a circle of radius 4 pass through the origin O, the points $A(-\sqrt{3}a, 0)$ and $B(0, -\sqrt{2}b)$, where a and b are real parameters and $ab \neq 0$. Then the locus of the centroid of ΔOAB is a circle of radius

Options
1. $\frac{11}{3}$
2. $\frac{5}{3}$
3. $\frac{7}{3}$
4. $\frac{8}{3}$

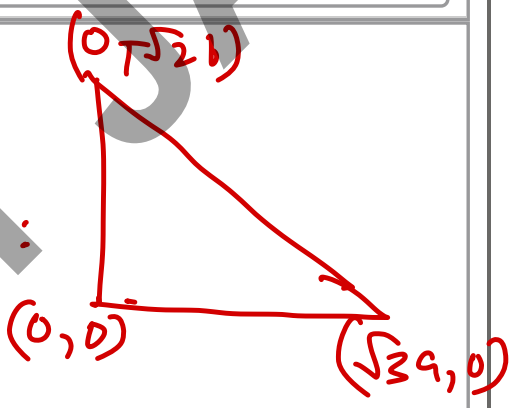
$3a^2 + 2b^2 = 64$

$3h = -\sqrt{3}a$

$9h^2 = 3a^2$

$9k^2 = 3b^2$

$9(h^2 + k^2) = 64$



Circle ✓ Medium

Question Type : **MCQ**
Question ID : **444792536**
Option 1 ID : **4447921829**
Option 2 ID : **4447921826**
Option 3 ID : **4447921827**
Option 4 ID : **4447921828**
Status : **Not Answered**
Chosen Option : --

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Q.20 Let A_1 be the bounded area enclosed by the curves $y = x^2 + 2$, $x + y = 8$ and y-axis that lies in the first quadrant. Let A_2 be the bounded area enclosed by the curves $y = x^2 + 2$, $y^2 = x$, $x = 2$ and y-axis that lies in the first quadrant. Then $A_1 - A_2$ is equal to

Options

1. $\frac{2}{3}\left(\sqrt{2}+1\right)$
2. $\frac{2}{3}\left(2\sqrt{2}+1\right)$
3. $\frac{2}{3}\left(3\sqrt{2}+1\right)$
4. $\frac{2}{3}\left(4\sqrt{2}+1\right)$

Question Type : **MCQ**
Question ID : **444792545**
Option 1 ID : **4447921862**
Option 2 ID : **4447921863**
Option 3 ID : **4447921864**
Option 4 ID : **4447921865**
Status : **Not Answered**
Chosen Option : **--**

Section : **Mathematics Section B**

Q.21 Let a differentiable function f satisfy the equation $\int_0^{36} f\left(\frac{tx}{36}\right) dt = 4\alpha f(x)$.

If $y = f(x)$ is a standard parabola passing through the points $(2, 1)$ and $(-4, \beta)$, then β^{α} is equal to _____.

Given -- Answer :

Question Type : **SA**
Question ID : **444792550**
Status : **Not Answered**

Q.22 The number of numbers greater than 5000, less than 9000 and divisible by 3, that can be formed using the digits 0, 1, 2, 5, 9, if the repetition of the digits is allowed, is _____

Given -- Answer :

Question Type : **SA**
Question ID : **444792547**
Status : **Not Answered**

PRASHANT JAIN

Q.23 Let a line L passing through the point P (1,1,1) be perpendicular to the lines $\frac{x-4}{4}=\frac{y-1}{1}=\frac{z-1}{1}$ and $\frac{x-17}{1}=\frac{y-71}{1}=\frac{z}{0}$. Let the line L intersect the yz-plane at the point Q. Another line parallel to L and passing through the point S (1, 0, − 1) intersects the yz-plane at the point R. Then the square of the area of the parallelogram PQRS is equal to _____

Given -- Answer : 6

3-D XII Tough

lengthy

$\begin{vmatrix} i & j & k \\ 4 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = -i+j+3k$

$\frac{x-1}{-1}=\frac{y-1}{1}=\frac{z-3}{1}$

$(0,2,4) \rightarrow yz$ plane

Question Type : **SA**
Question ID : **444792548**
Status : **Not Answered**

Q.24 Let $(2\alpha, \alpha)$ be the largest interval in which the function $f(t)=\frac{|t+1|}{t^2}$, $t<0$, is strictly decreasing. Then the local maximum value of the function $g(x) = 2\log_e(x-2)+\alpha x^2+4x-\alpha$, $x>2$, is _____

$\alpha=-1$

4

AOD XII Tough

Given -- Answer : 4

$g'(x)=\frac{2}{x-2}-2x+4=0$

$g(3)=-9+12+1=4$

$2=\frac{1}{x-2}=x-2$

$x=3$

$f(t) \rightarrow \frac{1}{t}+\frac{1}{t^2}$ $-2<t<0$

$-\frac{1}{t}-\frac{1}{t^2}$ $t<-1$

$f'(t) \rightarrow -\frac{1}{t^2}-\frac{2}{t^3}=-\frac{(t+2)}{t^3}$

$\frac{1}{t^2}+\frac{2}{t^3}=\frac{t+2}{t^3}$

$(-2,-1)$

Question Type : **SA**
Question ID : **444792549**
Status : **Not Answered**

Q.25 The number of 3×2 matrices A, which can be formed using the elements of the set $\{-2, -1, 0, 1, 2\}$ such that the sum of all the diagonal elements of A^TA is 5, is _____

Given -- Answer :

$\begin{bmatrix} a & c & e \\ b & d & f \end{bmatrix}\begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}=\begin{bmatrix} a^2+c^2+e^2 & \\ & b^2+d^2+f^2 \end{bmatrix}$

Question Type : **SA**
Question ID : **444792546**
Status : **Not Answered**

Section : **Physics Section A**

$a^2+b^2+c^2+d^2+e^2+f^2=5$

$4+1+0+0+0+0 \rightarrow 2\times2\times30=120$

$1+1+1+1+1+0 \rightarrow 6\times32=192$

312

MnD XII Medium

PRASHANT JAIN

NTA JEE Mains Jan 2026

Application No	
Candidate Name	
Roll No.	
Test Date	24/01/2026
Test Time	3:00 PM - 6:00 PM
Subject	B. Tech

Section : **Mathematics Section A**

Q.1 Let $[t]$ denote the greatest integer less than or equal to t. If the function

$$f(x)=\begin{cases} b^2\sin\left(\frac{\pi}{2}\left[\frac{\pi}{2}(\cos x+\sin x)\cos x\right]\right), & x<0 \\ \dfrac{\sin x-\frac{1}{2}\sin 2x}{x^3}, & x>0 \\ a, & x=0 \end{cases}$$

is continuous at $x=0$, then a^2+b^2 is equal to

Question Type : **MCQ**
Question ID : **444792616**
Option 1 ID : **4447922103**
Option 2 ID : **4447922102**
Option 3 ID : **4447922101**
Option 4 ID : **4447922104**
Status : **Answered**
Chosen Option : **3**

Options

1. $\frac{1}{2}$
2. $\frac{9}{16}$
3. $\frac{3}{4}$
4. $\frac{5}{8}$

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Q.2 The largest value of n, for which 40^n divides 60!, is

Options
1. 12
2. 14
3. 13
4. 11

Question Type : **MCQ**
Question ID : **444792607**
Option 1 ID : **4447922065**
Option 2 ID : **4447922067**
Option 3 ID : **4447922066**
Option 4 ID : **4447922068**
Status : **Answered**
Chosen Option : **1**

Q.3 The smallest positive integral value of a, for which all the roots of $x^4 - ax^2 + 9 = 0$ are real and distinct, is equal to

Options
1. 7
2. 3
3. 9
4. 4

Question Type : **MCQ**
Question ID : **444792603**
Option 1 ID : **4447922050**
Option 2 ID : **4447922049**
Option 3 ID : **4447922052**
Option 4 ID : **4447922051**
Status : **Answered**
Chosen Option : **1**

Q.4 The sum of all values of α, for which the shortest distance between the lines $\frac{x+1}{\alpha} = \frac{y-2}{-1} = \frac{z-4}{-\alpha}$ and $\frac{x}{\alpha} = \frac{y-1}{2} = \frac{z-1}{2\alpha}$ is $\sqrt{2}$, is

Options
1. 8
2. 6
3. -8
4. -6

Question Type : **MCQ**
Question ID : **444792613**
Option 1 ID : **4447922092**
Option 2 ID : **4447922090**
Option 3 ID : **4447922091**
Option 4 ID : **4447922089**
Status : **Answered**
Chosen Option : **4**

Q.5 If the domain of the function $f(x) = \sin^{-1}\left(\frac{1}{x^2 - 2x - 2}\right)$, is $(-\infty, \alpha] \cup [\beta, \gamma] \cup [\delta, \infty)$, then $\alpha + \beta + \gamma + \delta$ is equal to

Options
1. 3
2. 4
3. 2
4. 5

Question Type : **MCQ**
Question ID : **444792601**
Option 1 ID : **4447922042**
Option 2 ID : **4447922043**
Option 3 ID : **4447922041**
Option 4 ID : **4447922044**
Status : **Answered**
Chosen Option : **2**

PRASHANT JAIN

Q.6

Let the length of the latus rectum of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, (a>b)$, be 30. If its eccentricity is the maximum value of the function $f(t)=-\frac{3}{4}+2t-t^2$, then (a^2+b^2) is equal to

Options
1. 256
2. 496
3. 516
4. 276

Question Type : **MCQ**
Question ID : **444792610**
Option 1 ID : **4447922077**
Option 2 ID : **4447922078**
Option 3 ID : **4447922079**
Option 4 ID : **4447922080**
Status : **Answered**
Chosen Option : **2**

Conic XI easy

$\frac{2b^2}{a}=30$ $b^2=15a$ $f(t)=\frac{1}{4}-(t-1)^2$ $e=1/4$ $\frac{b^2}{a^2}=\frac{15}{16}$ $a=16$ 256 240

Q.7

Let $X=\{x\in\mathbb{N}:1\le x\le 19\}$ and for some $a, b\in\mathbb{R}$, $Y=\{ax+b:x\in X\}$. If the mean and variance of the elements of Y are 30 and 750, respectively, then the sum of all possible values of b is

Options
1. 60
2. 80
3. 20
4. 100

Question Type : **MCQ**
Question ID : **444792609**
Option 1 ID : **4447922074**
Option 2 ID : **4447922075**
Option 3 ID : **4447922073**
Option 4 ID : **4447922076**
Status : **Answered**
Chosen Option : **1**

$10a+b=30$ $a^2\left(\frac{19^2-1}{12}\right)=750$ $a^2=25$ $b=-20$ $(a=5)$ $b=80$ $(a=-5)$

Statistics XI easy

PRASHANT JAIN

Q.8

$\left(\frac{1}{3}+\frac{4}{7}\right)+\left(\frac{1}{3^2}+\frac{1}{3}\times\frac{4}{7}+\frac{4^2}{7^2}\right)+\left(\frac{1}{3^3}+\frac{1}{3^2}\times\frac{4}{7}+\frac{1}{3}\times\frac{4^2}{7^2}+\frac{4^3}{7^3}\right)+\ldots$ upto infinite terms, is equal to

Question Type : **MCQ**
Question ID : **444792606**
Option 1 ID : **4447922061**
Option 2 ID : **4447922063**
Option 3 ID : **4447922062**
Option 4 ID : **4447922064**
Status : **Answered**
Chosen Option : **2**

Options

1. $\frac{6}{5}$
2. $\frac{5}{2}$
3. $\frac{4}{3}$
4. $\frac{7}{4}$

Q.9

Let the angles made with the positive x-axis by two straight lines drawn from the point P(2, 3) and meeting the line $x + y = 6$ at a distance $\sqrt{\frac{2}{3}}$ from the point P be θ_1 and θ_2. Then the value of $(\theta_1 + \theta_2)$ is:

Question Type : **MCQ**
Question ID : **444792612**
Option 1 ID : **4447922088**
Option 2 ID : **4447922085**
Option 3 ID : **4447922086**
Option 4 ID : **4447922087**
Status : **Not Attempted and Marked For Review**
Chosen Option : **--**

Options

1. $\frac{\pi}{12}$
2. $\frac{\pi}{3}$
3. $\frac{\pi}{2}$
4. $\frac{\pi}{6}$

PRASHANT JAIN

Q.10 Consider the following three statements for the function $f:(0,\infty)\to\mathbb{R}$ defined by $f(x)=|\log_e x|-|x-1|$:

(I) f is differentiable at all $x>0$.

(II) f is increasing in $(0,1)$.

(III) f is decreasing in $(1,\infty)$.

Then.

Options

1. All (I), (II) and (III) are TRUE.
2. Only (I) and (III) are TRUE.
3. Only (I) is TRUE.
4. Only (II) and (III) are TRUE.

Question Type : **MCQ**
Question ID : **444792617**
Option 1 ID : **4447922108**
Option 2 ID : **4447922107**
Option 3 ID : **4447922105**
Option 4 ID : **4447922106**
Status : **Answered**
Chosen Option : **2**

Q.11 Let f be a function such that $3f(x)+2f\left(\frac{m}{19x}\right)=5x,\ x\neq 0$, where $m=\sum_{i=1}^{9}(i)^2$. Then $f(5)-f(2)$ is equal to

Options

1. 36
2. 18
3. -9
4. 9

Question Type : **MCQ**
Question ID : **444792602**
Option 1 ID : **4447922048**
Option 2 ID : **4447922047**
Option 3 ID : **4447922045**
Option 4 ID : **4447922046**
Status : **Answered**
Chosen Option : **2**

Q.12 Let a_1, a_2, a_3, a_4 be an A.P. of four terms such that each term of the A.P. and its common difference l are integers. If $a_1+a_2+a_3+a_4=48$ and $a_1 a_2 a_3 a_4 + l^4 = 361$, then the largest term of the A.P. is equal to

Options

1. 23
2. 21
3. 27
4. 24

Question Type : **MCQ**
Question ID : **444792605**
Option 1 ID : **4447922058**
Option 2 ID : **4447922057**
Option 3 ID : **4447922060**
Option 4 ID : **4447922059**
Status : **Not Attempted and Marked For Review**
Chosen Option : **--**

PRASHANT JAIN

$$3f(x) + 2f\left(\frac{m}{19x}\right) = 5x$$

$$3f\left(\frac{m}{19x}\right) + 2f(x) = \frac{5m}{19x}$$

$$m = \sum_{i=1}^{9} i^2 = \frac{9 \times 10 \times 19}{6} = 285$$

$$5f(x) = 15x - \frac{10m}{19x}$$

$$5f(x) = 15x - \frac{150}{x} \Rightarrow \boxed{f(x) = 3x - \frac{30}{x}}$$

$$f(5) = 15 - 6 = 9$$
$$f(2) = 6 - 15 = -9$$] (18)

PRASHANT JAIN

Q.13

Let $f(x)=\int \frac{7x^{10}+9x^{8}}{\left(1+x^{2}+2x^{9}\right)^{2}}dx$, $x>0$, $\lim_{x\to 0} f(x)=0$ and $f(1)=\frac{1}{4}$.

If $A=\begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{4} & f'(1) & 1 \\ \alpha^{2} & 4 & 1 \end{bmatrix}$ and $B = adj(adj\ A)$ be such that $|B| = 81$, then α^2 is equal to

Options
1. 2
2. 4
3. 3
4. 1

Question Type : **MCQ**
Question ID : **444792618**
Option 1 ID : **4447922110**
Option 2 ID : **4447922112**
Option 3 ID : **4447922111**
Option 4 ID : **4447922109**
Status : **Answered**
Chosen Option : **2**

Q.14

Let $f(\alpha)$ denote the area of the region in the first quadrant bounded by $x = 0, x = 1, y^2 = x$ and $y = |\alpha x - 5| - |1 - \alpha x| + \alpha x^2$. Then $(f(0) + f(1))$ is equal to

Options
1. 12
2. 7
3. 9
4. 14

Question Type : **MCQ**
Question ID : **444792619**
Option 1 ID : **4447922115**
Option 2 ID : **4447922113**
Option 3 ID : **4447922114**
Option 4 ID : **4447922116**
Status : **Not Attempted and Marked For Review**
Chosen Option : **--**

PRASHANT SAINI

Q.15 Let $\vec{a}=2\hat{i}-\hat{j}-\hat{k}$, $\vec{b}=\hat{i}+3\hat{j}-\hat{k}$ and $\vec{c}=2\hat{i}+\hat{j}+3\hat{k}$. Let $\vec{v}$ be the vector in the plane of the vectors $\vec{a}$ and $\vec{b}$, such that the length of its projection on the vector $\vec{c}$ is $\frac{1}{\sqrt{14}}$. Then $|\vec{v}|$ is equal to

Options
1. $\frac{\sqrt{35}}{2}$
2. $\frac{\sqrt{21}}{2}$
3. 13
4. 7

Question Type : **MCQ**
Question ID : **444792614**
Option 1 ID : **4447922096**
Option 2 ID : **4447922095**
Option 3 ID : **4447922094**
Option 4 ID : **4447922093**
Status : **Not Attempted and Marked For Review**
Chosen Option : **--**

Q.16 Let $\vec{a}=2\hat{i}-5\hat{j}+5\hat{k}$ and $\vec{b}=\hat{i}-\hat{j}+3\hat{k}$. If $\vec{c}$ is a vector such that $2(\vec{a}\times\vec{c})+3(\vec{b}\times\vec{c})=\vec{0}$ and $(\vec{a}-\vec{b})\cdot\vec{c}=-97$, then $\left|\vec{c}\times\hat{k}\right|^2$ is equal to

Options
1. 193
2. 233
3. 218
4. 205

Question Type : **MCQ**
Question ID : **444792615**
Option 1 ID : **4447922097**
Option 2 ID : **4447922100**
Option 3 ID : **4447922099**
Option 4 ID : **4447922098**
Status : **Answered**
Chosen Option : **3**

Q.17 The letters of the word "UDAYPUR" are written in all possible ways with or without meaning and these words are arranged as in a dictionary. The rank of the word "UDAYPUR" is

Options
1. 1578
2. 1579
3. 1580
4. 1581

Question Type : **MCQ**
Question ID : **444792608**
Option 1 ID : **4447922069**
Option 2 ID : **4447922070**
Option 3 ID : **4447922071**
Option 4 ID : **4447922072**
Status : **Answered**
Chosen Option : **3**

PRASHANT JAIN

Q.18 Let the image of parabola $x^2 = 4y$, in the line $x - y = 1$ be $(y + a)^2 = b(x - c)$, $a, b, c \in \mathbb{N}$. Then $a + b + c$ is equal to

Options
1. 8
2. 6
3. 4
4. 12

Question Type : MCQ
Question ID : 444792611
Option 1 ID : 4447922083
Option 2 ID : 4447922082
Option 3 ID : 4447922081
Option 4 ID : 4447922084
Status : Answered
Chosen Option : 2

Q.19 Let $y = y(x)$ be a differentiable function in the interval $(0, \infty)$ such that $y(1) = 2$, and $\lim_{t \to x} \left(\frac{t^2 y(x) - x^2 y(t)}{x - t} \right) = 3$ for each $x > 0$. Then $2y(2)$ is equal to

Options
1. 27
2. 23
3. 12
4. 18

Question Type : MCQ
Question ID : 444792620
Option 1 ID : 4447922120
Option 2 ID : 4447922117
Option 3 ID : 4447922118
Option 4 ID : 4447922119
Status : Answered
Chosen Option : 2

Q.20 Let $P = [p_{ij}]$ and $Q = [q_{ij}]$ be two square matrices of order 3 such that $q_{ij} = 2^{(i+j-1)} p_{ij}$ and $\det(Q) = 2^{10}$. Then the value of $\det(adj(adj\ P))$ is:

Options
1. 16
2. 124
3. 32
4. 81

Question Type : MCQ
Question ID : 444792604
Option 1 ID : 4447922053
Option 2 ID : 4447922056
Option 3 ID : 4447922055
Option 4 ID : 4447922054
Status : Not Answered
Chosen Option : --

Section : Mathematics Section B

PRASHANT JAIN

Q.21 Let S be a set of 5 elements and P(S) denote the power set of S. Let E be an event of choosing an ordered pair (A, B) from the set P(S) × P(S) such that $A \cap B = \emptyset$. If the probability of the event E is $\frac{3^p}{2^q}$, where p, q ∈ N, then p + q is equal to _____

Given Answer -- :

Question Type : **SA**
Question ID : **444792622**
Status : **Not Answered**

Q.22 Let (h, k) lie on the circle C : $x^2 + y^2 = 4$ and the point (2h + 1, 3k + 2) lie on an ellipse with eccentricity e. Then the value of $\frac{5}{e^2}$ is equal to _____.

Given Answer **9** :

Question Type : **SA**
Question ID : **444792623**
Status : **Answered**

Q.23 The number of elements in the set $\{x \in [0, 180^\circ] : \tan(x + 100^\circ) = \tan(x + 50^\circ)\tan x \tan(x - 50^\circ)\}$ is _____.

Given Answer **4** :

Question Type : **SA**
Question ID : **444792624**
Status : **Not Answered**

Q.24 If $f(x)$ satisfies the relation $f(x) = e^x + \int_0^1 \left(y + xe^x\right) f(y)\, dy$, then $e + f(0)$ is equal to _____

Given Answer -- :

Question Type : **SA**
Question ID : **444792625**
Status : **Not Answered**

Q.25 Let $z = (1 + i)(1 + 2i)(1 + 3i) \ldots (1 + ni)$, where $i = \sqrt{-1}$. If $|z|^2 = 44200$, then n is equal to ____

Given Answer -- :

Question Type : **SA**
Question ID : **444792621**
Status : **Not Answered**

Section : **Physics Section A**

PRASHANT JAIN

→ $\frac{t_{x+100}}{t_x} = t_{x+50} t_{x-50}$

$$\frac{s_{x+100} c_x}{s_x c_{x+100}} = \frac{s_{x+50} s_{x-50}}{c_{x+50} c_{x-50}}$$

C&D

$$\frac{s_{2x+100}}{s_{100}} = \frac{c_{100}}{-c_{2x}}$$

$$2 s_{2x+100} c_{2x} = s_{20}$$

$$s_{4x+100} + s_{100} = s_{20}$$

$$c_{10+4x} = s_{20} - s_{100} = -2 c_{60} s_{40}$$

$$c_{10+4x} = c_{130^\circ}$$

$4x + 10 = 2n\pi \pm 130^\circ$

$\frac{n\pi}{2} + 30^\circ$: $30^\circ, 120^\circ$

$\frac{n\pi}{2} - 35^\circ$: $55^\circ, 145^\circ$ (4) ★

→

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24) $f(x) = e^x + \int_0^1 (y + xe^x) f(y)\,dy$

$f(x) = e^x + \int_0^1 y f(y)\,dy + xe^x \int_0^1 f(y)$

$f(x) = (ax+1)e^x + b$

$\int_0^1 f(x) = b + ((a+1)e - 1) - ae = a$

$b + e - 1 = a$

$\int_0^1 x f(x) = \int_0^1 (x(ax+1)e^x + bx)$

$b = \frac{b}{2} + a(e-1) + 1$

$\frac{b}{2} + a(1-e) = 1$

$f(0) = 1 + b$

$f(0) + e = 2$

$b = 1 - e$

PRASHANT JAIN

NTA JEE Mains Jan 2026

Application No	
Candidate Name	
Roll No.	
Test Date	28/01/2026
Test Time	9:00 AM - 12:00 PM
Subject	B. Tech

Section : **Mathematics Section A**

Q.1 Let $S = \{x^3 + ax^2 + bx + c : a, b, c \in N \text{ and } a, b, c \le 20\}$ be a set of polynomials. Then the number of polynomials in S, which are divisible by $x^2 + 2$, is

Options 1. 6
2. 20
3. 10
4. 120

$x^2 = -2$, $x^3 = -2x$

$(b-2)x + (c-2a) = 0$ $\quad b=2$ $\quad c=2a$

FOM XI easy

Question Type : **MCQ**
Question ID : **444792676**
Option 1 ID : **4447922296**
Option 2 ID : **4447922298**
Option 3 ID : **4447922297**
Option 4 ID : **4447922299**
Status : **Answered**
Chosen Option : **2**

Q.2 The mean and variance of 10 observations are 9 and 34.2, respectively. If 8 of these observations are 2, 3, 5, 10, 11, 13, 15, 21, then the mean deviation about the median of all the 10 observations is

Options 1. 4
2. 5
3. 6
4. 7

statistics XI Medium

$x+y = 10$

$1152 = 2^2+3^2+5^2+10^2+11^2+13^2+15^2+21^2+x^2+y^2$

$x^2+y^2 = 1152 - 869 - 225$

$x^2+y^2 = 58$

$x=7, y=3$

Question Type : **MCQ**
Question ID : **444792683**
Option 1 ID : **4447922327**
Option 2 ID : **4447922326**
Option 3 ID : **4447922325**
Option 4 ID : **4447922324**
Status : **Not Answered**
Chosen Option : **--**

2, 3, 3, 5, 7, 10, 11, 13, 15, 21

8.5

$6.5 + 5.5 + 5.5 + 3.5 + 1.5 + 1.5 + 2.5 + 4.5 + 6.5 + 12.5$

$\bar{5} = \dfrac{\quad}{10}$

PRASHANT JAIN

Q.3 If $\frac{\tan(A-B)}{\tan A}+\frac{\sin^2 C}{\sin^2 A}=1, A, B, C \in \left(0, \frac{\pi}{2}\right)$, then

Options
1. tan A, tan B, tan C are in A.P.
2. tan A, tan C, tan B are in G.P.
3. tan A, tan B, tan C are in G.P.
4. tan A, tan C, tan B are in A.P.

Back

Trigo Tough XI

Question Type : **MCQ**
Question ID : **444792687**
Option 1 ID : **4447922341**
Option 2 ID : **4447922342**
Option 3 ID : **4447922340**
Option 4 ID : **4447922343**
Status : **Answered**
Chosen Option : **2**

Q.4 A bag contains 10 balls out of which k are red and $(10-k)$ are black, where $0 \le k \le 10$. If three balls are drawn at random without replacement and all of them are found to be black, then the probability that the bag contains 1 red and 9 black balls is:

Options
1. $\frac{7}{11}$
2. $\frac{7}{110}$
3. $\frac{14}{55}$
4. $\frac{7}{55}$

R	B	
7	3	$\rightarrow {}^3C_3/{}^{10}C_3$
6	4	$\rightarrow {}^4C_3/{}^{10}C_3$
5	5	
4	6	
3	7	
2	8	
1	9	
0	10	

$\frac{11 \times 10 \times 9}{3 \times 2 \times 1}$

Probability XII Medium

Question Type : **MCQ**
Question ID : **444792682**
Option 1 ID : **4447922320**
Option 2 ID : **4447922323**
Option 3 ID : **4447922321**
Option 4 ID : **4447922322**
Status : **Answered**
Chosen Option : **1**

$$\frac{{}^9C_3}{{}^3C_3+{}^4C_3-\cdots+{}^{10}C_3}=\frac{84}{{}^{11}C_4}=\frac{84}{330}$$

$=$

PRASHANT JAIN

$$\frac{S^2_C}{S^2_A} = 1 - \frac{\tan_{A-B}}{+_A} = \frac{S_A C_{A-B} - C_A S_{A-B}}{S_A C_{A-B}}$$

$$\frac{S^2_C}{S^2_A} = \frac{S_B}{S_A C_{A-B}}$$

$$S^2_C C_{A-B} = S_B$$

$$S^2_C (C_A C_B + S_A S_B) = S_B S_A$$

$$S_C^2 C_A C_B = S_A S_B C_C^2$$

$$\tan^2 C = \tan A \tan B$$

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Q.5 Let $y = y(x)$ be the solution of the differential equation

$$x\frac{dy}{dx} - \sin 2y = x^3\left(2 - x^3\right)\cos^2 y, \quad x \neq 0 .$$

If $y(2) = 0$, then $\tan(y(1))$ is equal to

Options

1. $-\frac{7}{4}$
2. $-\frac{3}{4}$
3. $\frac{7}{4}$
4. $\frac{3}{4}$

Question Type : **MCQ**
Question ID : **444792694**
Option 1 ID : **4447922368**
Option 2 ID : **4447922369**
Option 3 ID : **4447922370**
Option 4 ID : **4447922371**
Status : **Not Answered**
Chosen Option : **--**

Q.6 Let $y = x$ be the equation of a chord of the circle C_1 (in the closed half-plane $x \geq 0$) of diameter 10 passing through the origin. Let C_2 be another circle described on the given chord as its diameter. If the equation of the chord of the circle C_2, which passes through the point (2, 3) and is farthest from the center of C_2, is $x + ay + b = 0$, then $a - b$ is equal to

Options

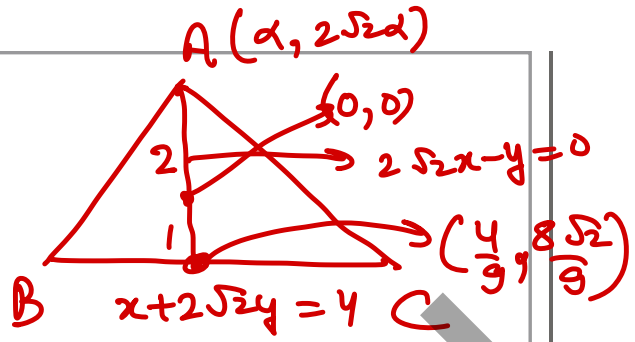
1. 6
2. -6
3. 10
4. 2

Question Type : **MCQ**
Question ID : **444792686**
Option 1 ID : **4447922338**
Option 2 ID : **4447922336**
Option 3 ID : **4447922339**
Option 4 ID : **4447922337**
Status : **Not Answered**
Chosen Option : **--**

PRASHANT JAIN

Q.7 Let ABC be an equilateral triangle with orthocenter at the origin and the side BC on the line $x+2\sqrt{2}y=4$. If the co-ordinates of the vertex A are (α, β), then the greatest integer less than or equal to $|\alpha+\sqrt{2}\beta|$ is

Options
1. 2
2. 5
3. 3
4. 4



straight line
XI
medium

Question Type : **MCQ**
Question ID : **444792685**
Option 1 ID : **4447922332**
Option 2 ID : **4447922335**
Option 3 ID : **4447922333**
Option 4 ID : **4447922334**
Status : **Not Answered**
Chosen Option : **--**

Q.8 The value of $\sum_{k=1}^{\infty}(-1)^{k+1}\left(\frac{k(k+1)}{k!}\right)$ is

Options
1. $\sqrt{e}$
2. 1/e
3. e/2
4. 2/e

LCD
XII
medium

Question Type : **MCQ**
Question ID : **444792690**
Option 1 ID : **4447922355**
Option 2 ID : **4447922354**
Option 3 ID : **4447922352**
Option 4 ID : **4447922353**
Status : **Not Answered**
Chosen Option : **--**

PRASHANT JAIN

Q.9 If α, β, where $\alpha < \beta$, are the roots of the equation $\lambda x^2 - (\lambda + 3)x + 3 = 0$ such that $\frac{1}{\alpha} - \frac{1}{\beta} = \frac{1}{3}$, then the sum of all possible values of λ is

Options

1. 6
2. 2
3. 8
4. 4

$(x-1)(\lambda x-3)=0$

$x=1 \quad x=\frac{3}{\lambda}$

$1-\frac{\lambda}{3}=\frac{1}{3} \Rightarrow \lambda=2$

$\frac{\lambda}{3}-1=\frac{1}{3} \Rightarrow \lambda=4$

Quadratic XI easy

Question Type : **MCQ**
Question ID : **444792678**
Option 1 ID : **4447922306**
Option 2 ID : **4447922304**
Option 3 ID : **4447922307**
Option 4 ID : **4447922305**
Status : **Answered**
Chosen Option : **1**

Q.10 If $g(x) = 3x^2 + 2x - 3, f(0) = -3$ and $4g(f(x)) = 3x^2 - 32x + 72$, then $f(g(2))$ is equal to:

Options

1. $\frac{7}{2}$
2. $-\frac{7}{2}$
3. $\frac{25}{6}$
4. $-\frac{25}{6}$

$f(x)$ is linear $f(x)=ax-3$

$4g(f(x)) = (3(ax-3)^2+2(ax-3)-3)4$

$= 3x^2-32x+72$

$12a^2=3$

$-64a=-32$

$a=\frac{1}{2}$

$f(x)=\frac{x}{2}-3$

$f(g(2))=f(13)$

$=\frac{13}{2}-3=\frac{7}{2}$

Functions XII Medium

Question Type : **MCQ**
Question ID : **444792677**
Option 1 ID : **4447922302**
Option 2 ID : **4447922301**
Option 3 ID : **4447922300**
Option 4 ID : **4447922303**
Status : **Not Attempted and Marked For Review**
Chosen Option : **--**

PRASHANT JAIN

Q.11 Let A, B and C be three 2×2 matrices with real entries such that $B = (I + A)^{-1}$ and $A + C = I$. If $BC = \begin{bmatrix} 1 & -5 \\ -1 & 2 \end{bmatrix}$ and $CB \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 12 \\ -6 \end{bmatrix}$, then $x_1 + x_2$ is

Options
1. 4
2. 0
3. -2
4. 2

$B(I+A) = (I+A)B$

$B + BA = B + AB$

$BA = AB$

$BA + BC = B = AB + CB$

$BC = CB$

$x_1 - 5x_2 = 12$

$-x_1 + 2x_2 = -6$

$\begin{bmatrix} 1 & -5 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 12 \\ -6 \end{bmatrix}$

$x_2 = -2$

$x_1 = 2$

Question Type : **MCQ**
Question ID : **444792680**
Option 1 ID : **4447922315**
Option 2 ID : **4447922312**
Option 3 ID : **4447922314**
Option 4 ID : **4447922313**
Status : **Not Attempted and Marked For Review**
Chosen Option : --

M&D XII Tough

Q.12 Let z be a complex number such that $|z - 6| = 5$ and $|z + 2 - 6i| = 5$. Then the value of $z^3 + 3z^2 - 15z + 141$ is equal to

Options
1. 61
2. 42
3. 50
4. 37

(-2,6) (6,0) (2,3)

$z = 2 + 3i$

$z^2 - 4z + 13 = 0$

$z^3 + 3z^2 - 15z + 141$

$= (z^2 - 4z + 13)(z + 7) + 50$

$= 0 + 50$

Question Type : **MCQ**
Question ID : **444792679**
Option 1 ID : **4447922311**
Option 2 ID : **4447922309**
Option 3 ID : **4447922310**
Option 4 ID : **4447922308**
Status : **Not Attempted and Marked For Review**
Chosen Option : --

Complex No. XII Medium

Reduction of

PRASHANT JAIN

Q.13 Let S = {1, 2, 3, 4, 5, 6, 7, 8, 9}. Let x be the number of 9-digit numbers formed using the digits of the set S such that only one digit is repeated and it is repeated exactly twice. Let y be the number of 9-digit numbers formed using the digits of the set S such that only two digits are repeated and each of these is repeated exactly twice. Then,

Options

1. $21x = 4y$
2. $29x = 5y$
3. $45x = 7y$
4. $56x = 9y$

$x = {}^9C_1 \times {}^8C_7 \times \frac{9!}{2!} = 36 \times 9!$

$y = {}^9C_2 \times {}^7C_5 \times \frac{9!}{2!2!} = \frac{9}{4} \times 36 \times 2 \times 9!$

$\frac{x}{36} = \frac{y}{189}$ 21

$21x = 4y$

PnC easy

Question Type : **MCQ**
Question ID : **444792684**
Option 1 ID : **4447922328**
Option 2 ID : **4447922331**
Option 3 ID : **4447922329**
Option 4 ID : **4447922330**
Status : **Answered**
Chosen Option : **1**

Q.14 If $\int \left(\frac{1 - 5\cos^2 x}{\sin^5 x \cos^2 x} \right) dx = f(x) + C$, where C is the constant of integration, then $f\left(\frac{\pi}{6}\right) - f\left(\frac{\pi}{4}\right)$ is equal to

Options

1. $\frac{1}{\sqrt{3}}\left(26 + \sqrt{3}\right)$
2. $\frac{1}{\sqrt{3}}\left(26 - \sqrt{3}\right)$
3. $\frac{2}{\sqrt{3}}\left(4 + \sqrt{6}\right)$
4. $\frac{4}{\sqrt{3}}\left(8 - \sqrt{6}\right)$

$\int \frac{\sec^2 x}{\sin^5 x} - \int \frac{5}{\sin^5 x}$

$\frac{\tan x}{\sin^5 x} + \int \frac{5 \tan x \cos x}{\sin^6 x} - \frac{5}{\sin^5 x} + C$

$f(x) = \frac{\tan x}{\sin^5 x}$

$f\left(\frac{\pi}{6}\right) = \frac{32}{\sqrt{3}}$

$f\left(\frac{\pi}{4}\right) = 4\sqrt{2}$

$\frac{32}{\sqrt{3}} - 4\sqrt{2}$

$\frac{4}{\sqrt{3}}(8 - \sqrt{6})$

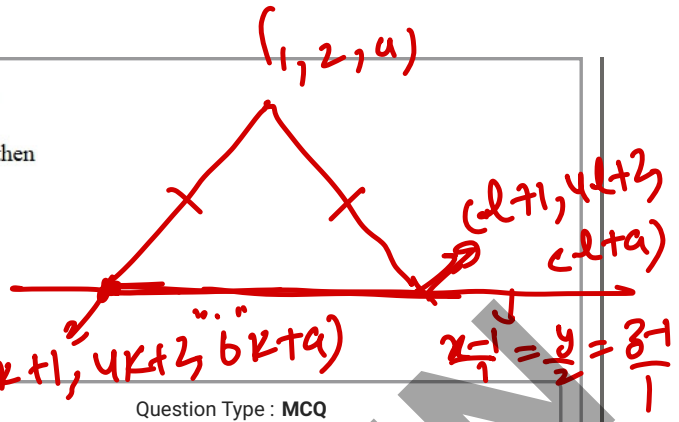
I T Tough

Question Type : **MCQ**
Question ID : **444792695**
Option 1 ID : **4447922372**
Option 2 ID : **4447922375**
Option 3 ID : **4447922374**
Option 4 ID : **4447922373**
Status : **Not Attempted and Marked For Review**
Chosen Option : **--**

PRASHANT JAIN

Q.15 If the distances of the point $(1, 2, a)$ from the line $\frac{x-1}{1}=\frac{y}{2}=\frac{z-1}{1}$ along the lines $L_1: \frac{x-1}{3}=\frac{y-2}{4}=\frac{z-a}{b}$ and $L_2: \frac{x-1}{1}=\frac{y-2}{4}=\frac{z-a}{c}$ are equal, then $a+b+c$ is equal to

Options 1. 7
2. 4
3. 5
4. 6



$\frac{3k}{1}=\frac{4k+2}{2}=\frac{bk+a-1}{1}$

$k=1$

$3=a+b-1$

$a+b=4$

$\Rightarrow a=c$

$\frac{x-1}{1}=\frac{y}{2}=\frac{z-1}{1}$

3D ✗17 tough

then for other line lengthy

Question Type : **MCQ**
Question ID : **444792688**
Option 1 ID : **4447922344**
Option 2 ID : **4447922347**
Option 3 ID : **4447922346**
Option 4 ID : **4447922345**
Status : **Not Attempted and Marked For Review**
Chosen Option : --

Q.16 For three unit vectors $\vec{a}, \vec{b}, \vec{c}$ satisfying $|\vec{a}-\vec{b}|^2+|\vec{b}-\vec{c}|^2+|\vec{c}-\vec{a}|^2=9$ and $|2\vec{a}+k\vec{b}+k\vec{c}|=3$, the positive value of k is

Options 1. 6
2. 3
3. 4
4. 5

$\hookrightarrow ab+bc+ca=-\frac{3}{2} \Rightarrow a+b+c=0$

$|2a+k(b+c)|=3$

$|2-k|=3 \quad k=5,-1$

Vector medium ✗17

Question Type : **MCQ**
Question ID : **444792689**
Option 1 ID : **4447922351**
Option 2 ID : **4447922348**
Option 3 ID : **4447922349**
Option 4 ID : **4447922350**
Status : **Not Attempted and Marked For Review**
Chosen Option : --

PRASHANT JAIN

Q.17 The value of

$$\lim_{x\to 0}\frac{\log_e\left(\sec(ex)\cdot\sec\left(e^2x\right)\cdot\ldots\cdot\sec\left(e^{10}x\right)\right)}{e^2-e^{2\cos x}}$$

is equal to

Options

1. $\frac{\left(e^{10}-1\right)}{2\left(e^2-1\right)}$
2. $\frac{\left(e^{10}-1\right)}{2e^2\left(e^2-1\right)}$
3. $\frac{\left(e^{20}-1\right)}{2e^2\left(e^2-1\right)}$
4. $\frac{\left(e^{20}-1\right)}{2\left(e^2-1\right)}$

$\sec(e^2x)=\frac{1}{\cos(e^2x)}=\frac{1}{1-\frac{e^{2r}x^2}{2!}}=1+\frac{e^{2r}x^2}{2!}$

$\sum_{r=1}^{10}\ln\left(1+\frac{e^{2r}x^2}{2!}\right)$

$e^2-e^{2-x^2}$

$=\frac{\frac{x^2}{2}\left(e^2+e^4+\ldots+e^{20}\right)}{e^2\left(e^{x^2}-1\right)}$

$=\frac{e^2\left(e^{20}-1\right)}{2\left(e^2-1\right)e^{x^2}}$

Limits Tough XII

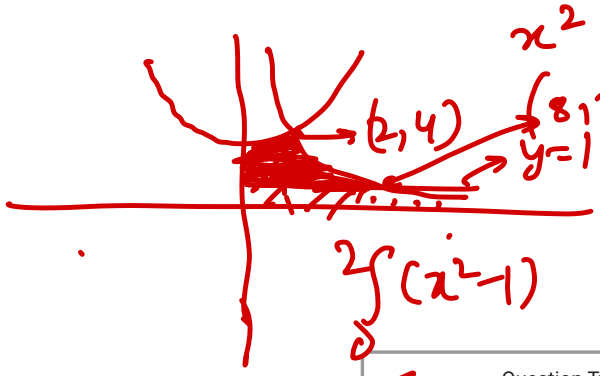
Question Type :	**MCQ**
Question ID :	**444792693**
Option 1 ID :	**4447922366**
Option 2 ID :	**4447922367**
Option 3 ID :	**4447922365**
Option 4 ID :	**4447922364**
Status :	**Not Attempted and Marked For Review**
Chosen Option :	**--**

PRASHANT JAIN

Q.18 The area of the region $R = \{(x, y) : xy \le 8, 1 \le y \le x^2, x \ge 0\}$ is

Options

1. $\frac{1}{3}\left(40\log_e(2)+27\right)$
2. $\frac{2}{3}\left(24\log_e(2)-7\right)$
3. $\frac{2}{3}\left(20\log_e(2)+9\right)$
4. $\frac{1}{3}\left(49\log_e(2)-15\right)$



$x^2 = \frac{8}{x}$

$\int_0^2 (x^2-1) + \int_2^8 \left(\frac{8}{x}-1\right)$

$\frac{8}{3} - 2 + [8(\ln 4) - 6]$

$16\ln 2 - \frac{14}{3}$

AUC Medium JEE

$\frac{2}{3}(24\ln 2 - 7)$

Question Type : MCQ
Question ID : 444792692
Option 1 ID : 4447922363
Option 2 ID : 4447922361
Option 3 ID : 4447922360
Option 4 ID : 4447922362
Status : Not Attempted and Marked For Review
Chosen Option : --

Q.19 Let f be a polynomial function such that $f(x^2+1) = x^4 + 5x^2 + 2$, for all $x \in \mathbb{R}$.

Then $\int_0^3 f(x)dx$ is equal to

Options

1. $\frac{41}{3}$
2. $\frac{33}{2}$
3. $\frac{5}{3}$
4. $\frac{27}{2}$

$f(x^2+1) = (x^2+1)^2 + 3(x^2+1) - 2$

$f(t) = t^2 + 3t - 2$

$\int_0^3 (t^2+3t-2) = \frac{t^3}{3} + \frac{3t^2}{2} - 2t \Big|_0^3$

$= 9 + \frac{27}{2} - 6$

$= \frac{33}{2}$

DI (XII) easy

Question Type : MCQ
Question ID : 444792691
Option 1 ID : 4447922358
Option 2 ID : 4447922357
Option 3 ID : 4447922359
Option 4 ID : 4447922356
Status : Not Attempted and Marked For Review
Chosen Option : --

PRASHANT JAIN

Q.20 The common difference of the A.P.: $a_1, a_2, \ldots, a_m$ is 13 more than the common difference of the A.P.: $b_1, b_2, \ldots, b_n$. If $b_{31} = -277$, $b_{43} = -385$ and $a_{78} = 327$, then a_1 is equal to

$d_b = \frac{-385+277}{12} = -9$ $\quad d_a = 4$

$77 \times 4 + a = 327$

$a = 19$

Options
1. 24
2. 16
3. 21
4. 19

S&S XI easy

Question Type : **MCQ**
Question ID : **444792681**
Option 1 ID : **4447922319**
Option 2 ID : **4447922316**
Option 3 ID : **4447922318**
Option 4 ID : **4447922317**
Status : **Answered**
Chosen Option : **4**

Section : **Mathematics Section B**

Q.21 In a G.P., if the product of the first three terms is 27 and the set of all possible values for the sum of its first three terms is $\mathbb{R} - (a, b)$, then $a^2 + b^2$ is equal to _____.

Given Answer : 90

$\frac{3}{a} + 3 + 3a = S$

$3\left(a + \frac{1}{a} + 1\right) = S$

$S \geq 9 \quad S \leq -3$

$R - (-3, 9)$

S&S medium XI

Question Type : **SA**
Question ID : **444792696**
Status : **Not Attempted and Marked For Review**

Q.22 If $k = \tan\left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1}\left(\frac{2}{3}\right)\right) + \tan\left(\frac{1}{2}\sin^{-1}\left(\frac{2}{3}\right)\right)$, then the number of solutions of the equation $\sin^{-1}(kx - 1) = \sin^{-1}x - \cos^{-1}x$ is _____

Given Answer : 1

$k = \tan\left(\frac{\pi}{4} + \frac{1}{2}\sin^{-1}\frac{2}{3}\right) + \tan\left(\frac{\pi}{4} - \frac{1}{2}\sin^{-1}\frac{2}{3}\right)$

$= \tan\theta + \cot\theta$

$= \frac{2}{\sin 2\theta} = \frac{2}{\sin\left(\frac{\pi}{2} + \frac{1}{2}\sin^{-1}\frac{2}{3}\right)} = \frac{2}{2/3} = 3$

$\sin^{-1}(3x-1) = \frac{\pi}{2} - 2\sin^{-1}x$

$3x - 1 = 2x^2 - 1$

$x = 0, \quad x = \frac{3}{2}$ ✗

ITP Tough XII

Question Type : **SA**
Question ID : **444792697**
Status : **Not Attempted and Marked For Review**

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Q.23 For some $\theta \in \left(0, \frac{\pi}{2}\right)$, let the eccentricity and the length of the latus rectum of the hyperbola $x^2 - y^2\sec^2\theta = 8$ be e_1 and l_1, respectively, and let the eccentricity and the length of the latus rectum of the ellipse $x^2\sec^2\theta + y^2 = 6$ be e_2 and l_2, respectively. If $e_1^2 = e_2^2\left(\sec^2\theta + 1\right)$, then $\left(\frac{l_1 l_2}{e_1 e_2}\right)\tan^2\theta$ is equal to _____

Given --
Answer :

Question Type : SA
Question ID : 444792700
Status : Not Attempted and Marked For Review

Q.24 The value of $\sum_{r=1}^{20}\left(\left|\sqrt{\pi\left(\int_0^r x\left|\sin \pi x\right| dx\right)}\right|\right)$ is_____

Given --
Answer :

Question Type : SA
Question ID : 444792699
Status : Not Attempted and Marked For Review

Q.25 Let PQR be a triangle such that $\overrightarrow{PQ} = -2\hat{i} - \hat{j} + 2\hat{k}$ and $\overrightarrow{PR} = a\hat{i} + b\hat{j} - 4\hat{k}$, $a, b \in \mathbb{Z}$. Let S be the point on QR, which is equidistant from the lines PQ and PR. If $\left|\overrightarrow{PR}\right| = 9$ and $\overrightarrow{PS} = \hat{i} - 7\hat{j} + 2\hat{k}$, then the value of $3a - 4b$ is _____

Given --
Answer :

Question Type : SA
Question ID : 444792698
Status : Not Attempted and Marked For Review

Section : Physics Section A

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$$\sum_{r=1}^{20} \left| \sqrt{\pi \int_0^r x|\sin \pi x|\,dx} \right|$$

$$I_1 = \int_0^r x|\sin \pi x|$$

$$I_1 = \int_0^r (r-x)\,|\sin \pi x|$$

$$2I_1 = \int_0^r |\sin \pi x| = r^2 \int_0^1 |\sin \pi x|$$

$$2I_1 = r^2 \cdot \frac{2}{\pi} \qquad I_1 = \frac{r^2}{\pi}$$

$$\sum_{r=1}^{20} \sqrt{r^2} = 210$$

PRASHANT JAIN

NTA JEE Mains Jan 2026

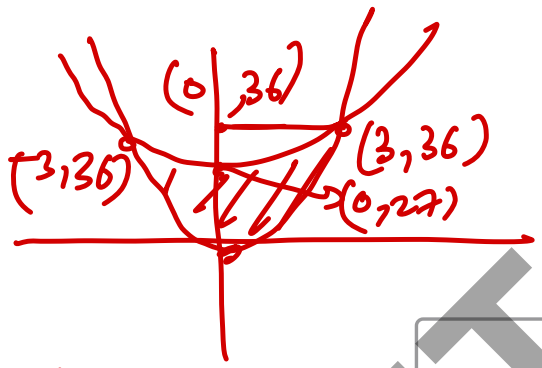
Application No	
Candidate Name	
Roll No.	
Test Date	28/01/2026
Test Time	3:00 PM - 6:00 PM
Subject	B. Tech

Section : **Mathematics Section A**

Q.1 Let $P_1 : y = 4x^2$ and $P_2 : y = x^2 + 27$ be two parabolas. If the area of the bounded region enclosed between P_1 and P_2 is six times the area of the bounded region enclosed between the line $y = \alpha x$, $\alpha > 0$ and P_1, then α is equal to :

Options

1. 15
2. 6
3. 8
4. 12



$\frac{2}{3}(3\times36 - 9\times3)$

$2\times27 = 54$

Question Type : **MCQ**
Question ID : **8606541670**
Option 1 ID : **8606545690**
Option 2 ID : **8606545687**
Option 3 ID : **8606545688**
Option 4 ID : **8606545689**
Status : **Not Answered**
Chosen Option : --

$g = \alpha x_1^2$ $\left(\frac{1}{2}, \frac{1}{3}\right)$

$54 = \alpha x_1^2 \Rightarrow \alpha x_1 = 4x_1^2$

$54 = 4x_1^2$

$x_1 = 3, -3$

Conic Medium ✓

PRASHANT JAIN

Q.2

Let the arithmetic mean of $\frac{1}{a}$ and $\frac{1}{b}$ be $\frac{5}{16}$, $a > 2$. If α is such that $a, 4, \alpha, b$ are in A.P., then the equation $\alpha x^2 - ax + 2(\alpha - 2b) = 0$ has :

Options

1. one root in $(1, 4)$ and another in $(-2, 0)$
2. one root in $(0, 2)$ and another in $(-4, -2)$
3. complex roots of magnitude less than 2
4. both roots in the interval $(-2, 0)$

Question Type : **MCQ**
Question ID : **8606541654**
Option 1 ID : **8606545625**
Option 2 ID : **8606545626**
Option 3 ID : **8606545623**
Option 4 ID : **8606545624**
Status : **Not Attempted and Marked For Review**
Chosen Option : **--**

Q.3

Given below are two statements :

Statement I : The function $f : \mathbf{R} \to \mathbf{R}$ defined by $f(x) = \frac{x}{1 + |x|}$ is one-one.

Statement II : The function $f : \mathbf{R} \to \mathbf{R}$ defined by $f(x) = \frac{x^2 + 4x - 30}{x^2 - 8x + 18}$ is many-one.

In the light of the above statements, choose the **correct answer** from the options given below :

Options

1. Both **Statement I** and **Statement II** are true
2. **Statement I** is true but **Statement II** is false
3. **Statement I** is false but **Statement II** is true
4. Both **Statement I** and **Statement II** are false

Question Type : **MCQ**
Question ID : **8606541652**
Option 1 ID : **8606545615**
Option 2 ID : **8606545617**
Option 3 ID : **8606545618**
Option 4 ID : **8606545616**
Status : **Answered**
Chosen Option : **2**

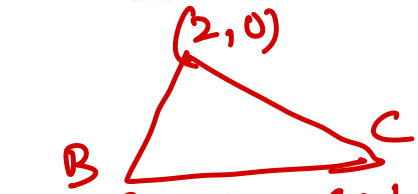
PRASHANT JAIN

Q.4 Let A be the focus of the parabola $y^2 = 8x$. Let the line $y = mx + c$ intersect the parabola at two distinct points B and C. If the centroid of the triangle ABC is $\left(\frac{7}{3}, \frac{4}{3}\right)$, then $(BC)^2$ is equal to :

Options

1. 32
2. 80
3. 89
4. 41

Conic
XI
medium



$7 = 2(t_1^2 + t_2^2 + 1)$

$4 = 4(t_1 + t_2) \Rightarrow t_1 + t_2 = 1$

$t_1^2 + t_2^2 = \frac{5}{2}$

$(2t_1^2, 4t_1)$ $(2t_2^2, 4t_2)$

$BC^2 = 4[(t_1 - t_2)(t_1 + t_2)]^2 + 16(t_1 - t_2)^2$

$= 4(t_1 - t_2)^2 + 16(t_1 - t_2)^2$

$= 20[t_1^2 + t_2^2 - 2t_1t_2]$

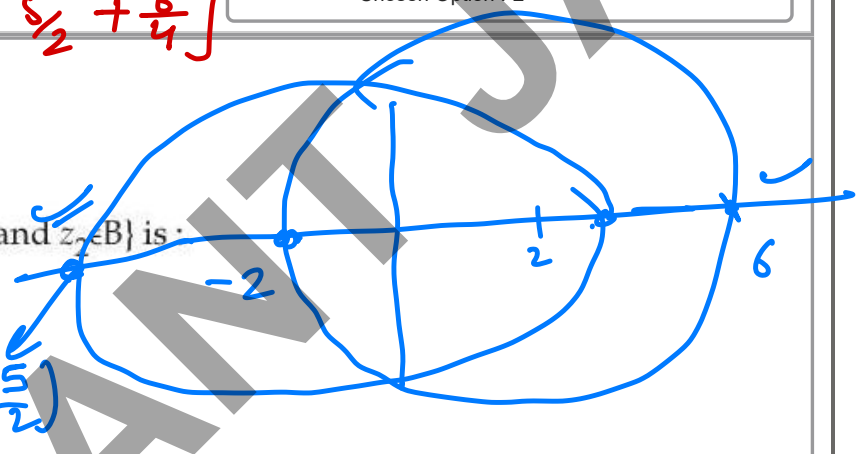
$= 20\left[\frac{5}{2} + \frac{6}{4}\right]$

Question Type : **MCQ**
Question ID : **8606541663**
Option 1 ID : **8606545659**
Option 2 ID : **8606545661**
Option 3 ID : **8606545662**
Option 4 ID : **8606545660**
Status : **Answered**
Chosen Option : **2**

Q.5 Let
$A = \{z \in \mathbb{C} : |z - 2| \leq 4\}$ and
$B = \{z \in \mathbb{C} : |z - 2| + |z + 2| = 5\}$.
Then the max $\{|z_1 - z_2| : z_1 \in A$ and $z_2 \in B\}$ is :

Options

1. 8
2. $\frac{15}{2}$
3. $\frac{17}{2}$
4. 9



$6 + \frac{5}{2} = \frac{17}{2}$

Complex No.
XI
Easy

Question Type : **MCQ**
Question ID : **8606541653**
Option 1 ID : **8606545620**
Option 2 ID : **8606545619**
Option 3 ID : **8606545621**
Option 4 ID : **8606545622**
Status : **Not Attempted and Marked For Review**
Chosen Option : **--**

PRASHANT JAIN

Q.6

Let the ellipse $E: \frac{x^2}{144}+\frac{y^2}{169}=1$ and the hyperbola $H: \frac{x^2}{16}-\frac{y^2}{\lambda^2}=-1$ have the same foci. If e and L respectively denote the eccentricity and the length of the latus rectum of H, then the value of $24(e+L)$ is :

Options

1. 148
2. 296
3. 126
4. 67

Question Type : **MCQ**
Question ID : **8606541662**
Option 1 ID : **8606545657**
Option 2 ID : **8606545658**
Option 3 ID : **8606545656**
Option 4 ID : **8606545655**
Status : **Not Attempted and Marked For Review**
Chosen Option : **--**

Q.7

The sum of the coefficients of x^{499} and x^{500} in $(1+x)^{1000}+x(1+x)^{999}+x^2(1+x)^{998}+\ldots+x^{1000}$ is :

Options

1. $^{1001}C_{501}$
2. $^{1002}C_{501}$
3. $^{1000}C_{501}$
4. $^{1002}C_{500}$

Question Type : **MCQ**
Question ID : **8606541656**
Option 1 ID : **8606545634**
Option 2 ID : **8606545631**
Option 3 ID : **8606545633**
Option 4 ID : **8606545632**
Status : **Answered**
Chosen Option : **3**

PRASHANT JAIN

Q.8 The sum of all the elements in the range of $f(x)=\text{Sgn}(\sin x)+\text{Sgn}(\cos x)+\text{Sgn}(\tan x)+\text{Sgn}(\cot x)$, $x \neq \frac{n\pi}{2}$, $n \in Z$, where $\text{Sgn}(t)=\begin{cases}1, & \text{if } t>0\\ -1, & \text{if } t<0\end{cases}$, is :

→ 4, 0, −2 = (2)

Options

1. 2
2. 4
3. 0
4. -2

Observation
easy
Functions
XII

Question Type : **MCQ**
Question ID : **8606541651**
Option 1 ID : **8606545611**
Option 2 ID : **8606545612**
Option 3 ID : **8606545614**
Option 4 ID : **8606545613**
Status : **Answered**
Chosen Option : **3**

Q.9 Considering the principal values of inverse trigonometric functions, the value of the expression $\tan\left(2\sin^{-1}\left(\frac{2}{\sqrt{13}}\right)-2\cos^{-1}\left(\frac{3}{\sqrt{10}}\right)\right)$ is equal to :

$\sin\theta_1 = \frac{2}{\sqrt{13}}$ $\cos\theta_2 = \frac{3}{\sqrt{10}}$

$\tan\theta_1 = \frac{2}{3}$ $\tan\theta_2 = \frac{1}{3}$

$\tan(\theta_1-\theta_2) = \frac{\frac{3}{9}}{1+\frac{2}{9}} = \frac{3}{11}$

$\tan(2\theta_1-\theta_2) = \frac{\frac{6}{11}}{1-\frac{9}{121}} = \frac{66}{112} = \frac{33}{56}$

Options

1. $\frac{33}{56}$
2. $\frac{16}{63}$
3. $-\frac{16}{63}$
4. $-\frac{33}{56}$

ITF
Medium
XII

Question Type : **MCQ**
Question ID : **8606541661**
Option 1 ID : **8606545652**
Option 2 ID : **8606545651**
Option 3 ID : **8606545653**
Option 4 ID : **8606545654**
Status : **Answered**
Chosen Option : **1**

PRASHANT JAIN

Q.10 Let the circle $x^2+y^2=4$ intersect x-axis at the points A(a, 0), a > 0 and B(b, 0). Let P(2 cosα, 2 sinα), $0<\alpha<\frac{\pi}{2}$ and Q(2 cosβ, 2 sinβ) be two points such that $(\alpha-\beta)=\frac{\pi}{2}$. Then the point of intersection of AQ and BP lies on :

Options

1. $x^2+y^2-4x-4y-4=0$
2. $x^2+y^2-4y-4=0$
3. $x^2+y^2-4x-4=0$
4. $x^2+y^2-4x-4y=0$

Question Type : **MCQ**
Question ID : **8606541660**
Option 1 ID : **8606545649**
Option 2 ID : **8606545648**
Option 3 ID : **8606545647**
Option 4 ID : **8606545650**
Status : **Not Answered**
Chosen Option :

Q.11 Let $[\cdot]$ denote the greatest integer function. Then $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{12(3+[x])}{3+[\sin x]+[\cos x]}\right)dx$ is equal to :

Options

1. $12\pi+5$
2. $11\pi+2$
3. $15\pi+4$
4. $13\pi+1$

Question Type : **MCQ**
Question ID : **8606541667**
Option 1 ID : **8606545678**
Option 2 ID : **8606545677**
Option 3 ID : **8606545675**
Option 4 ID : **8606545676**
Status : **Answered**
Chosen Option : **3**

PRASHANT JAIN

Q.12

Let $f(x)=\lim\limits_{\theta\to 0}\left(\dfrac{\cos\pi x-x^{\left(\frac{2}{\theta}\right)}\sin(x-1)}{1+x^{\left(\frac{2}{\theta}\right)}(x-1)}\right)$, $x\in\mathbf{R}$. Consider the following two statements :

(I) $f(x)$ is discontinous at $x=1$.

(II) $f(x)$ is continous at $x=-1$.

Then,

Options

1. Only (II) is True
2. Neither (I) nor (II) is True
3. Both (I) and (II) are True
4. Only (I) is True

LCD XII Medium

Question Type : **MCQ**
Question ID : **8606541666**
Option 1 ID : **8606545672**
Option 2 ID : **8606545674**
Option 3 ID : **8606545673**
Option 4 ID : **8606545671**
Status : **Answered**
Chosen Option : **2**

PRASHANT JAIN

Q.13 The probability distribution of a random variable X is given below :

X	4k	$\frac{30}{7}$k	$\frac{32}{7}$k	$\frac{34}{7}$k	$\frac{36}{7}$k	$\frac{38}{7}$k	$\frac{40}{7}$k	6k
P(X)	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{1}{15}$

If $E(X) = \frac{263}{15}$, then $P(X < 20)$ is equal to :

Options

1. $\frac{8}{15}$
2. $\frac{3}{5}$
3. $\frac{14}{15}$
4. $\frac{11}{15}$

$\frac{263}{15} = \frac{k(8 + \frac{30}{7} + \frac{64}{7} -)}{15}$

$12 = \frac{7}{2}$

$P(X < 20) = \frac{11}{15}$

No elegant

Probability

XII

medium

Question Type : **MCQ**
Question ID : **8606541658**
Option 1 ID : **8606545639**
Option 2 ID : **8606545640**
Option 3 ID : **8606545642**
Option 4 ID : **8606545641**
Status : **Not Answered**
Chosen Option : **--**

PRASHANT JAIN

Q.14

Let $y = y(x)$ be the solution of the differential equation $x\frac{dy}{dx} - y = x^2 \cot x$, $x \in (0, \pi)$. If $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$, then

$6y\left(\frac{\pi}{6}\right) - 8y\left(\frac{\pi}{4}\right)$ is equal to :

Options

1. -3π
2. 3π
3. $-\pi$
4. π

Question Type : **MCQ**
Question ID : **8606541669**
Option 1 ID : **8606545685**
Option 2 ID : **8606545686**
Option 3 ID : **8606545683**
Option 4 ID : **8606545684**
Status : **Answered**
Chosen Option : **3**

Q.15

Let P be a point in the plane of the vectors $\vec{AB} = 3\hat{i} + \hat{j} - \hat{k}$ and $\vec{AC} = \hat{i} - \hat{j} + 3\hat{k}$ such that P is equidistant from the lines AB and AC. If $\left|\vec{AP}\right| = \frac{\sqrt{5}}{2}$, then the area of the triangle ABP is :

Options

1. $\frac{\sqrt{26}}{4}$
2. $\frac{\sqrt{30}}{4}$
3. $\frac{3}{2}$
4. 2

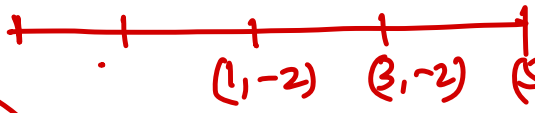
Question Type : **MCQ**
Question ID : **8606541664**
Option 1 ID : **8606545664**
Option 2 ID : **8606545665**
Option 3 ID : **8606545663**
Option 4 ID : **8606545666**
Status : **Answered**
Chosen Option : **2**

PRASHANT JAIN

Q.16 An ellipse has its center at $(1, -2)$, one focus at $(3, -2)$ and one vertex at $(5, -2)$. Then the length of its latus rectum is :

Options

1. $6\sqrt{3}$
2. $4\sqrt{3}$
3. 6 ✓
4. $\frac{16}{\sqrt{3}}$



$ae = 2$

$a = 4$

$e = 1/2$

$b^2 = 12$

$a^2 = 4$

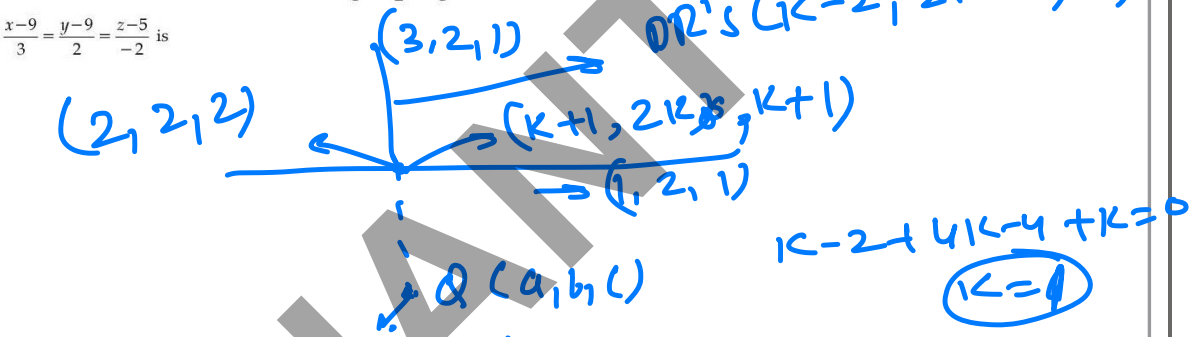
easy conic XI

Question Type : **MCQ**
Question ID : **8606541659**
Option 1 ID : **8606545643**
Option 2 ID : **8606545645**
Option 3 ID : **8606545646**
Option 4 ID : **8606545644**
Status : **Answered**
Chosen Option : **3**

Q.17 Let Q(a, b, c) be the image of the point P(3, 2, 1) in the line $\frac{x-1}{1} = \frac{y}{2} = \frac{z-1}{1}$. Then the distance of Q from the line $\frac{x-9}{3} = \frac{y-9}{2} = \frac{z-5}{-2}$ is

Options

1. 5
2. 6
3. 7 ✓
4. 8



$K - 2 + 4K - 4 + K = 0$

$K = 1$

DR's $(3K+8, 2K+7, -2K+2)$

$3 \quad 2 \quad -2$

$9K + 24 + 4K + 14 + 4K - 4 = 0$

$K = -2$

$(2, 3, 6)$ → 7

3-D medium XII

Question Type : **MCQ**
Question ID : **8606541665**
Option 1 ID : **8606545668**
Option 2 ID : **8606545669**
Option 3 ID : **8606545670**
Option 4 ID : **8606545667**
Status : **Answered**
Chosen Option : **3**

PRASHANT JAIN

Q.18

Let $f(x)=\int \frac{dx}{x^{\left(\frac{2}{3}\right)}+2x^{\left(\frac{1}{2}\right)}}$ be such that $f(0)=-26+24\log_e(2)$. If $f(1)=a+b\log_e(3)$, where $a,b\in \mathbf{Z}$, then $a+b$ is equal to :

Options

1. -5
2. -26
3. -11
4. -18

$x = t^6$

$6\int \frac{t^5\,dt}{t^4+2t^3}$

$6\int \frac{t^2-4+4}{t+2}\,dt$

$6\left(t-2+\frac{24}{t+2}\right)$

$3t^2-12t+24\ln(t+2)+C$

$24\ln 3 - 35 = 0$

II
medium
XII

Question Type : **MCQ**
Question ID : **8606541668**
Option 1 ID : **8606545682**
Option 2 ID : **8606545679**
Option 3 ID : **8606545681**
Option 4 ID : **8606545680**
Status : **Not Answered**
Chosen Option : **--**

$C = -26$

Q.19

$\frac{6}{3^{26}}+\frac{10\cdot 1}{3^{25}}+\frac{10\cdot 2}{3^{24}}+\frac{10\cdot 2^2}{3^{23}}+\ldots+\frac{10\cdot 2^{24}}{3}$ is equal to :

Options

1. 2^{26}
2. 2^{25}
3. 3^{26}
4. 3^{25}

$\frac{6}{3^{26}}+\frac{10}{3^{25}}\left(1+6+6^2+\ldots+6^{24}\right)$

$\frac{10}{3^{26}}\left(\frac{6^{25}-1}{5}\right) = 2^{26}$

S&S
XI
easy

Question Type : **MCQ**
Question ID : **8606541655**
Option 1 ID : **8606545630**
Option 2 ID : **8606545629**
Option 3 ID : **8606545628**
Option 4 ID : **8606545627**
Status : **Not Answered**
Chosen Option : **--**

PRASHANT JAIN

Q.20 Given below are two statements :

Statement I : $25^{13} + 20^{13} + 8^{13} + 3^{13}$ is divisible by 7.

Statement II : The integral part of $(7 + 4\sqrt{3})^{25}$ is an odd number.

In the light of the above statements, choose the **correct answer** from the options given below :

Options

1. Both **Statement I** and **Statement II** are true
2. **Statement I** is false but **Statement II** is true
3. Both **Statement I** and **Statement II** are false
4. **Statement I** is true but **Statement II** is false

Question Type : **MCQ**
Question ID : **8606541657**
Option 1 ID : **8606545635**
Option 2 ID : **8606545638**
Option 3 ID : **8606545636**
Option 4 ID : **8606545637**
Status : **Not Answered**
Chosen Option : **--**

Section : **Mathematics Section B**

Q.21 If $\sum_{r=1}^{25}\left(\frac{r}{r^4 + r^2 + 1}\right) = \frac{p}{q}$, where p and q are positive integers such that gcd(p, q) = 1, then p + q is equal to _____.

Given Answer : **976**

Question Type : **SA**
Question ID : **8606541672**
Status : **Answered**

Q.22 If the distance of the point P(43, α, β), β < 0, from the line $\vec{r} = 4\hat{i} - \hat{k} + \mu(2\hat{i} + 3\hat{k})$, $\mu \in \mathbf{R}$ along a line with direction ratios 3, −1, 0 is $13\sqrt{10}$, then $\alpha^2 + \beta^2$ is equal to _____

Given Answer : **--**

Question Type : **SA**
Question ID : **8606541674**
Status : **Not Answered**

PRASHANT JAIN

Q.23 Let f be a differentiable function satisfying $f(x) = 1 - 2x + \int_0^x e^{(x-t)} f(t)\,dt, x \in \mathbf{R}$ and let $g(x) = \int_0^x (f(t)+2)^{15}(t-4)^6(t+12)^{17}\,dt, x \in \mathbf{R}$. If p and q are respectively the points of local minima and local maxima of g, then the value of $|p+q|$ is equal to _____.

Given --
Answer :

Question Type : **SA**
Question ID : **8606541675**
Status : **Not Answered**

Q.24 Let $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ and B be two matrices such that $A^{100} = 100B + I$. Then the sum of all the elements of B^{100} is _____

Given --
Answer :

Question Type : **SA**
Question ID : **8606541671**
Status : **Not Answered**

Q.25 Three persons enter in a lift at the ground floor. The lift will go upto 10th floor. The number of ways, in which the three persons can exit the lift at three different floors, if the lift does not stop at first, second and third floors, is equal to _____.

Given --
Answer :

Question Type : **SA**
Question ID : **8606541673**
Status : **Not Answered**

Section : **Physics Section A**

Options

1.

2.

3.

4.

Question Type : **MCQ**
Question ID : **8606541693**
Option 1 ID : **8606545764**
Option 2 ID : **8606545767**
Option 3 ID : **8606545765**
Option 4 ID : **8606545766**
Status : **Answered**
Chosen Option : **4**

PRASHANT JAIN